

MAGNETOM

Startup System

System Start-up

Aera
Skyra
Avanto fit
Skyra fit

Document Version

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1.1 General



The specification for the air conditioning system must be met in the examination room. If the specifications are not met, condensed water will damage the TX-box and RFCEL.

1.1.1 Prerequisites

- Room light must be functioning in every room of the MR site.
- The system components have been positioned and installed according to the Planning Guide and Installation Instructions.
- The RF integrity of the RF-room was measured after installing the RF-room and its components.

1.1.2 Required Aids and Tools

The service tools required for starting up the MR system include:

- Water pump (if no on-site water supply is available, manual water pump part # 8115136)
- Service tool kit (complete)
 - RF-cable 2-m BNC (part no.: 8733479, included in the kit) is required for the tune-up procedure.
- Digital multimeter
- Protective conductor test device
- Helium syphon (dewar side)
- Torque wrench 20-50 Nm (part no.: 8395803)
- Torque wrench 5-20 Nm (part no.: 8395829)
- Accessory for torque wrench (part no.: 8396116)
- Set of standard tools
- ESD service kit

1.1.2.1 Magnet system

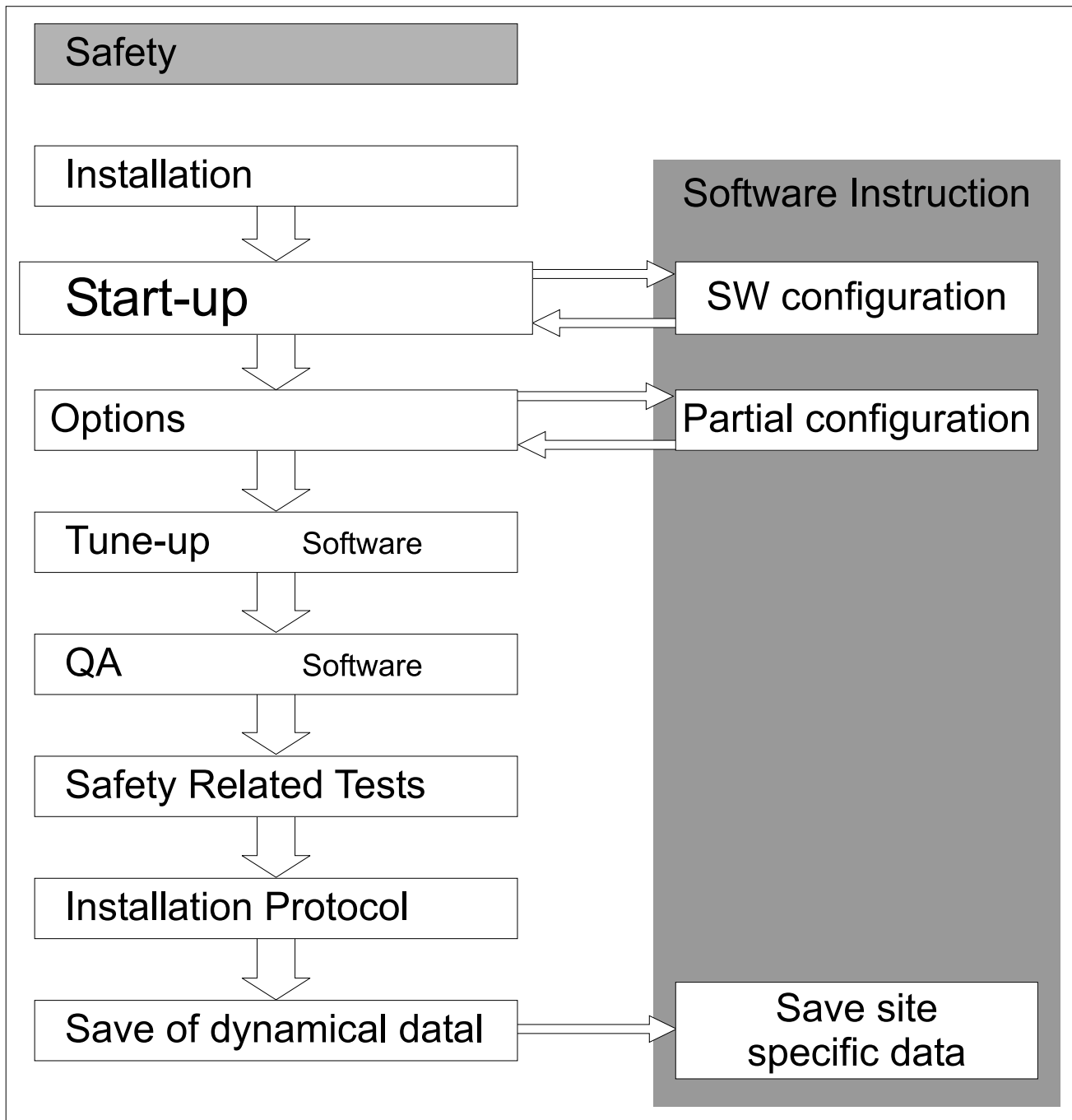
- Array shim device
- Magnet power supply
- Helium leak tester (part no.:9900978)
- Heat gun
- Helium service syphon (magnet side)
- Protective clothing (face protection shield, gloves, protective shoes, etc.)

1.1.3 Overall flowchart



System start-up should be performed in the order of the steps described. Complete each step before you start working on the next one.

Fig. 1: Startup flowchart



1.1.4 Safety

Observe the safety notes in the register "General Safety Notes" and "MR-Specific Safety Information" of "Technical Documents" when performing work on the system. This is important for your own safety and for the safety of third parties, as well as to protect the MR system and parts.

Read the special safety notes regarding:

- General safety for service personnel.
- The "Strong Magnetic Field" warning sign.
- The "Cardiac Pacemaker" warning sign.
- Bringing ferromagnetic materials into the magnet area.
- Safety Information for working when the magnet is under field.
- Hazards involved in cryogen filling.
- Working with heavy loads.
- Protective conductor test prior to switch-on and after repair work.
- Reduction of the magnetic field to 20 mT when quenching the magnet.

Using the service platform

- Before using the service platform, perform the relevant safety checks.
- Assemble the platform as described in the TSG.

2.1 Safety

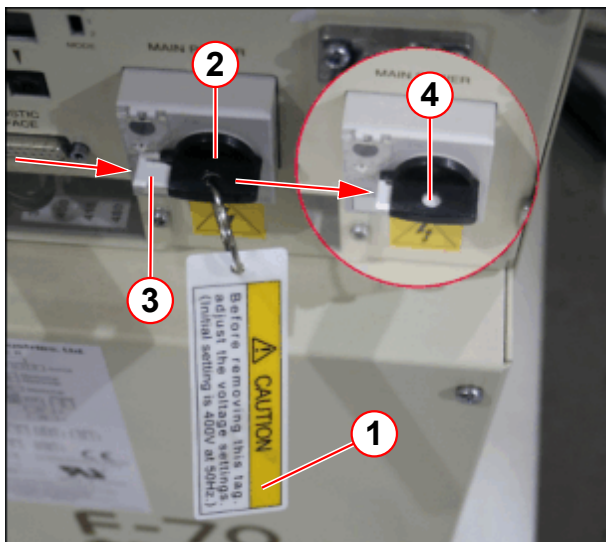


For Safety Information, refer to the MR-Specific Safety Information.

2.2 Checking the adaptation to line voltage

- Switch off the helium compressor.

Fig. 2: F70 Main power switch - locking device



- (1) Voltage caution label
- (2) Main power switch
- (3) White locking device
- (4) Device fully engaged (view through hole)

- Engage the **white switch locking device** fully **before** attempting to **turn off** the compressor.
- Push in the white locking device fully.
 - » The **device** should be **visible** through the **blocking hole** (pos. 4).

- Switch off F102 SEP/MREF.
- Switch off F100 EPC.



WARNING

High voltage

Observe the warning. Otherwise death or serious injury may occur.

- » Switch OFF the on-site line voltage (EPC: F100, F102) and secure it with a lock to prevent anyone from switching it on inadvertently.
- » Verify that no line voltage is present at the EPC, GPA, SEP and LHe compressor unit.

- Ask the project manager about the on-site line voltage or perform your own measurement.

The on-site line voltage is:	VAC
The on-site line frequency is:	Hz

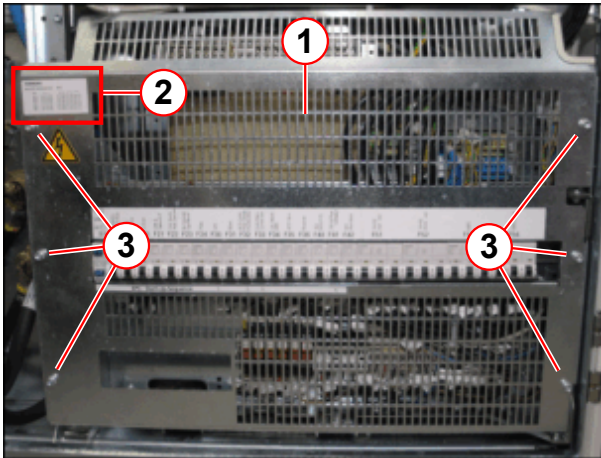


For detailed information, refer to the "Installation Manual, Starting the magnet cooling".

Checking the line voltage adjustments of the components.

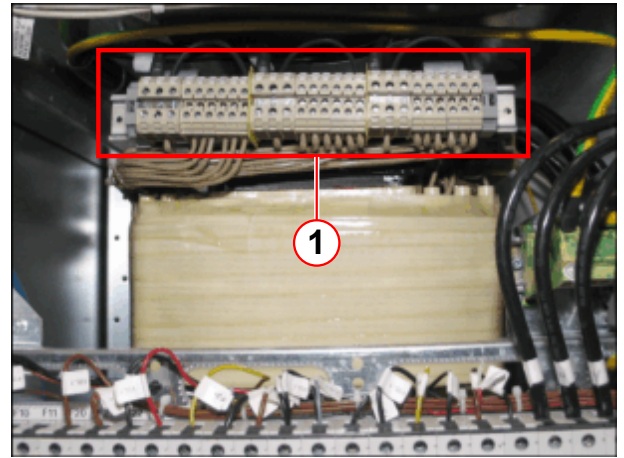
- Check to ensure that the line voltage adjustment of the EPC is selected correctly.

Fig. 3: Protective cover of EPC line voltage



- (1) Protective cover
- (2) Line voltage adaption label
- (3) Screws

Fig. 4: EPC line voltage adaption

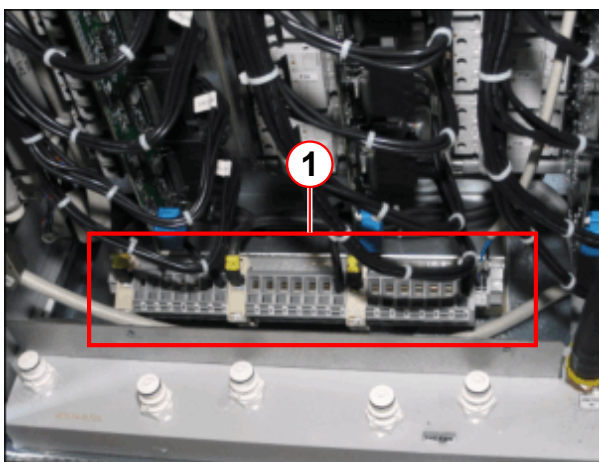


- (1) Terminal blocks

- If required, select it correctly.

- Check to ensure that the line voltage adjustment of the GPA is selected correctly.

Fig. 5: Terminal blocks of GPA line voltage

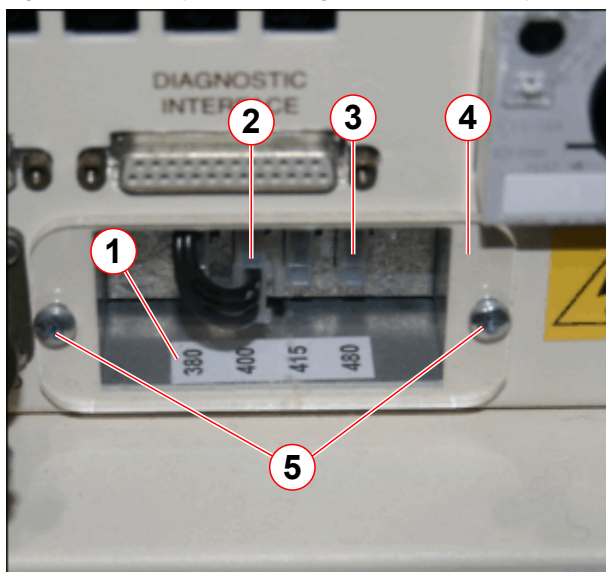


- (1) Terminal blocks

- If required, select it correctly.

- Check to ensure that the line voltage adjustment of the helium compressor is selected correctly.

Fig. 6: F70 Compressor voltage selection access panel

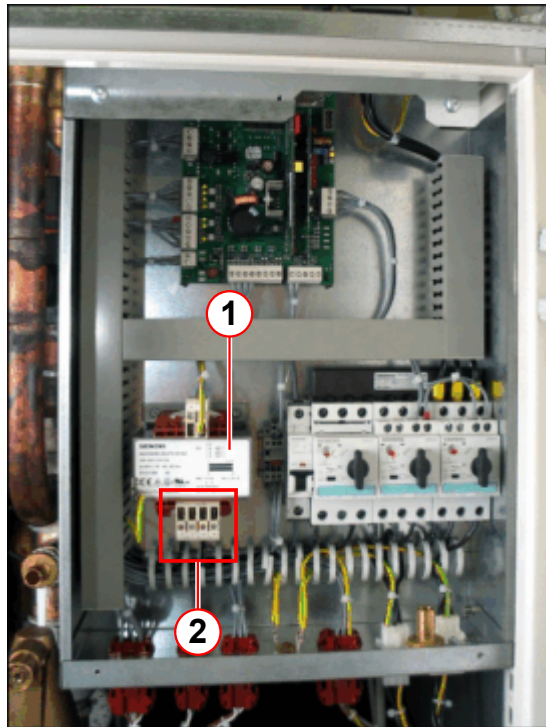


- If required, select it correctly.

- (1) Voltage label
- (2) Voltage selection connectors
- (3) Voltage tap
- (4) Clear plastic cover
- (5) Screws (2 off)

- Check to ensure that the line voltage adjustment of the SEP is selected correctly.

Fig. 7: SEP line voltage adaption



- (1) T1
(2) Terminal blocks

- If required, select it correctly.

- » Refer to Installation Instructions / Starting the magnet cooling /Line power adaptation

2.3 Gradient Coil position check



CAUTION

Gradient coil (GC) movement during transportation.

Problems with shimming.

Problems with GC cable connections.

- » Check the alignment and z-position of the GC.
- » Document the check in the log file.

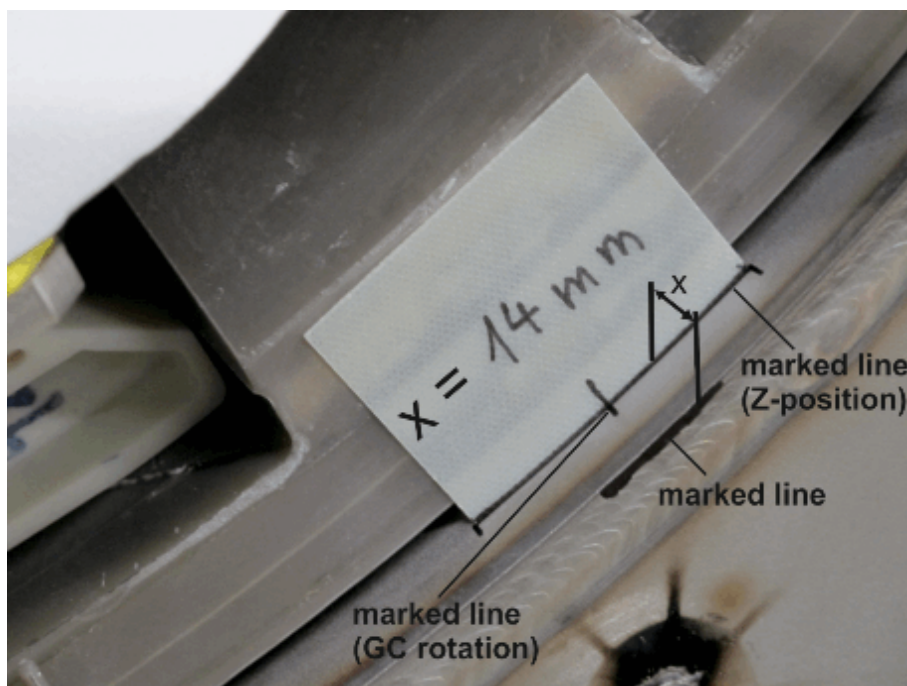
The position of the GC is marked before leaving the factory.

- Marked line indicates the z-position of the GC.
- Marked line indicates the alignment of the GC (GC rotation).
- Distance (X), noted on the label.

This value is written in the log-file (with SW VE11):

C:\MedCom\log\GC_Positionindicator.rtf

Fig. 8: Gradient coil position



Procedure of the gradient coil position check:

- Check the marked position of the GC
 - » GC rotation
 - » Z-position

- Measure the distance (x) from the marked line to the GC.
 - » Open the logfile with WordPad.
 - » Write the value (x) into the logfile: c:\ MedCom \ log \ GC_Positionindicator.rtf



Do not change the file name!
The file will be transferred automatically with IB_sample.

Service actions if movement of the GC during transportation was detected.

Deviation between Value (x) in the factory and distance (x) at customer	Service action
Deviation $\leq$ +/- 3mm	No activities
Deviation $>$ +/- 3mm	■ Check the air cushions ■ Loosen the gradient connection at the connection plate. ■ Loosen the coax connection. ■ Re-align the gradient coil. ■ Re-connect the gradient cables.

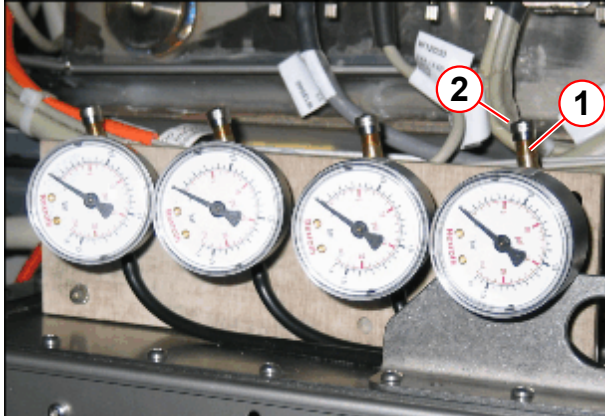
2.4 Deflating of the Air-Cushions



The gradient coil air cushions are no longer required during the operation of the MR-system.

In principle the air cushions will be only used as transport lock during the delivery.

Fig. 9: Pressure gauges



- (1) Deflating valve
- (2) Cap

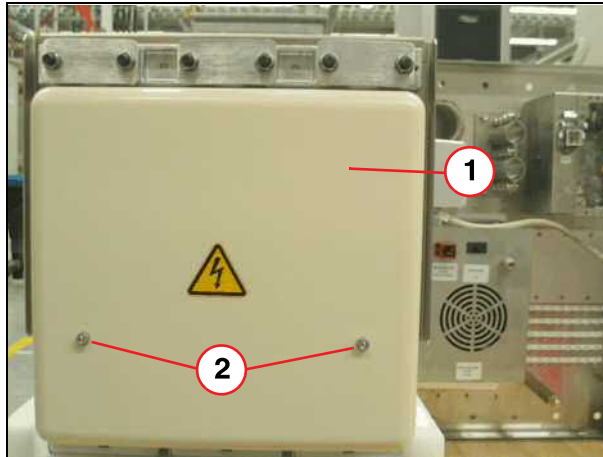
- Remove the caps of the deflating valves.
- Deflate the air cushions using the 4 deflating valves.

2.5 Checking the gradient connections

2.5.1 Gradient cables outside the RF-room

2.5.1.1 Cable connection at the RF-filter plate

Fig. 10: Protective cover, gradient filter (technical room)



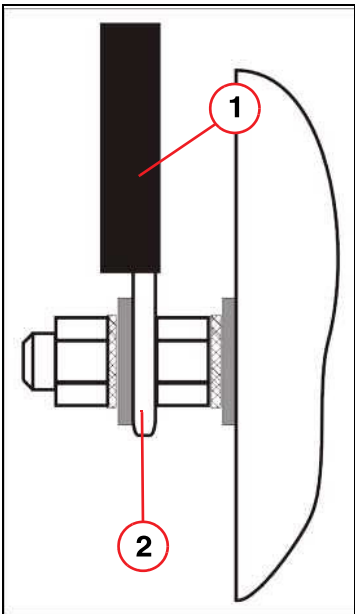
- (1) Protective cover
- (2) Screws

- Remove the protective cover (1) of the gradient filters at the RF-filter plate.

Fig. 11: Gradient cable connection at RF-filter plate

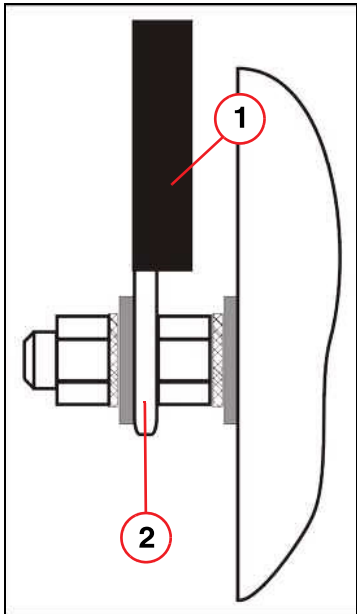


Fig. 12: Filter plate 1/1



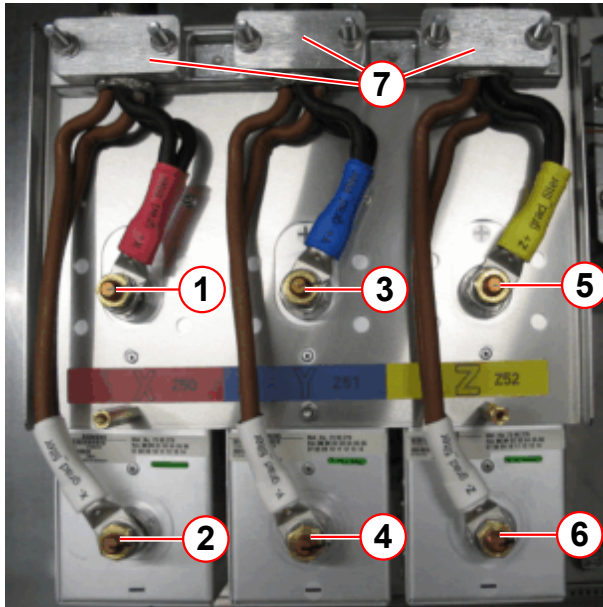
- (1) Gradient cable
- (2) Cable lug

Fig. 13: Filter plate 2/1



- (1) Gradient cable
- (2) Cable lug

Fig. 14: Gradient cable connection at RF-filter plate (equipment room)



- (1) X+
- (2) X-
- (3) Y+
- (4) Y-
- (5) Z+
- (6) Z-
- (7) Cable clamps

- Do a polarity check of the gradient cables (X, Y, and Z) to the gradient filters.
- Check the torque of the cable connections. (X+ / X-, Y+ / Y-, Z+ / Z-) (→ 1, 2, 3, 4, 5, 6/ Fig. 14 Page 19)



Use a torque of 22 ± 1 Nm.

- Check whether the cable clamps are tightened. (→ 7/ Fig. 14 Page 19)
- Attach the protective cover.



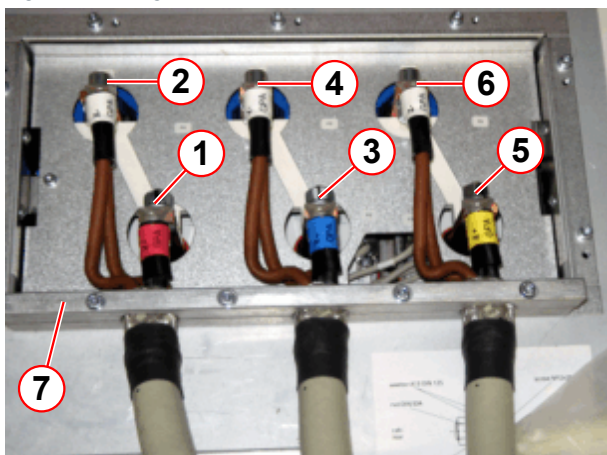
The cable clamps (→ 7/ Fig. 14 Page 19) should clamp the shielding only. Ensure that the insulation is not clamped.

2.5.1.2 Check the connection at the GPA



Ensure that the gradient cables are connected as shown in the figure. (→ Fig. 15 Page 20) The shield must be clamped between the rails.

Fig. 15: GPA gradient cable connection



- (1) X+
- (2) X-
- (3) Y+
- (4) Y-
- (5) Z+
- (6) Z-
- (7) Rail

- Remove the protective cover.
- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (X+ / X-, Y+ / Y-, Z+ / Z-) (→ 1, 2, 3, 4, 5, 6/Fig. 15 Page 20)



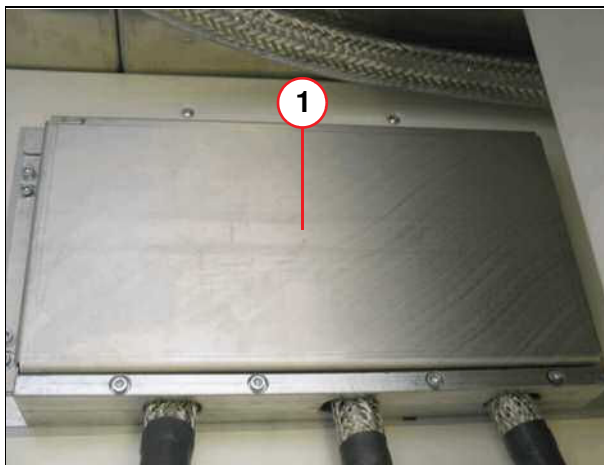
Use a torque of 40 ± 4 Nm.

- Check whether the cable clamps are tightened. (→ 7/Fig. 15 Page 20)



The cable clamps (→ 7/Fig. 15 Page 20) should clamp the shielding only. Ensure that the insulation is not clamped.

Fig. 16: GPA gradient cover 1



- (1) Gradient cover

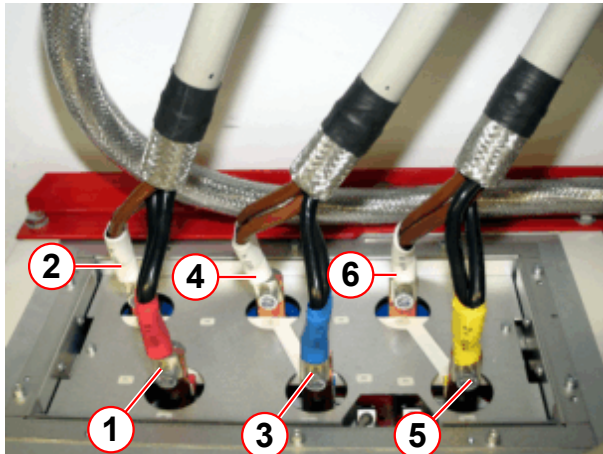
- Install the protective cover (1).

2.5.1.3 Connecting the GPA using a cable kit that is longer than 1.5 m



The cable lugs can point toward the back or front; do not rotate them through 180 degrees.

Fig. 17: GPA gradient cable connection



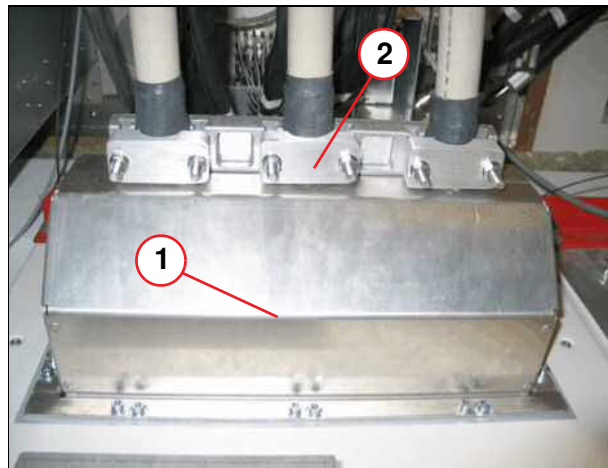
- (1) X+
- (2) X-
- (3) Y+
- (4) Y-
- (5) Z+
- (6) Z-

- Remove the protective cover.
- Do a polarity check of the gradient cables (X, Y, Z)
- Check the torque of the cable connections (X+ / X-, Y+ / Y-, Z+ / Z-). (→ 1, 2, 3, 4, 5, 6/Fig. 17 Page 21)



Use a torque of 40 ± 4 Nm.

Fig. 18: GPA gradient cover



- (1) Gradient cover
- (2) Clamps

- Check whether the cable clamps are tightened. (→ 2/Fig. 18 Page 21)
- Attach the protective cover.



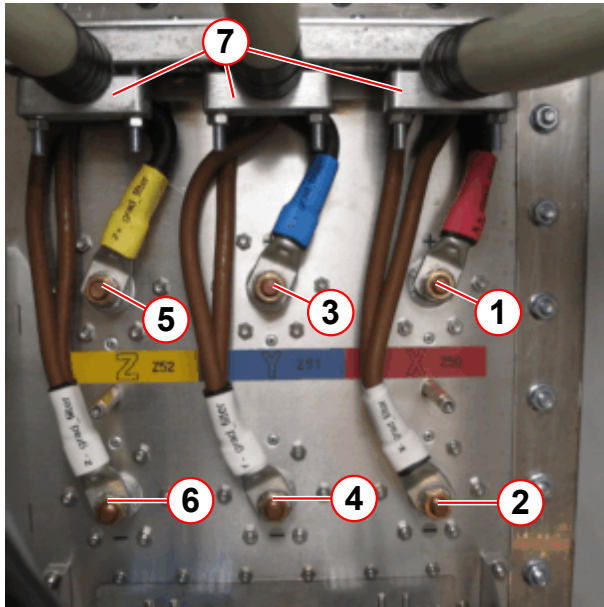
The cable clamps ((→ 2/Fig. 18 Page 21) should clamp the shielding only. Ensure that the insulation is not clamped.

2.5.2 Checking the gradient cables in the RF-room (Aera / Skyra / Skyra fit)



The cable lugs can point toward the back or front; do not rotate them through 180 degrees.

Fig. 19: Gradient connection at RF-filter plate (examination room)



- (1) X+
- (2) X-
- (3) Y+
- (4) Y-
- (5) Z+
- (6) Z-
- (7) Clamp

RF-filter plate:

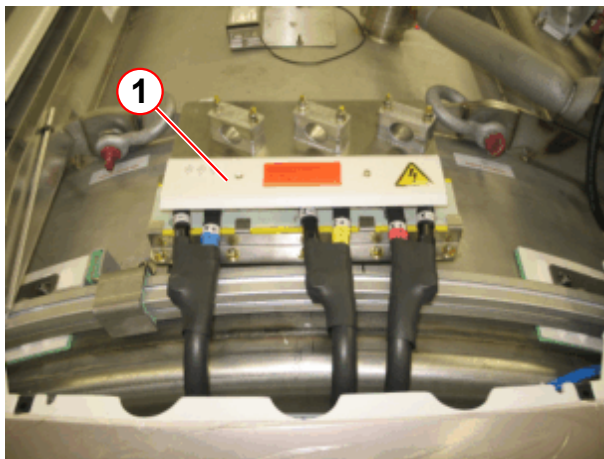
- Remove the protective cover.
- Do a polarity check of the gradient cables (X, Y, and Z) at the RF-filter plate.
- Check the torque of the cable connections (X+ / X-, Y+ / Y-, Z+ / Z-). (→ 1, 2, 3, 4, 5, 6/ Fig. 19 Page 22)



Use a torque of 22 ± 1 Nm.

- Check whether the cable clamps are tightened. (→ 7/ Fig. 19 Page 22)
- Install the protective cover.

Fig. 20: Magnet gradient connection plate

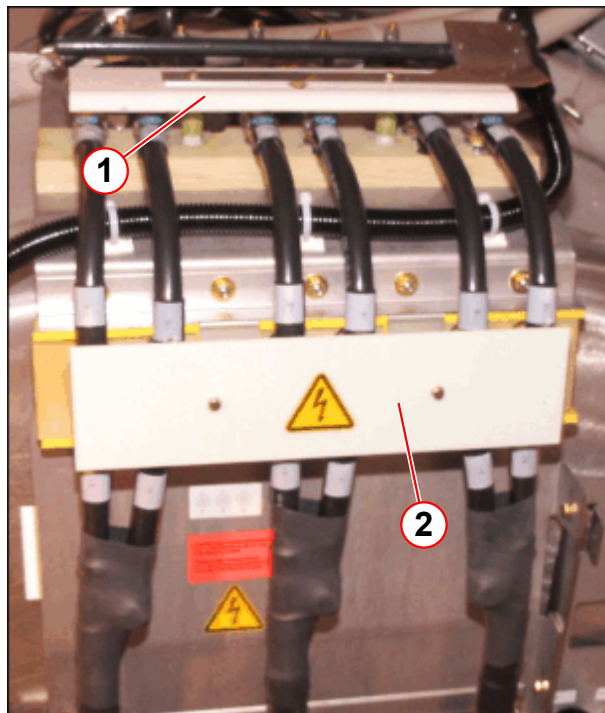


- (1) Protective cover

Aera, Skyra

- Remove the protective cover (1) at the magnet gradient connection plate.

Fig. 21: Skyra fit, Gradient connection plate

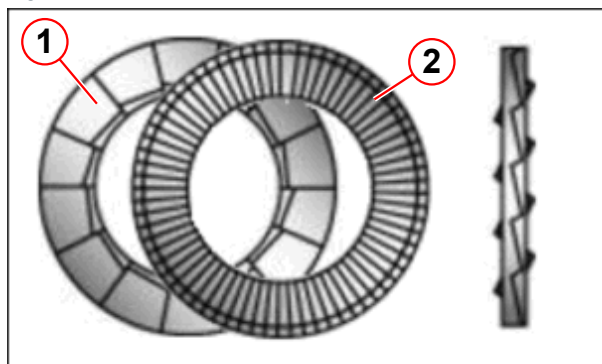


- (1) Protective cover 1
- (2) Protective cover 2

Skyra fit

- Remove the protective cover (1) and protective cover (2) at the gradient connection plate.

Fig. 22: NORD-LOCK washer



- (1) Cam side
- (2) Ridged side

A Nord-Lock washer (comprising two parts) must be used between the lug and the nut. Align the washer parts correctly (see the figure on the left).

At the connection plate:**Check the GC mounting bolts with the correct torque.**

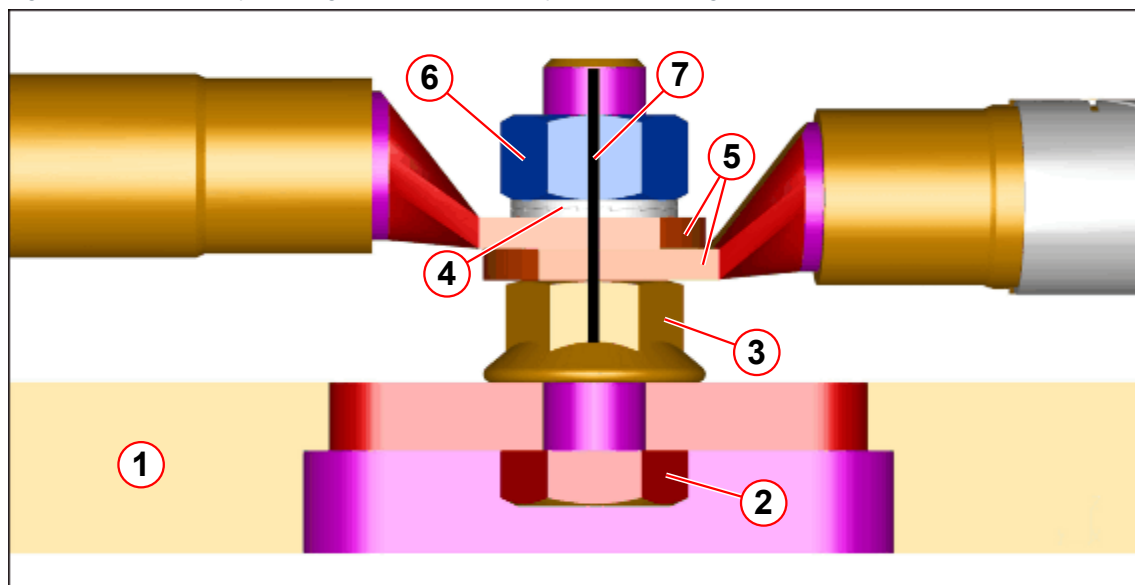
- Check the correct configuration of the connection.
- Fasten the flange nut (3) with a non-magnetic torque wrench (SW18).

For better access remove nut (6) and cable (5)



- Apply a torque of 21 ± 2 Nm.

Fig. 23: Correct set-up of the gradient connection plate at the magnet



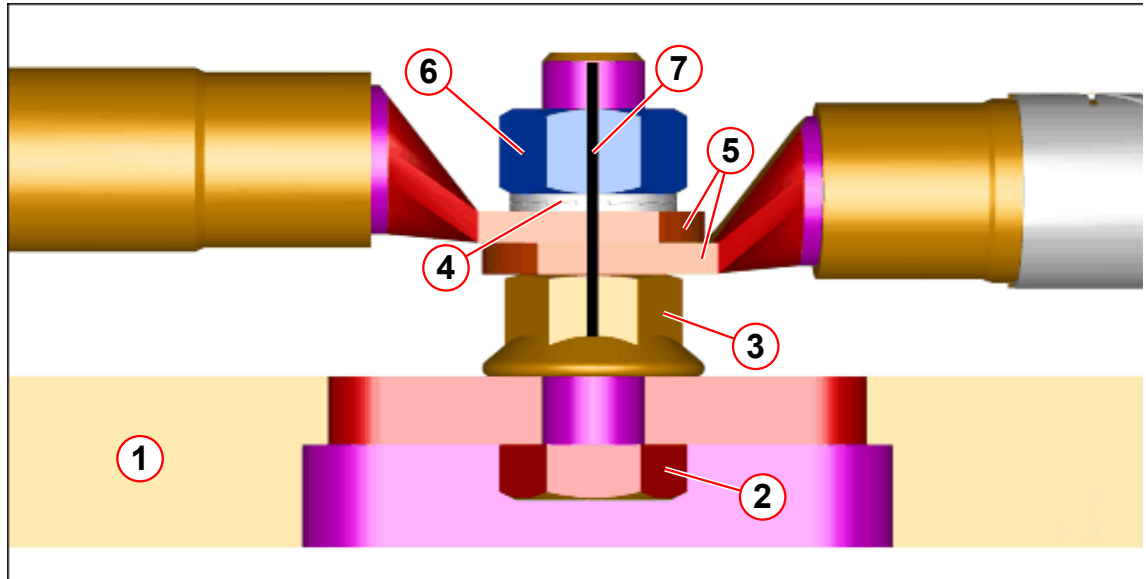
- (1) Connection plate
- (2) 12 mm bolt (SW 19)
- (3) Flange nut (SW 18) (torque 21 ± 2 Nm)
- (4) NORD-LOCK washers
- (5) Gradient cable lugs
- (6) Nut (SW 19) (torque 40 ± 4 Nm)
- (7) Mark with waterproof pen

- Fasten the cables with NORD-LOCK washer and nut (6) using a non-magnetic torque wrench (SW19).



- Apply a torque of 40 ± 4 Nm.
- With a waterproof pen, mark the positions of all nuts with a line over the connections.

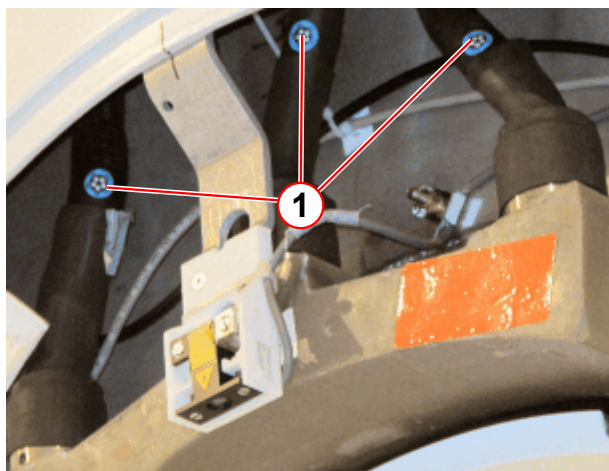
Fig. 24: Correct set-up of the gradient connection plate at the magnet



- (1) Connection plate
- (2) 12 mm bolt (SW 19)
- (3) Flange nut (SW 18) (torque 21 ± 2 Nm)
- (4) NORD-LOCK washers
- (5) Gradient cable lugs
- (6) Nut (SW 19) (torque 40 ± 4 Nm)
- (7) Mark with waterproof pen

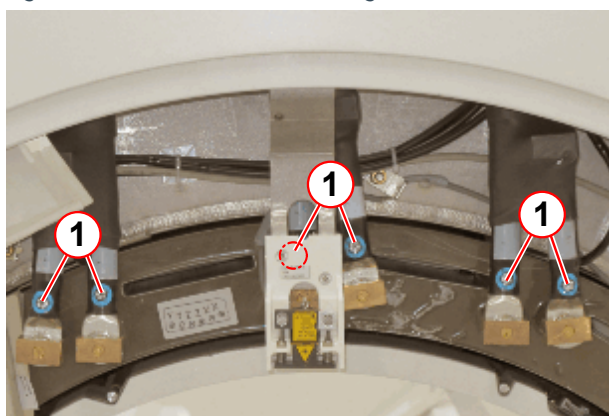
2.5.2.1 Temperature Indicator Attachment (Clock Stickers)

Fig. 25: GC cable connection (XQ-gradient)



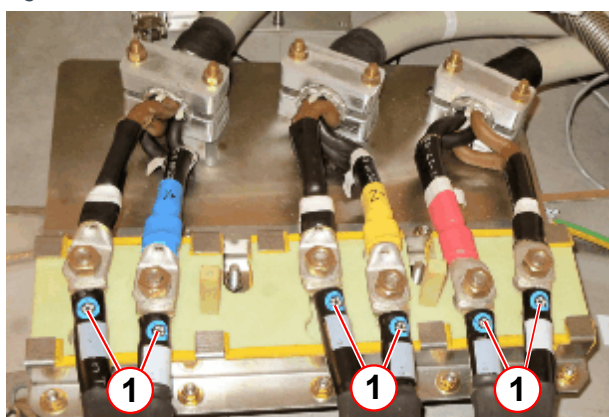
(1) Temperature Indicators (Clock Stickers)

Fig. 26: GC cable connection (XJ-gradient)



(1) Temperature Indicators (Clock Stickers)

Fig. 27: Gradient Connection Plate



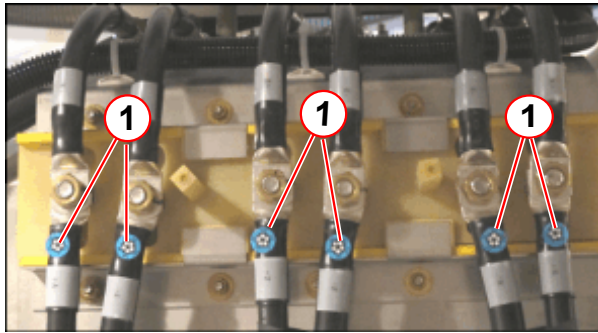
(1) Temperature Indicators (Clock Stickers)

Check or attach Temperature Indicators close to the screwed connections:

- at each GC connection.

- Check or attach temperature indicators at each connection of the gradient connection plate.
- Install the protective cover.

Fig. 28: Gradient cable connection



(1) Temperature indicators

Skyra fit

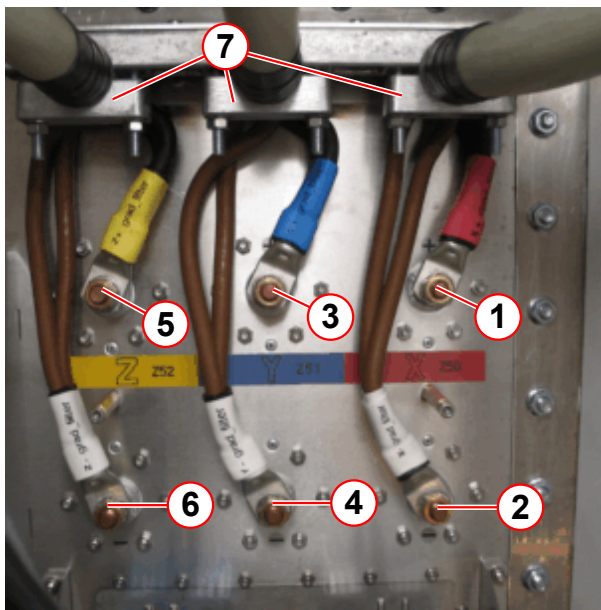
- Check or attach temperature indicators at each connection of the additional gradient connection plate.
- Install the protective cover.

2.5.3 Checking the gradient cables in the RF-room (Avanto fit)



The cable lugs can point toward the back or front; do not rotate them through 180 degrees.

Fig. 29: Gradient connection at RF-filter plate (examination room)



- (1) X+
- (2) X-
- (3) Y+
- (4) Y-
- (5) Z+
- (6) Z-
- (7) Clamp

RF-filter plate

- Remove the protective cover.
- Do a polarity check of the gradient cables (X, Y, and Z) at the RF-filter plate.
- Check the torque of the cable connections (X+ / X-, Y+ / Y-, Z+ / Z-). (→ 1, 2, 3, 4, 5, 6/Fig. 29 Page 27)



Use a torque of 22 ± 1 Nm.

- Check whether the cable clamps are tightened. (→ 7/Fig. 29 Page 27)
- Install the protective cover.

Fig. 30: Fasten GC connectors with 30 Nm



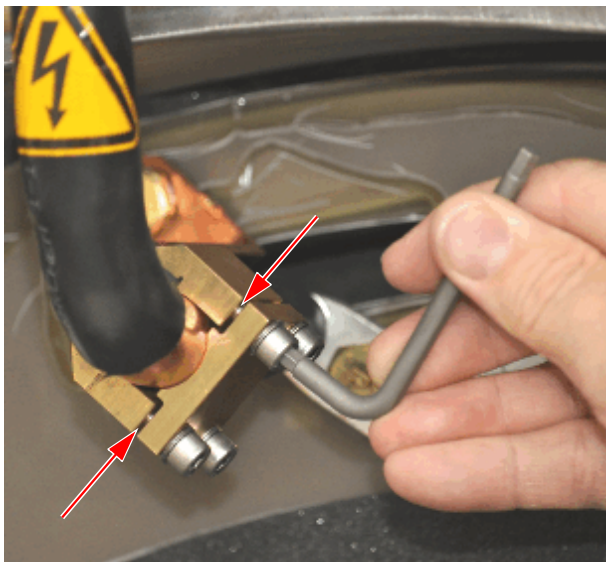
Inner cable lug fastening:

- Fasten the inner cable lug screw with a non-magnetic torque wrench (SW17).



- Apply a torque of 30 ± 3 Nm.

Fig. 31: Gap between clamp halves



(1) Check for equal gaps

- Check the gap on both sides of the clamp
 - » The gap must be equal on both sides.
- Tighten the clamp firmly.



- Apply a torque of **6 - 8 Nm**.

Fig. 32: Fasten cable bracket hex screw with 30 Nm



- Fasten the hex cable bracket screw with the non-magnetic wrench and the jaw wrench insert SW17 (material number 105 63 219).



- Apply a torque of 30 ± 3 Nm.

Fig. 33: Fasten hex cable bracket screw with 30 Nm



- Fasten the bracket hex brass screw at the GC with a non-magnetic torque wrench (SW17).



- Apply a torque of 30 ± 3 Nm.

At the Gradient Connection Plate:

- At the gradient connection plate, check again that the gradient cable lugs are not exposed to any mechanic stress. Open the **nuts (6)** and remove the **NORD-LOCK washers (4)**; the **cable lugs (5)** must be easily movable on the bolt. In rare cases, when not ok, loosen the **flange nut (3)** again and re-adjust the bolt position within the long holes.
- Fasten the flange nut (3).



Apply a torque of 21 ± 2 Nm.

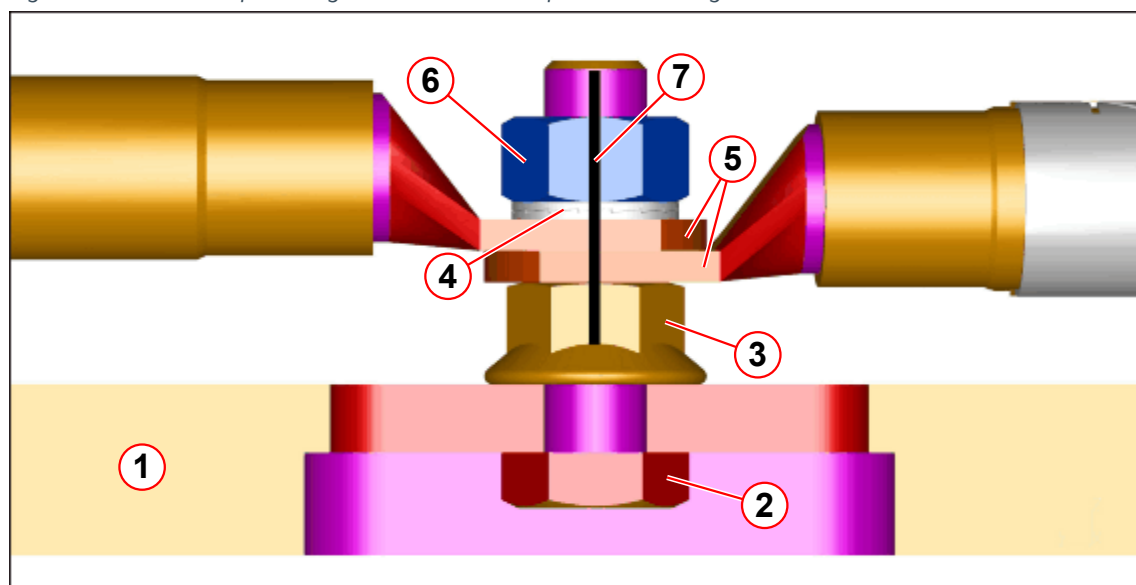
- Fasten the cables with NORD-LOCK washer (4) and nut (6) using a non-magnetic torque wrench (SW19).



Apply a torque of 40 ± 4 Nm.

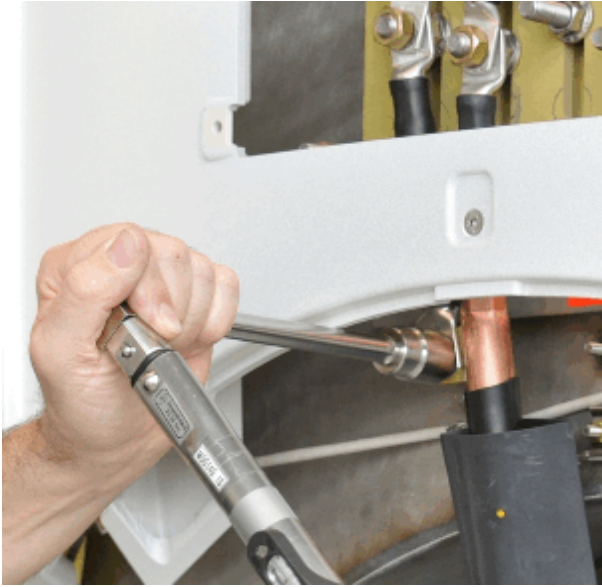
- With a waterproof pen, mark the positions of all nuts with a line over the connections.

Fig. 34: Correct set-up of the gradient connection plate at the magnet



- (1) Connection plate
- (2) 12 mm bolt (SW 19)
- (3) Flange nut (SW 18) (torque 21 ± 2 Nm)
- (4) NORD-LOCK washers
- (5) Gradient cable lugs
- (6) Nut (SW 19) (torque 40 ± 4 Nm)
- (7) Mark with waterproof pen

Fig. 35: Fastening of gradient cable joint screw with 40 Nm - Example



At the coax cable joints:

- Fasten the coax cable joint screws with a non-magnetic torque wrench (SW17).



- Apply a torque of 40 ± 4 Nm.
- With a waterproof pen, mark the positions of the nuts.

2.5.3.1 Shrink Tube Fastening

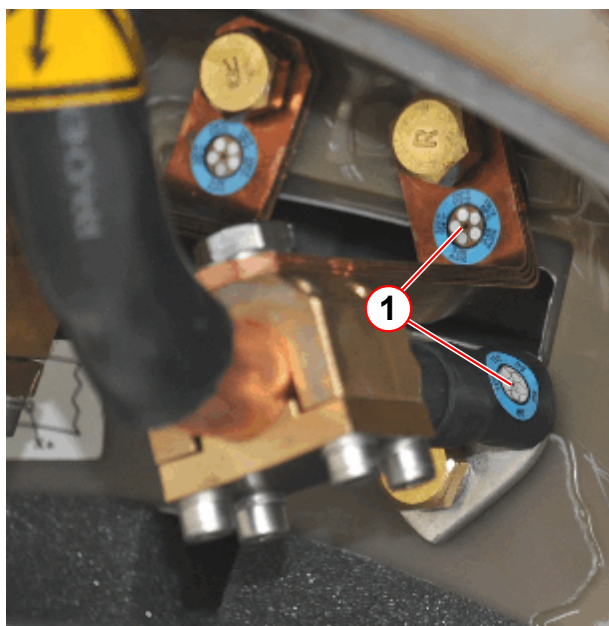
Fig. 36: Fastening of shrink tube with a cable tie



- Feed a cable tie through the hole in the shrink tube.
- Fasten the shrink tube at the coax cable joint.
 - » The cable tie should be mounted above the cable joint screw.
 - » Shrinking the tube with a heat gun is not necessary here.

2.5.3.2 Temperature Indicator Attachment (Clock Stickers)

Fig. 37: GC cable connection

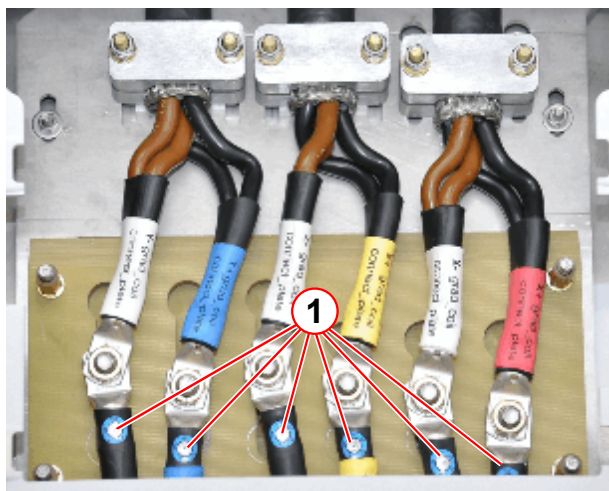


(1) Temperature Indicators (Clock Stickers)

Attach Temperature Indicators close to the screwed connections:

- at each GC connection (inner and outer lug).

Fig. 38: Gradient Connection Plate



(1) Temperature Indicators (Clock Stickers)

- attach temperature indicators at each connection of the gradient connection plate
- in total, 12 stickers have to be placed.

- The figure below shows all gradient cables completely assembled.

Fig. 39: Gradient cables completely assembled



2.6 Checking the Water Circuit



WARNING

The specifications for the air condition system have to be achieved in the examination room.

If the specifications are not kept, condensed water will damaged the TX-box and the RFCEL.

» Check the environmental conditions. Refer to (→ Tab. 1 Page 34).

Tab. 1 Environmental conditions

	Examination room
Room temperature	+18 C to +22 C
Relative air humidity without condensation	40% - 60%
Heat dissipation to air	≤ 3.0 kW
Absolute humidity	< 11.0 g/kg
Air pressure	700 - 1060 mbar



It is possible that the MBB water circuit was not filled with water during the installation because of missing environmental conditions.

If the environmental conditions are in specification, open the valves and fill the MBB water circuit.

Refer to the "System Cooling" installation instructions.

Perform the following steps if the MBB water circuit was not filled during the installation:

System with SEP

- Open the valves (MBB return and supply) on top of the SEP.
- Refill water.

System with IFP

- Open the valves (MBB return and supply) at the IFP.
- Switch on the booster pump with switch 4Q1 in the IFP electronic box.
- Refill water.

Do a visual check of the correct water connections:

- On top of the ACS.
- At the RF-filter plate.
- At the MBB.
- On top of the SEP or at the IFP.

Check that the shutoff valves are open:

- Top of the SEP or at the IFP.
- Shutoff valves for the compressor.

2.7 Checking the water pressure at the TX-box manometer



The default factory setting of the regulator is 5 bar.

This should make it possible to supply water between 2.5 - 3.5 bar to the TX-Box with water pump in normal mode.

Do not change the factory setting of the pressure regulator at the TX-Box to get the correct water pressure at the TX-Box.



Do not use any connection to a drinking water system if it becomes necessary to refill water.

Use distilled water.



Refer to the TSG for water filling!

System with SEP

- Check the static pressure at the manometer in the SEP

Tab. 2 Water pressure

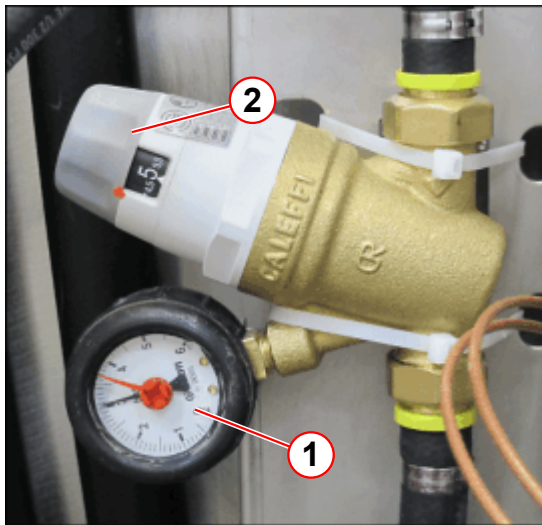
	Pressure manometer	
Static pressure	1.0 - 0.5 bar	



If necessary adjust the water pressure to the correct value.

- Check the pressure at the TX-box manometer.
 - » The pressure should be approximately 0.1 bar higher than the pressure at the SEP manometer.

Fig. 40: TX-box pressure limiter



- (1) Manometer
- (2) Pressure limiter

System with IFP

- Switch off the chiller
- Check the static pressure at the manometer in the chiller
- Check the pressure at the IFP manometer
 - Chiller **lower** than the IFP or on the **same level**:
 - » static water pressure: approx. 1.5 bar.
 - Chiller **higher** than the IFP panel (the limitation is 10m above the IFP):
 - » static water pressure: approx. 1.5 bar.



If necessary adjust the water pressure to the correct value.

- Check the pressure at the TX-box manometer
 - » The pressure should be approximately 0.1 bar higher than at the IFP manometer.

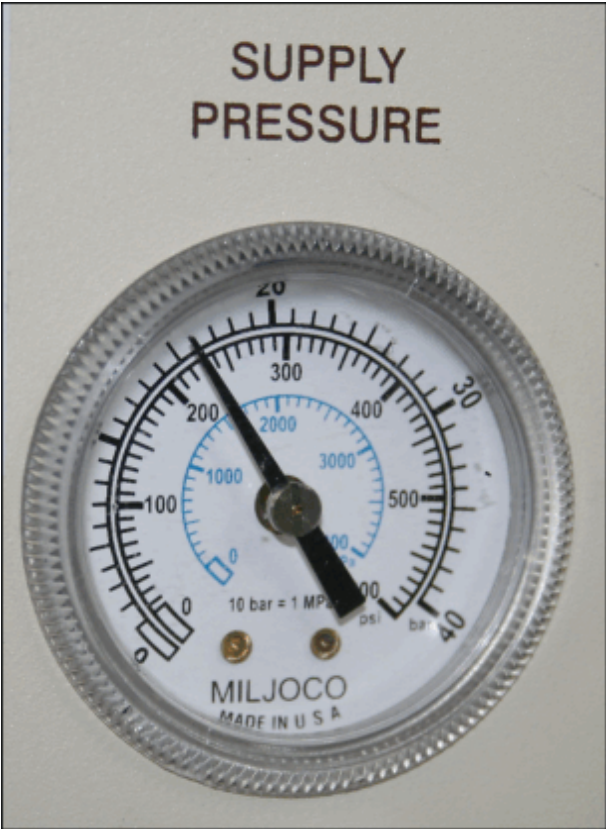
2.8 Checks on the helium compressor

2.8.1 Checking the static pressure of the compressor unit

Do a check of the correct static helium pressure of the compressor unit.

- i** Compressor unit should be off for about 30 minutes before checking the helium gas pressure.
- i** If necessary adjust the helium gas pressure to the correct value.

Fig. 41: F70 Static pressure reading



- Check the **helium gas pressure** at the compressor unit. The correct value is listed in the table (→ Tab. 3 Page 38).

Tab. 3 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
4K - 50/60 Hz	13.5 - 14.0 bar (13.6 nominal value) 196 - 203 psig 1350 - 1400 kPa	21 - 22 bar 304 - 319 psig 2100 - 2200 kPa



Refill the compressor unit with helium (grade 99.999%) if the gas pressure does not match the pressure shown in the table above.

2.8.2 Check the configuration of the configuration mode switch

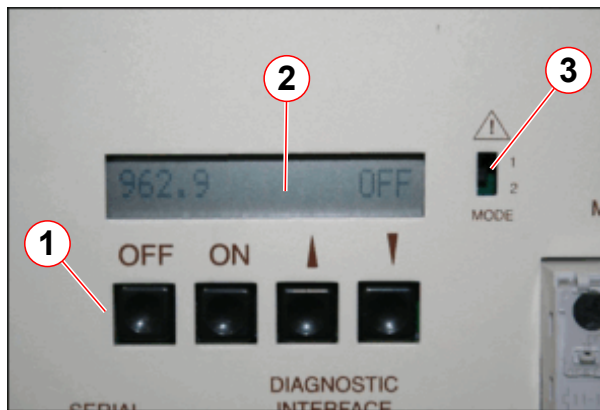
Before the compressor unit's **main power switch** is switched on, the **configuration mode switch** must be set to configuration 2.

The **ON** and **OFF** buttons are **disabled** in configuration mode 2, as the compressor unit is controlled via the pressure within the magnet.

Tab. 4 Configuration mode switch

Configuration	Switch position	Magnet system type	ON/OFF
Mode 1	UP	10 KGM shield cooling	Enabled
Mode 2	DOWN	4 KGM recondensing	Disabled

Fig. 42: F70 System status display



- (1) Control buttons
- (2) LCD display
- (3) Configuration selector switch

- Use a flat-blade screwdriver to move the switch (3) up or down.
 - Switch **up** = **configuration 1**.
 - Switch **down** = **configuration 2**.



Do not change the switch position after the main power has been switched on.

2.9 Switching on the system

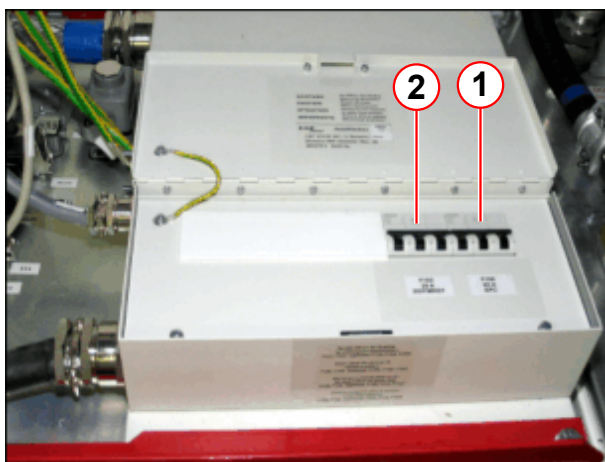
2.9.1 Circuit breakers

2.9.1.1 Circuit breakers EPC

Switch on the following circuit breakers in the EPC mains box:

- F100 50A EPC
- F102 25A SEP/MREF

Fig. 43: Circuit breakers in EPC mains box



- (1) F100 63A EPC
(2) F102 25A SEP/IFP - MREF

Switch on the the following circuit breakers first:

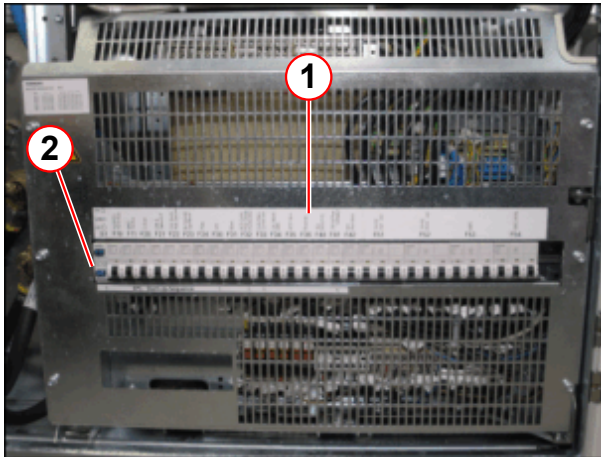
1. F30 UPS
2. F32 ACS Control
3. F33 Coils Main
4. F41 24V / CAN24/7
5. Switch on S1
6. F42 36V Alarm box
7. F40 36V MSUP PS

Switch on the following circuit breakers in the EPC:

- F10 PS RFCEL / PS TX-Box
- F11 MKO/TX-Box
- F20 E-Shim
- F21 GPA & Vacuum pump
- F22 Control room
- F23 IR-Console
- F24 PTab
- F31 Mars

- F34 Ctrl. Room MRC
- F35 ACS Fans
- F36 Reserve
- F51 THYRE star coil
- F52 THYRE delta coil
- F54 MKO-RFPA

Fig. 44: Protective cover of EPC line voltage



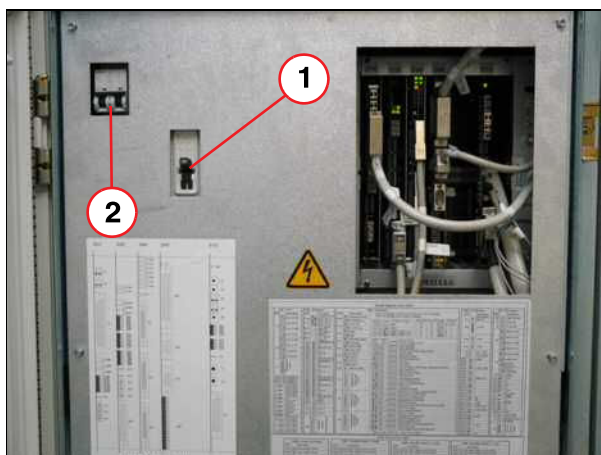
- (1) Circuit breakers
(2) S1

2.9.1.2 Circuit breakers GPA

Switch on the following circuit breakers in the GPA:

- F1
- F2

Fig. 45: GPA_fuses



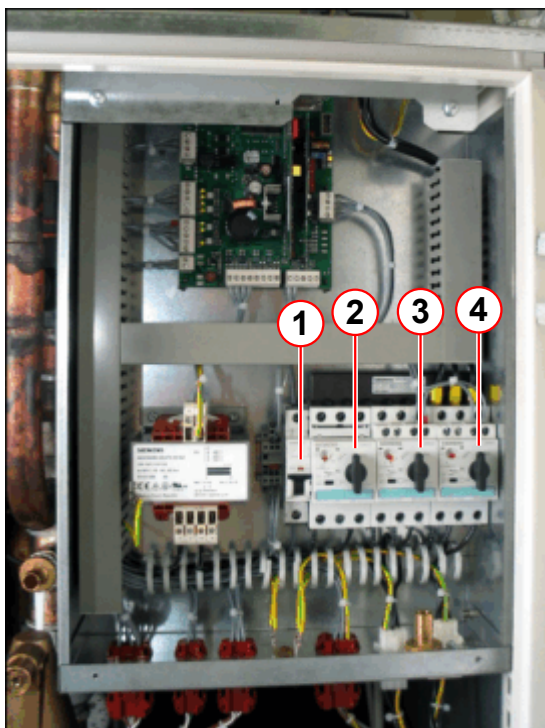
- (1) F1
(2) F2

2.9.1.3 Circuit breakers SEP (Optional)

Check to see that the following circuit breakers in the SEP are switched on:

- F1
- Q1, Q2, Q3

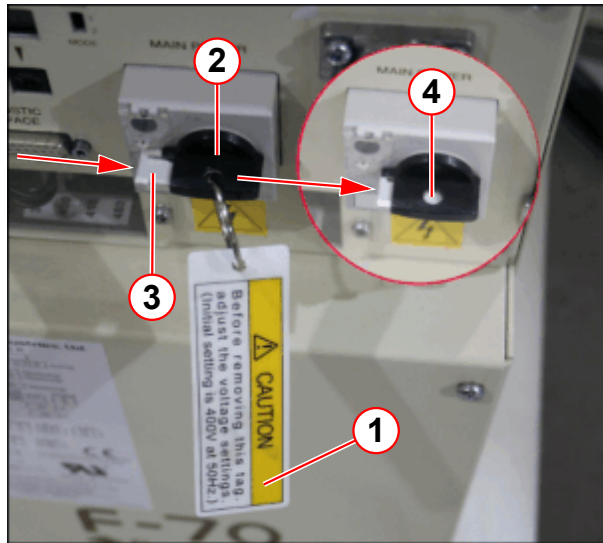
Fig. 46: SEP circuit breakers



- (1) F1
- (2) Q1 (T1)
- (3) Q2 (MREF)
- (4) Q3 (PUMP)

2.9.1.4 Switch on the helium compressor unit

Fig. 47: F70 Main power switch - locking device



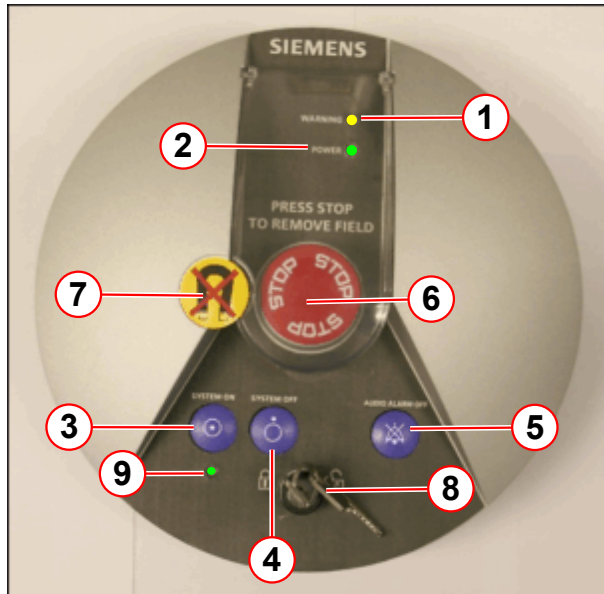
- (1) Voltage caution label
- (2) Main power switch
- (3) White locking device
- (4) Device fully engaged (view through hole)

- **Push in** the white locking device fully.
 - » The **device** should be **visible** through the **blocking hole** (pos. 4).
- Switch on the compressor unit.

2.9.2 The alarm box

2.9.2.1 Status indicator at the alarm box

Fig. 48: Alarm box - LEDs and buttons



- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

After starting the compressor unit, the **alarm box** will indicate:-

- **LINE POWER (1):**
 - » LED lights up green:



Check the alarm box status and LEDs. If the red LED is illuminated, refer to the Troubleshooting Guide.

2.9.2.2 LEDs

Tab. 5 OR98/99 alarm box LEDs

LED	Description
LINE VOLTAGE	Power is supplied to the alarm box
SYSTEM WARNING	Liquid helium level below warning thresholds Quench/switch heater error or ERDU button/battery error Magnet pressure too high or low Compressor unit stopped (pressure switch activated) ERDU battery voltage < 22 V EIS drive error

2.9.2.3 Final Checks

Ensure that the '**SYSTEM ON**' LED is lit when the alarm box is operational

2.10 Emergency shutdown buttons

For safety reasons, all emergency shutdown buttons must be tested prior to starting up the MR system.



If a UPS system is installed, the UPS system and the MR system must be shut off by pushing either one of the EPO buttons.



Refer to local codes/guidelines for the use of EPO buttons.

- Check that all system components are voltage-free.
- Check that the "Line" LED at the alarm box is off.
- Release the emergency power off switch and reactivate the system by switching on the main power.
- Perform this test on all emergency power off switches in the equipment room, control room and examination room.

EPO	Function	
	o.k.	not o.k.
1		
2		
3		
4		
5		

2.10.1 Only for Magnetom Aera



WARNING

Malfunction of the Implant Suite supervision (FPO)!

Risk of personal injury or death (e.g., malfunction or failure of implant)

- » In case the Emergency Power Off (EPO) test fails during start-up, a system with installed Implant Suite (FPO) license must not be handed over.

EGA15.210

2.11 Posting warning labels

Prior to initial Start-up, the project manager defines the 0.5 mT area.

- Post the warning labels identifying the controlled access area (0.5 mT zone) at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone. The sign has to be posted in a way that it is visible even when the door to the controlled area is opened.



Prior starting up the magnet system make sure that the warning labels are posted!

The warning labels are available in different languages. Pick the correct language and the correct field strength (→ 1/Fig. 49 Page 48), (→ 1/Fig. 50 Page 48). The following pictures are examples.



WARNING

Strong magnetic field

Error to observe the following Safety Information may result in physical injury and equipment damage.

- » Ensure that the warning labels identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.
- » The sign has to be posted in a way that it is visible even when the door to the controlled area is opened!

Fig. 49: Warning sign 1.5T



(1) Field strength

Fig. 50: Warning sign 3T



(1) Field strength

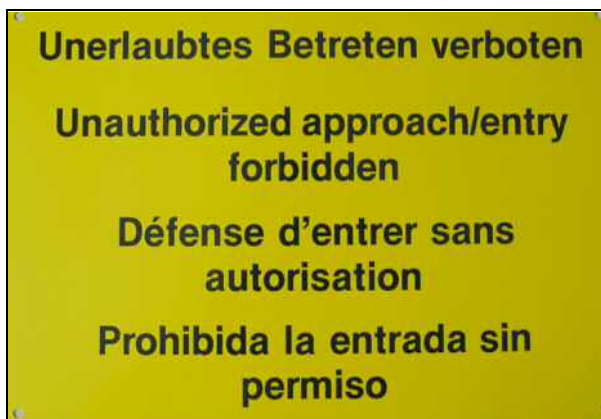
Fig. 51: Warning sign "Pacemaker"



Fig. 52: Warning sign "Filling-up"



Fig. 53: Warning sign "Unauthorized"



2.11.1 Hearing protection sign

The hearing protection sign is shipped together with the warning labels.



The hearing protection sign must be posted at the main entrance door of the examination room. The sign should be positioned at eye level height as shown in (→ 1/ Fig. 55 Page 50).

Fig. 54: Hearing protection sign



Fig. 55: Warning Signs



(1) Hearing protection sign

2.11.2 Sign for bruising hazards



If the distance from the fully extended patient table from the service end to the wall is less than 500 mm, a label or warning label has to be installed to identify the hazardous area. This can be done e.g. with a color-coordinated tape attached to the floor.



The bruising hazard sign must be posted at the rear of the patient table if the distance between patient table and wall is less than 500 mm.

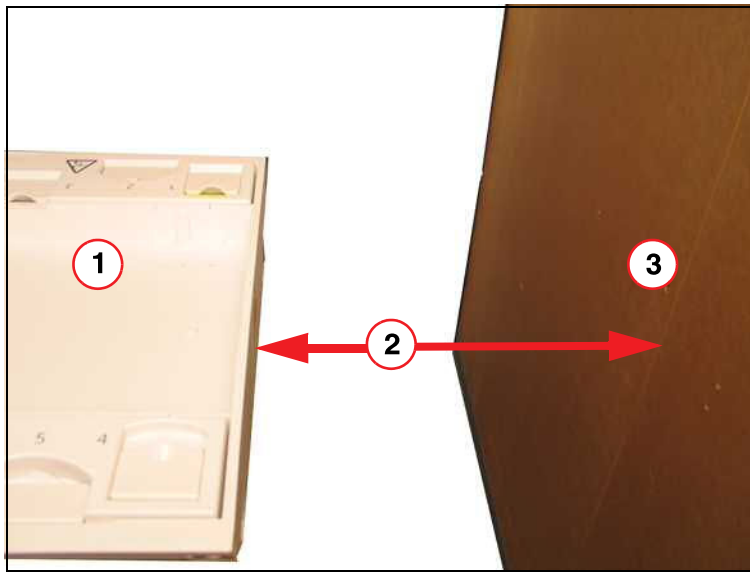
Fig. 56: Bruising hazard sign



Distance check

- Pull the emergency release handle on the patient table.
- Manually move the table all the way back (toward the magnet service end) until the table stops in the horizontal end position.
 - » If the distance between the patient table and the wall (service end) of the RF room is **greater than** 500 mm, no further actions are necessary. Perform final steps.
 - » If the distance between the patient table and the wall (service end) of the RF-room is **less than** 500 mm, the bruising hazard sign has been posted at the rear of the patient table and the controlled access area has been protected with yellow and black tape .

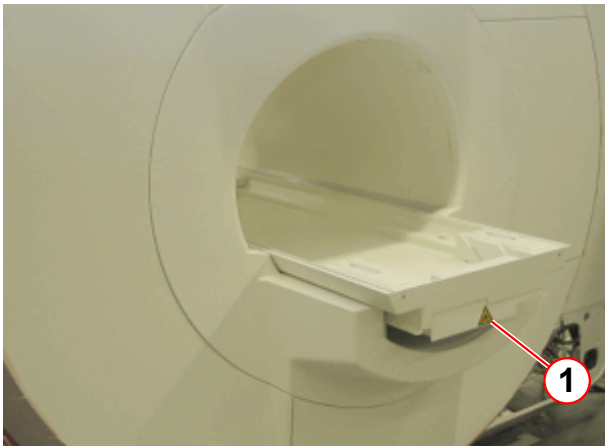
Fig. 57: Bruising hazard - controlled access area



- (1) Patient table back in horizontal end position
- (2) Distance
- (3) Wall of the RF room

- Post the bruising hazard sign at the rear of the patient table.

Fig. 58: Magnet service end



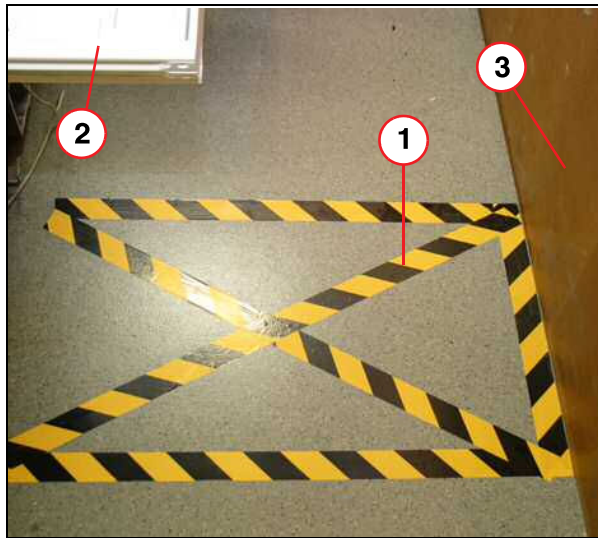
- (1) Sign for bruising hazards

Fig. 59: Sign for bruising hazards



- Protect the controlled area with yellow and black tape.

Fig. 60: Service end area



- (1) Marking on the floor
- (2) Patient table
- (3) Wall of the RF room

2.11.3 Warning Label - Quench Pipe

2.11.3.1 Quench Pipe Inspection

- Visually inspect the condition of the quench pipe.
- Check the **quench pipe** and **insulation** for **damage**.
- Check to ensure that the **connection screws** to the magnet quench valve are all fitted and tight.
- Check to ensure that the warning **label or tape** is attached to the **pipe insulation**.
 - **On long pipe sections** the warning labels or tape should be attached at regular distances , **that is, between 2 - 3 meters**.

Fig. 61: Warning tape- Quench tube



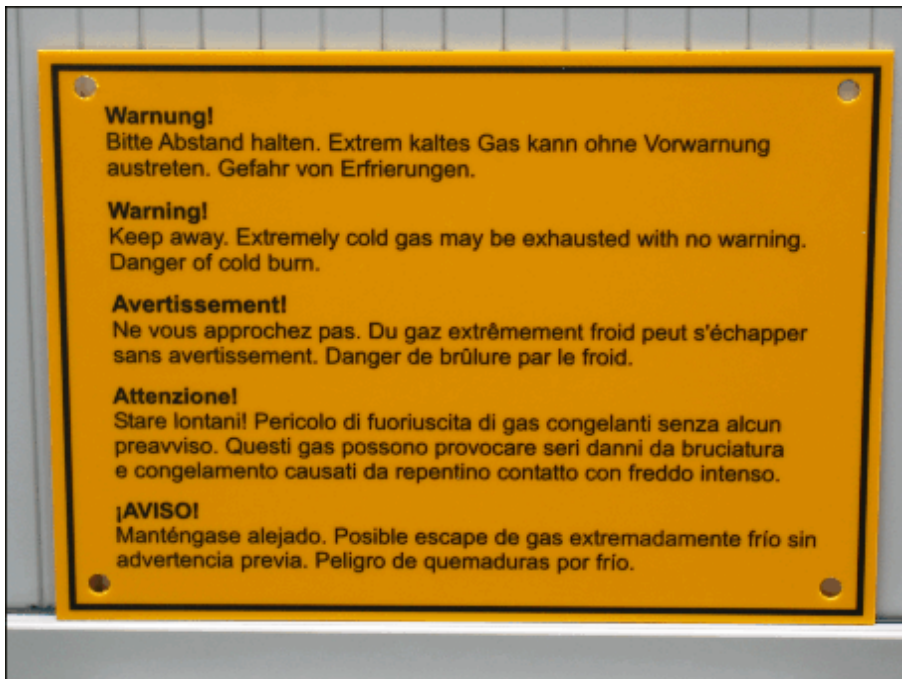
2.11.3.2 External quench pipe



The warning label (→ Fig. 62 Page 54) must be permanently attached next to the external quench pipe outlet.

- Check to ensure that the warning label is attached next to the external quench pipe outlet.

Fig. 62: Warning label - Quench line outlet



3.1 General



Software configuration information is not published as hardcopy documentation. For more information, refer to the Online Help.

Performing the software configuration of the system (Options/Service/Local Service/Configuration). Most of the software configuration steps performed are completely automatic. Follow the instructions shown on-screen. The pop-up windows will prompt you to perform the actions required. Refer to the Online Help for detailed information.

- Do a check of the MeasSettings in the Service Software/Configuration/Menu.

3.1.1 ERDU Buttons

- Check the right number of ERDU button configurations.
 - » **SeSo / Configuration / Measurement / Meas. Settings**

3.1.2 Emergency Power Off (EPO)

The EPO must be configured when the hardware option is installed.

This will enable the maintenance check for EPO.

Configure the EPO via SESO / Configuration.

Refer to online help / Configuration / Measurement.

3.1.3 Remote Diagnostics



It is important to install and configure the Rdiag to get support from HSC in case you need it during installation.

For detailed information regarding the rdiag configuration, refer to CB-DOC / Siemens Remote Service / Installation and Start-up.

3.1.3.1 Introduction

Remote Diagnostics allow you to establish a connection from a remote location in order to provide remote service to the system: online technical assistance, automated reporting of system Quality Assurance Data (QDB, EDB), magnet monitoring data, and system utilization data (customer option).

Only router connections are supported for new installations equipped with the syngo MR system.

3.1.3.2 Supported connections

- Direct connection via Internet Endpoint Security (VPN tunneling) Authentication: SHA1 or MD5 supported encryption: DES or 3DES

- VPN router from Siemens.

3.1.3.3 Responsibility

The project manager is responsible for planning a remote connection in cooperation with the customer.

3.2 TestTools

Mandatory testing with the TestTools is required at the end of the installation.

Instead of running all tests in one run, perform them individually over a period of time (e.g., during wait times, lunch break, etc.).

To get an idea of how much time is needed for the tests, please refer to the following table. Please plan to perform the tests accordingly:

Test time in hh:mm:ss	Fast	Normal	Stability	Interactive
Status	0:10			
Control	4:30	1:15		
TX	0:20	3:42	10:20	
RX	0:30	9:07	25:50	
Gradient	0:30	8:45		
Components		0:11		
LC Plug				30:00
PMU				0:40
PTAB				0:40
Sum	6:00	23:00	36:10	32:20

System test time : 01:37:30



Before handing over the system, please ensure that all tests of the TestTools have been performed and are within specification.



Do the LC-Plug tests prior to shimming the magnet. Perform them when the patient table is removed for moving the system into the site.

4.1 System with SEP



It is important to set the water pressure to the correct value. Otherwise the TX-Box can be damaged.

- Check the pressure on the manometer in the SEP and via SeSo.

Tab. 6 Water pressure

	Pressure manometer	
ECO mode¹		
Operating pressure	approx 2.5 - 0.5 bar	

1. MR system is switched off. Only SEP and the helium compressor are on.

Normal mode	Pressure manometer in the SEP	SeSo: Magnet & Cooling / Cooling Status
Pump off	1.0 - 0.5 bar (Static pressure)	1.0 - 0.5 bar (Static pressure)
Pump on	Static pressure + 3.5 bar ¹	Static pressure - 0.15 bar = Return pressure
		Static pressure + 3.5 bar = (Supply pressure)
		Differential pressure = 3.5 bar (N4_VD11D) Differential pressure = 3.3 bar (N4_VD13) (Differential pressure = Sup- ply pressure - Return pres- sure)

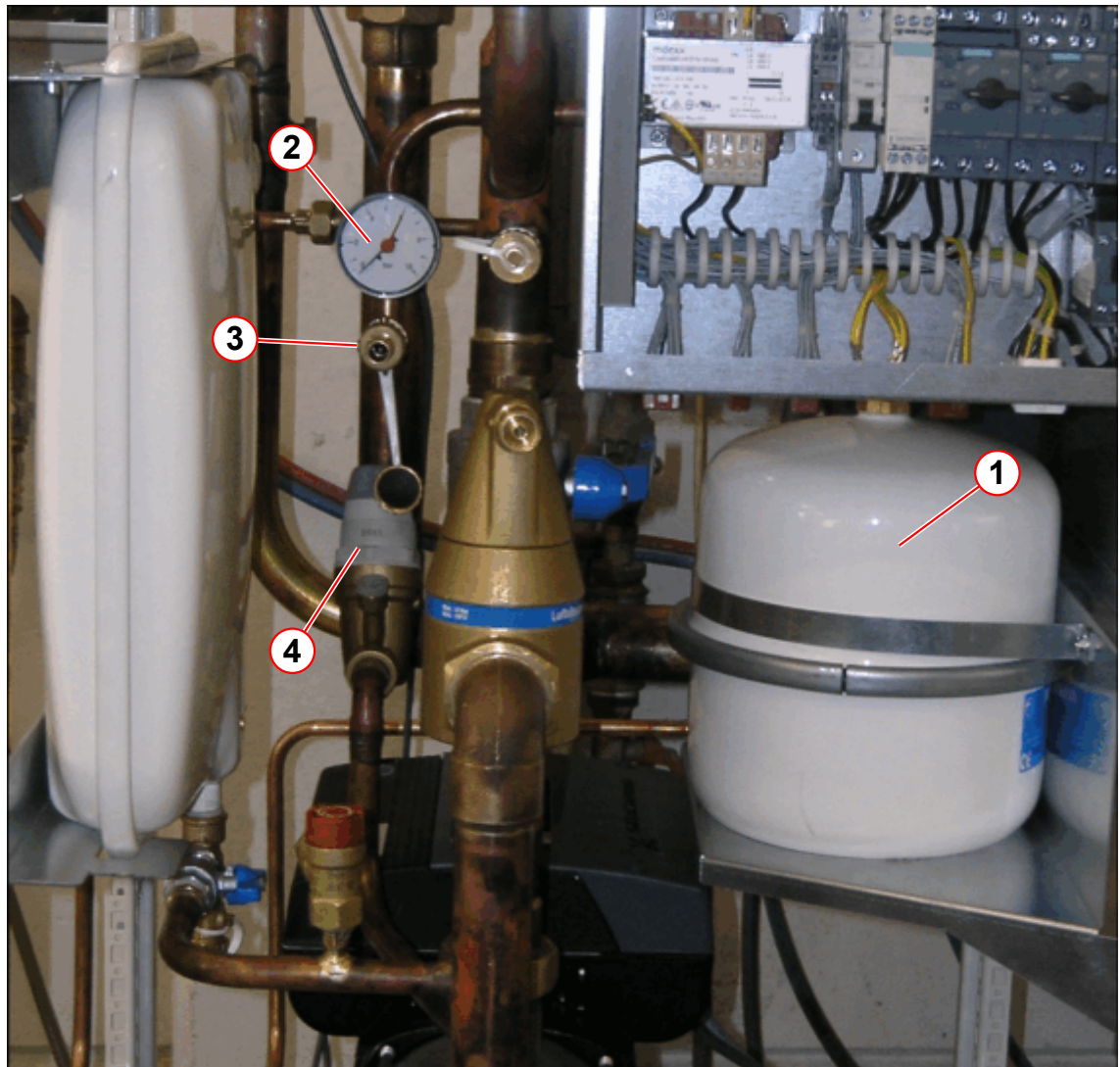
1. MR system is switched on.

4.1.1 SEP with auto-refill device



The auto-refill device maintains the secondary water return pressure constant at 1.0 bar. Water is refilled automatically if water loss appears.
Prerequisite: The auto-refill device is filled with water and the pressure regulator is adjusted to 1.0 bar.

Fig. 63: Auto-refill device



- (1) Expansion vessel
- (2) Pressure manometer
- (3) Fill port of auto-refill device
- (4) Pressure regulator

- Connect a water hose to the fill port of the auto-refill device. (→ 3/ Fig. 63 Page 59)
- Open the fill port slowly.
- Fill the auto-refill device to a range between 4.0 bar and 6.0 bar.
- Check the pressure via the pressure manometer. (→ 2/ Fig. 63 Page 59)

4.2 System with Chiller Type ECO122/ECO133



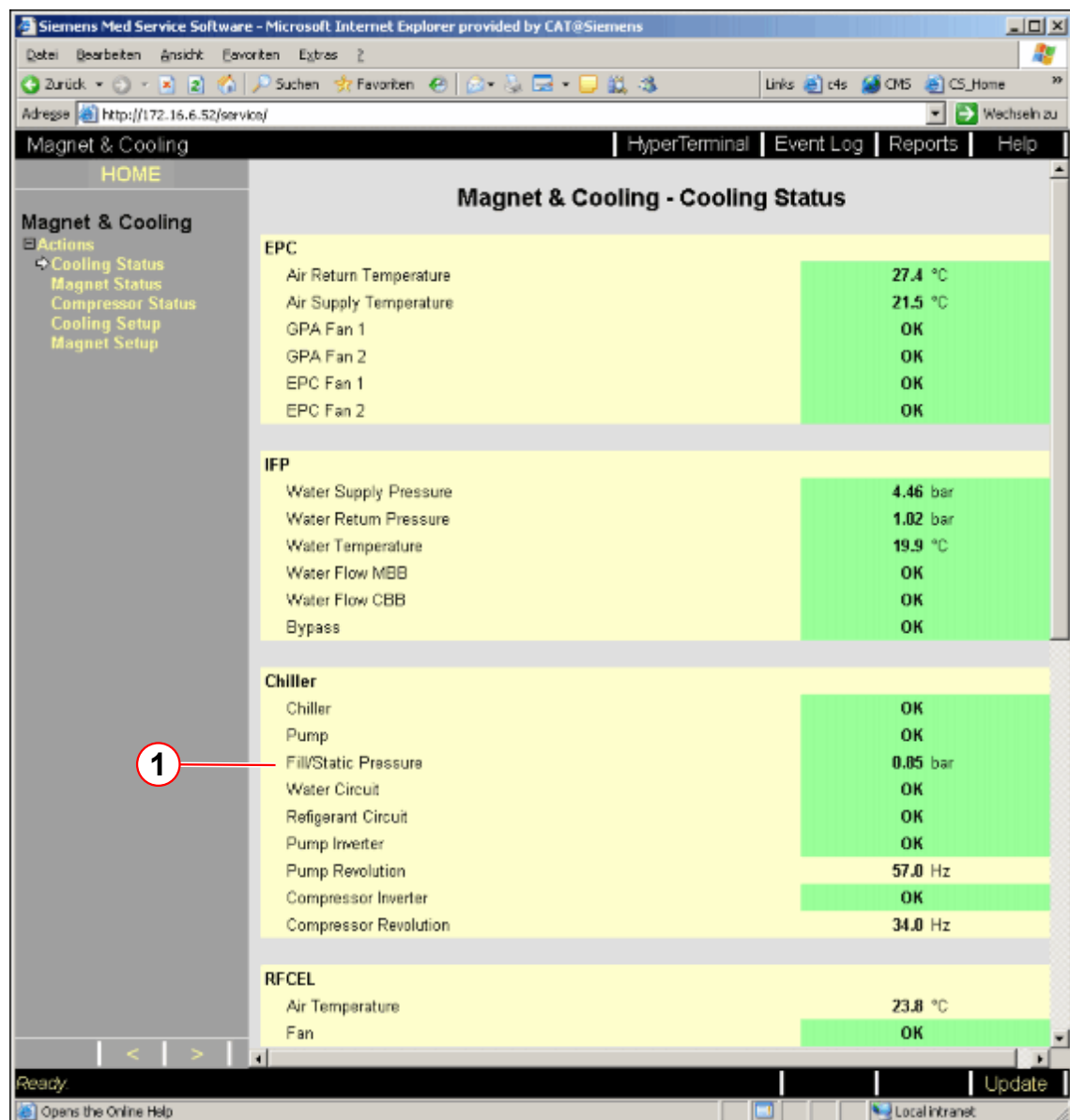
It is important to set the water pressure to the correct value.
Otherwise the TX-Box can be damaged.



If necessary adjust the water pressure to the correct value.

Check the water pressure via SeSo / Magnet & Cooling / Cooling Status

Fig. 64: Cooling status



(1) Fill/static pressure

Chiller on the same level as the IFP:

- Check the fill/static pressure of the chiller.
 - » Maximum pressure is 1.5 bar.
 - » If fill/static pressure < 1.5 bar, refill with water.

Chiller higher than the IFP (limitation is 10m above the IFP)

- Check the fill/static pressure of the chiller.
 - » Maximum pressure is 0.5 bar.



Do not use any connection to a drinking water system if it becomes necessary to refill water.



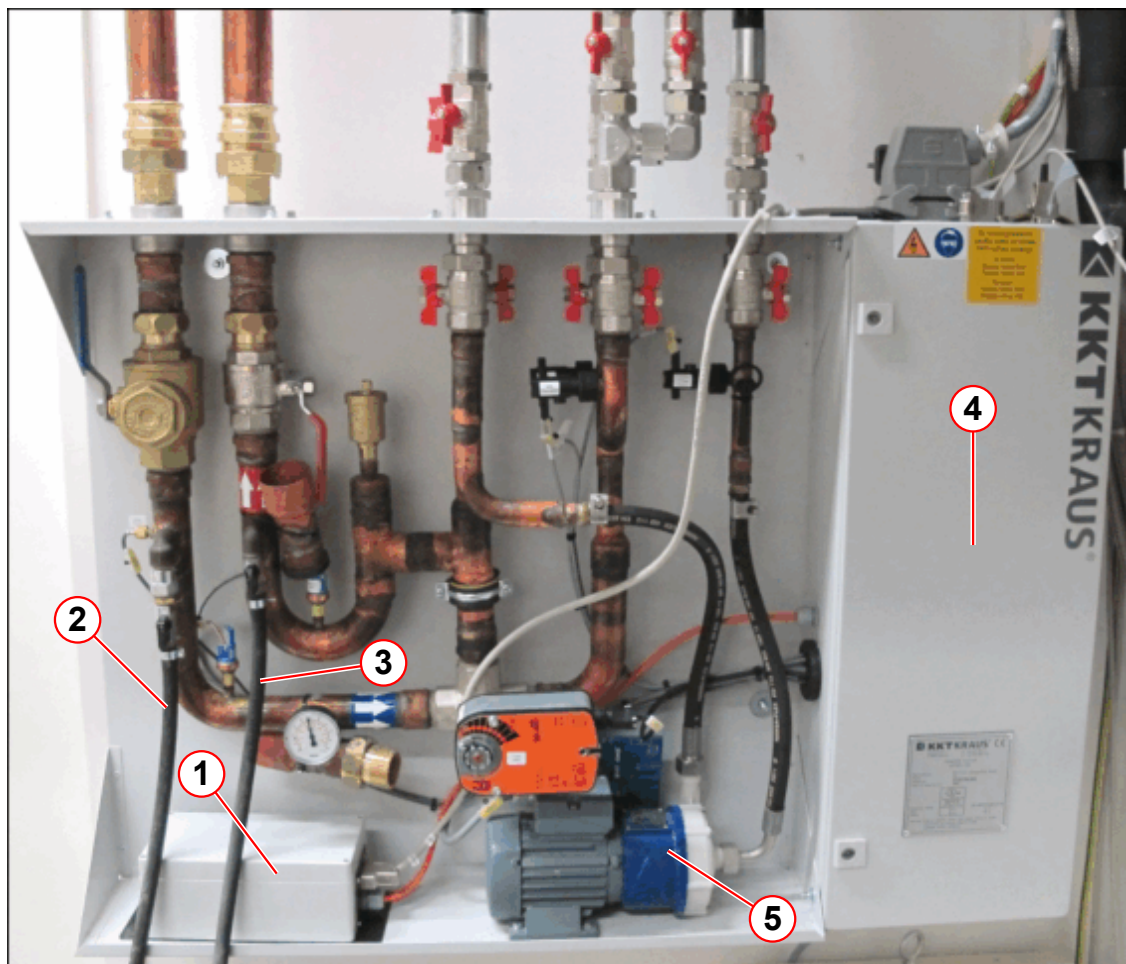
Refer to the TSG for the water filling procedure.

4.3 Avanto^{fit}, Skyra^{fit} with Chiller type KSC215/KCC215 and IFP K2268



Only necessary for the Avanto^{fit} and Skyra^{fit} upgrade with Chiller, type KSC215 or KCC215.

Fig. 65: IFP K2268 with signal converter IFP



- (1) Signal converter IFP
- (2) To compressor "supply"
- (3) From compressor "return"
- (4) Electronics box
- (5) Booster pump

Procedure

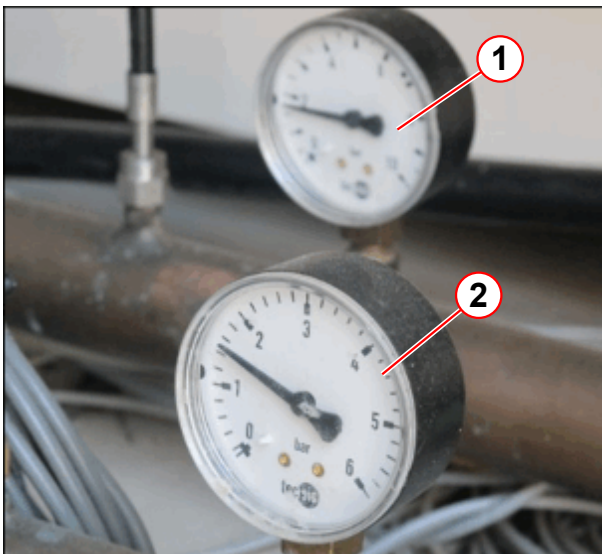
- Switch off the chiller

Fig. 66: Chiller KSC215/KCC215



- Check to ensure that the valves at the IFP to MBB circuit are closed (MBB supply and TX-Box return).
- Check to ensure that the static pressure is at 1.5bar. (→ 2/ Fig. 67 Page 63)

Fig. 67: Water pressure



- (1) High pressure circuit
(2) Low pressure circuit

Fill the water circuit to 1.5 bar, if necessary.

Adjusting the differential pressure to 3.5 bar for Skyra fit and 4.0 bar for Avanto fit

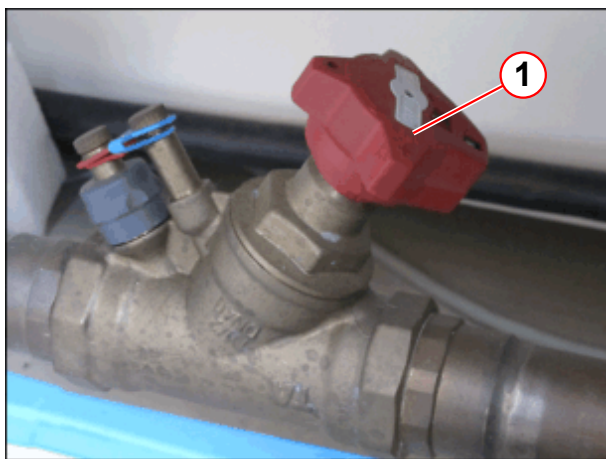
- Switch on the chiller.



The water pressure (p_{high}) at the manometer in the chiller can be high if the chiller is on. (→ 1/ Fig. 67 Page 63)

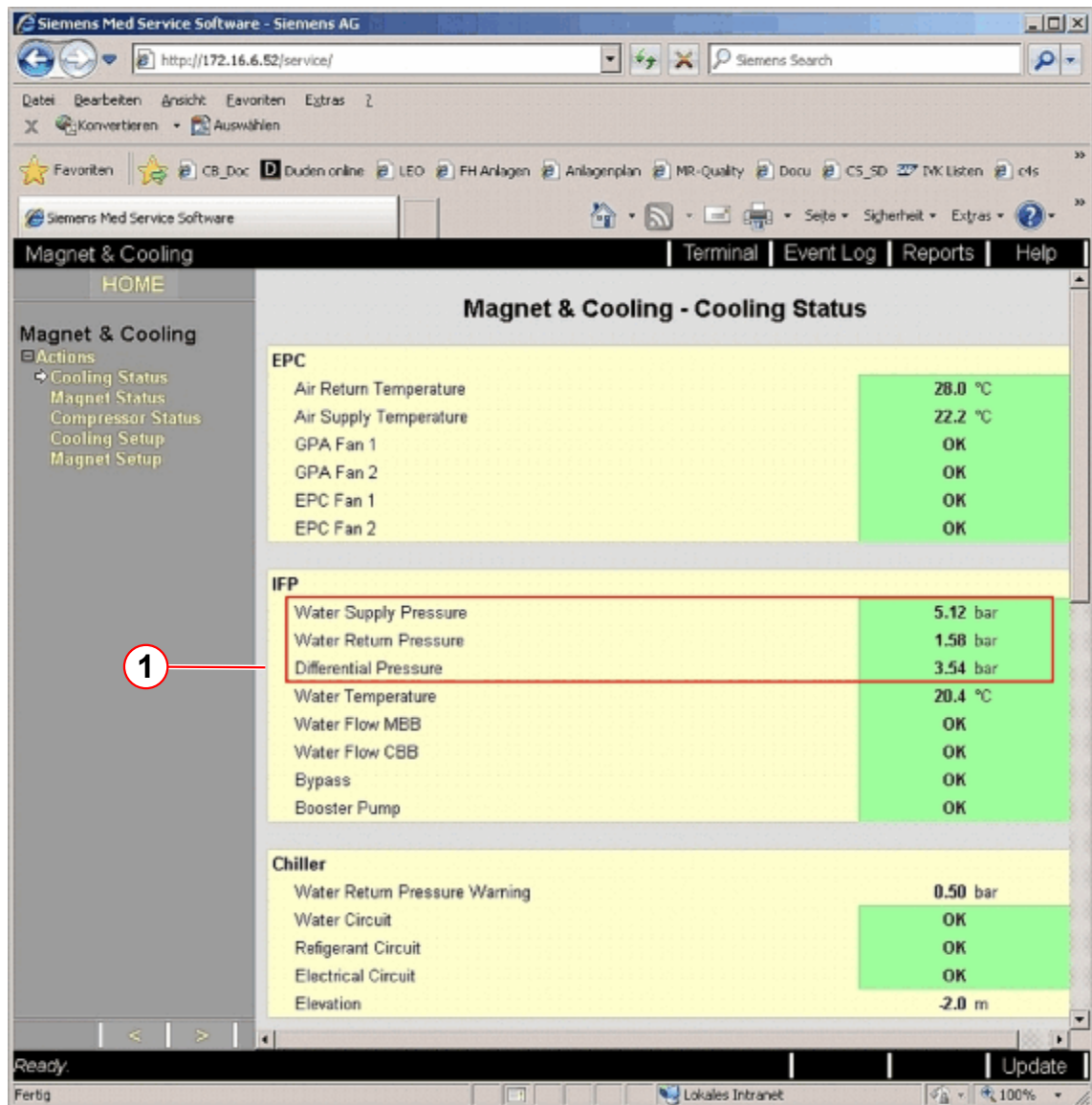
- Check the water pressure p_{high} and p_{low} via SeSo/Magnet&Cooling Staus. (→ Fig. 64 Page 60)
 - » Check the difference in pressure via SeSo/Magnet&Cooling/Cooling Status ($p_{\text{high}} - p_{\text{low}}$). (→ 1/ Fig. 69 Page 65)
- Adjust to the correct pressure difference with the TA valve of the chiller. (→ 1/ Fig. 68 Page 64)

Fig. 68: Chiller KSC/KCC



(1) TA - valve

Fig. 69: Cooling status



(1) Difference in pressure

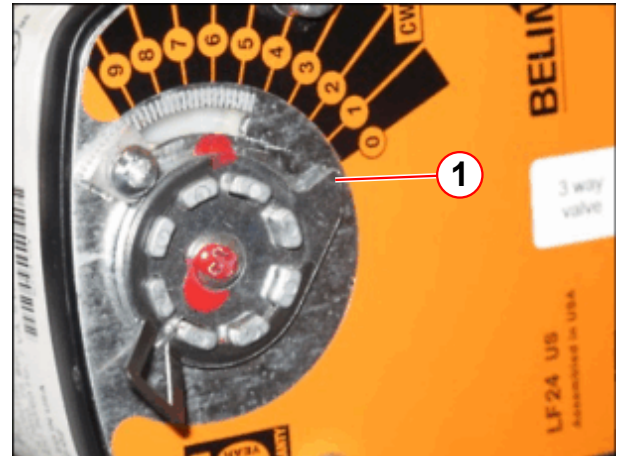
Faults during adjustment:

Chiller switched to bypass mode (position 0). No water flow to the MR system!
 (→ 1/ Fig. 71 Page 66)

Fig. 70: Bypass valve



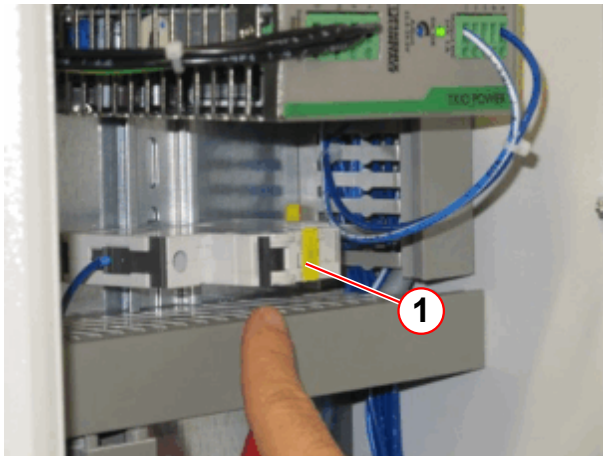
Fig. 71: Bypass valve



(1) Position "0" = "Closed"

- » Reset the chiller: Switch on/off circuit breaker 4F2.

Fig. 72: Electronics box IFP



(1) 4F2

Open the MBB water circuit in the following order:

1. Slowly open the "Return from TX-Box" valve.
 2. Slowly open the "MBB supply" valve.
- Check the difference in pressure.
 - Re-adjust the water pressure, using the TA valve.



The water pressure will drop due to escaping air!
Refill with water and readjust the difference in pressure.

Setting the warning level for the water return pressure.

- Check the installation site.
- Calculate the difference in elevation. (→ 1/ Fig. 75 Page 68)(→ 1/ Fig. 77 Page 70)
(→ 1/ Fig. 76 Page 69)

- Enter the difference in meters via the **Cooling Setup**. (→ 1/Fig. 73 Page 67)
- Check the values via the **Cooling Status**. (→ 1, 2/Fig. 74 Page 68)



The warning level threshold depends on the installation site of the chiller.
Refer to (→ Fig. 75 Page 68), (→ Fig. 77 Page 70), (→ Fig. 76 Page 69)

Fig. 73: Cooling setup

The screenshot shows the 'Magnet & Cooling - Cooling Setup' interface. On the left is a navigation menu with 'Magnet & Cooling' expanded, showing 'Cooling Status', 'Magnet Status', 'Compressor Status', 'Cooling Setup' (selected), and 'Magnet Setup'. The main area is divided into sections: 'EPC' with 'Air Return Temperature' (Max. 38.0 °C) and 'Air Supply Temperature' (Max. 27.0 °C); 'IFP' with 'Differential Pressure Warning' (3.8 bar); and 'Chiller' with 'Elevation' (0.0 m). A red circle with the number '1' points to the 'Elevation' field. At the bottom, there is a 'Save values as default' checkbox and a status bar with 'Ready', 'Save', 'Default', and 'Update' buttons.

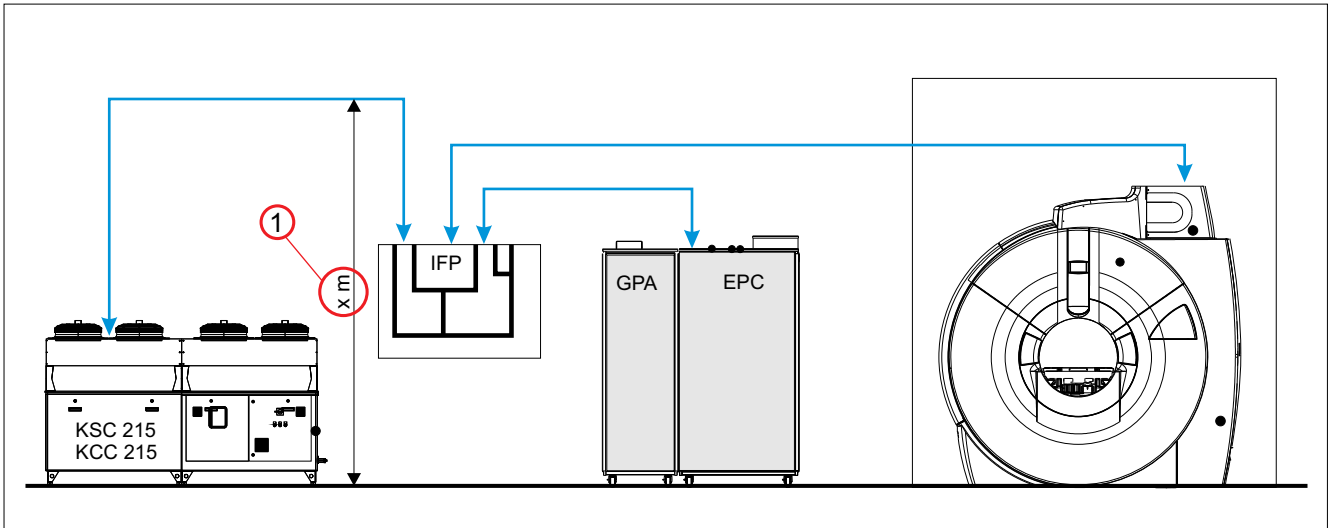
(1) Elevation

Fig. 74: Cooling setup



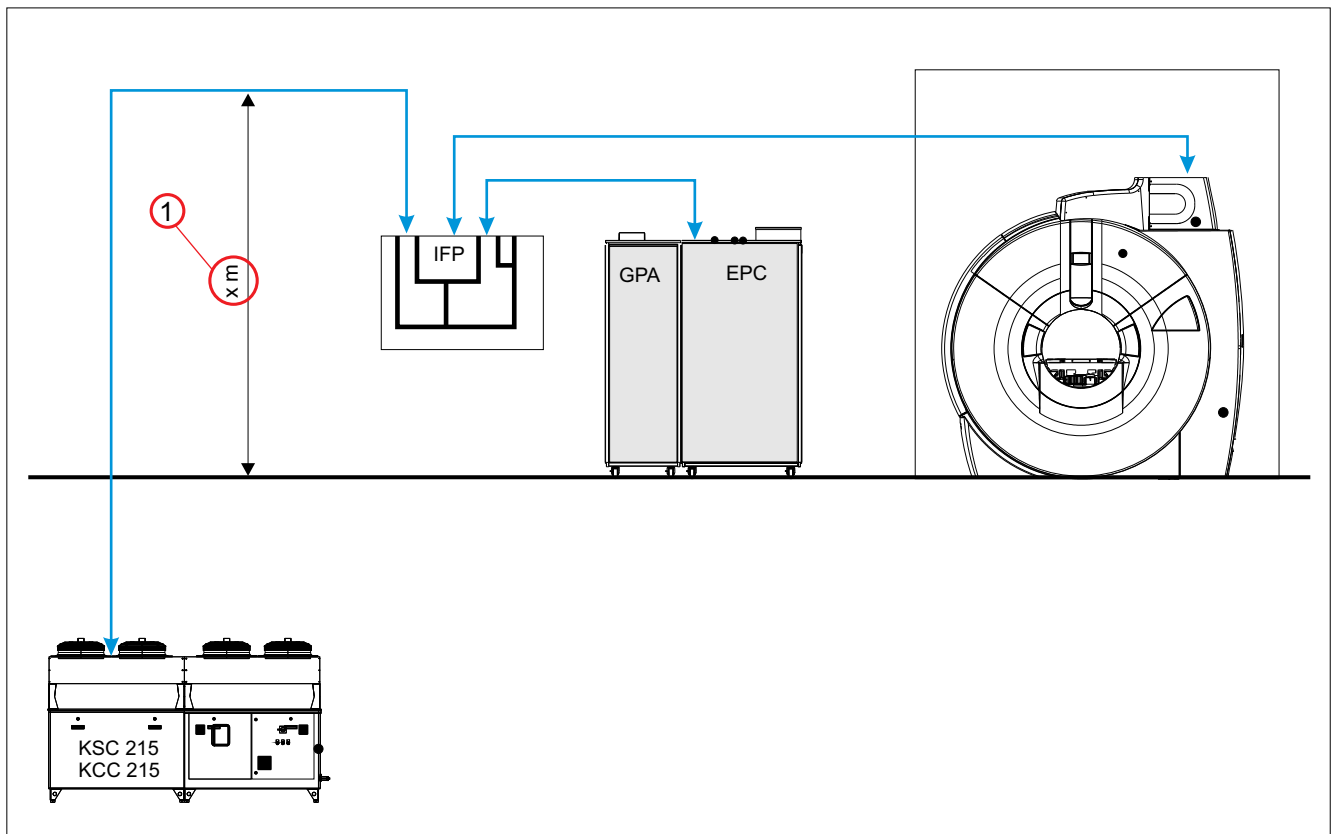
- (1) Elevation value
- (2) Return pressure threshold

Fig. 75: Chiller - same level



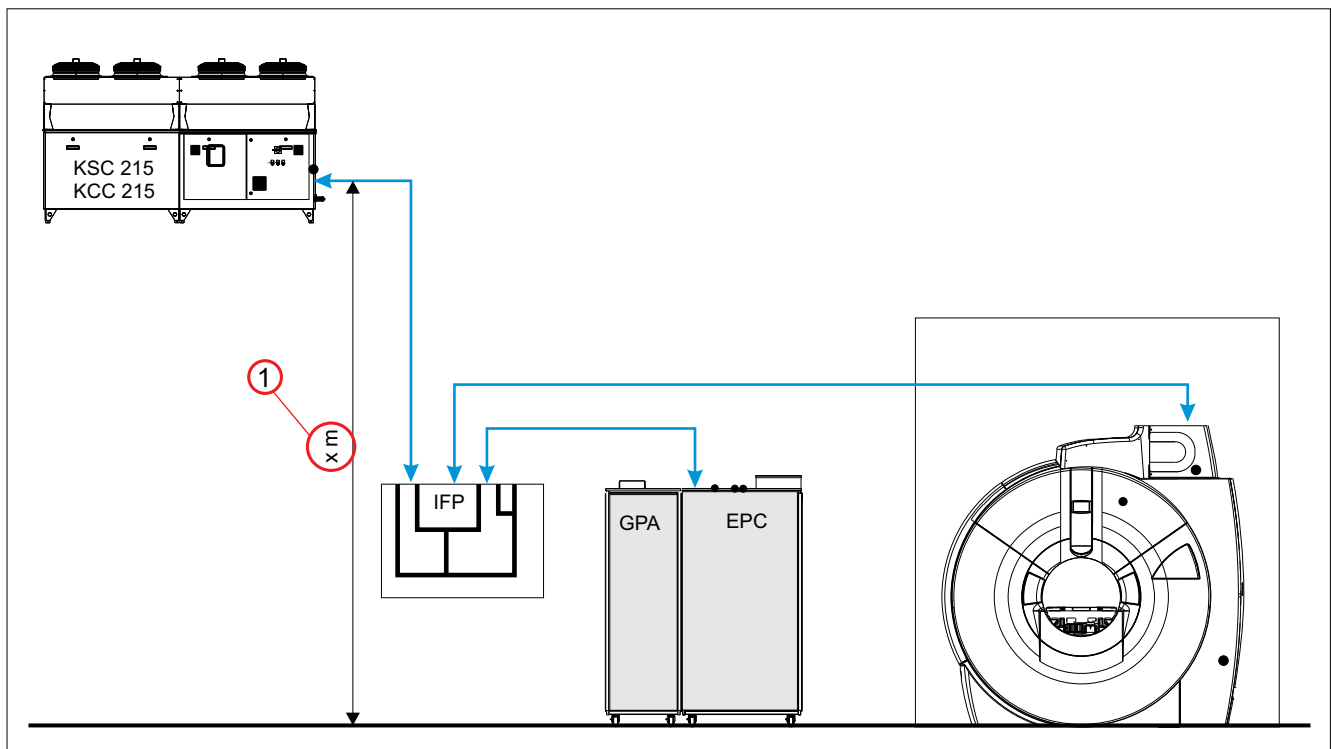
- (1) Elevation

Fig. 76: Chiller - lower level



(1) Elevation

Fig. 77: Chiller - higher level



(1) Elevation

4.4 Pressure at the TX-Box



The default factory setting of the regulator is 5 bar.

This should make it possible to supply water between 2.5 - 3.5 bar to the TX-Box with water pump in normal mode.

Do not change the factory setting of the pressure regulator at the TX-Box to get the correct water pressure at the TX-Box.

- Check for correct water pressure at the TX-Box manometer.
 - » MR system is on and water pump is **not** in ECO mode:
The pressure has to be **2.5 bar to 3.5 bar** to avoid damage to the TX-Box.

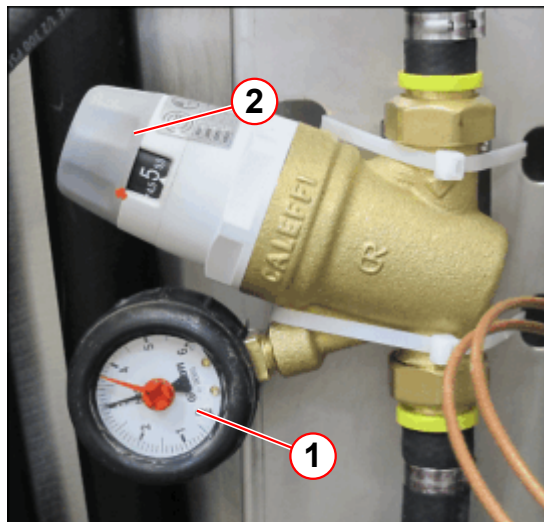


High water pressure will damage the TX-Box.



The pressure is lower if the SEP or chiller is in ECO mode.

Fig. 78: TX-box pressure limiter



- (1) Manometer
- (2) Pressure limiter

System with SEP

If the pressure at the TX-Box is not in specification perform the following checks:

If the pressure at the manometer is < 2.5 bar:

1. Check if the water pump is in normal mode.
 - » Check if the supply water pressure is correct under **SeSo / Magnet & Cooling / Cooling Status**

2. Switch off the water pump in the SEP.
 - » Check the water pressure on the manometer in the SEP.
 - » Pressure should be 0.9 bar to 1.1 bar.
 - » Release or pressurize the water pressure to the correct value.
3. Check if the water flow direction is correct.
4. Check the water lines for bends.
5. Check if the valves are fully open.
6. Clean the strainer of the secondary water circuit in the SEP.
7. Change the TX-Box pressure regulator.

If the pressure at the manometer is >3.5 bar:

1. Check if the water pump is in normal mode.
2. Check if the supply water pressure is correct under **SeSo / Magnet & Cooling / Cooling Status**
 - » Replace the pressure regulator if supply pressure <2.5 bar.
3. Switch off the water pump in the SEP.
 - » Check the water pressure on the manometer in the SEP.
 - » Pressure should be 0.9 bar to 1.1 bar.
 - » Release or pressurize the water pressure to the correct value.
4. Check if the water flow direction is correct.
5. Check the water lines for bends.
6. Check if the valves are fully open.
7. Change the TX-Box pressure regulator.



A drop of the water pressure due to air in the water circuit should be normally compensated by the autorefill device (which should be filled with distilled water).

System with Chiller and IFP

If the pressure at the TX_box is not in specification perform the following checks:

If the pressure at the manometer is < 2.5 bar:

1. Check if the water pump is in normal mode.
 - » Check if the supply water pressure is correct under **SeSo / Magnet & Cooling / Cooling Status**
2. Switch off the water pump in the chiller
 - » Check the water pressure on the **IFP manometer** (refer to table)

Tab. 7 Water pressure

Chiller switched off	Chiller manometer	IFP manometer	TX_Box manometer	Chiller low pressure warning (adjustable SeSo)	IFP low pressure warning (fix threshold)
Chiller on the same level or below the IFP	<1.5 bar	1.1 - 1.5 bar	1.1 - 1.5 bar	0.5 bar	0.5 bar
Chiller above the IFP (max 10 m)	<1.0 bar	1.7 - 1.9 bar	1.7 - 1.9 bar	0.5 bar	0.5 bar

» Release or pressurize the water pressure to the correct value.

3. Check if the water flow direction is correct.
4. Check the water lines for bends.
5. Check if the valves are fully open.
6. Clean the strainer of the IFP and strainer of the chiller.
7. Change the TX-Box pressure regulator.

If the pressure at the manometer is >3.5 bar:

1. Check if the water pump is in normal mode
2. Check if the supply water pressure is correct under **SeSo / Magnet & Cooling / Cooling Status**
 - » Replace the pressure regulator if supply pressure <2.5 bar.
3. Switch off the water pump in the chiller.
 - » Check the water pressure on the IFP manometer (refer to the table).
 - » Release or pressurize the water pressure to the correct value.
4. Check if the water flow direction is correct.
5. Check the water lines for bends.
6. Check if the valves are fully open.
7. Clean the strainer of the IFP and the strainer of the chiller.
8. Change the TX-Box pressure regulator.



A drop of the water pressure due to air in the water circuit should be normally compensated by the autorefill device (which should be filled with distilled water glycol mixture).

Please refer to "Start-up / Magnet System". Print number M7-010.815.20.

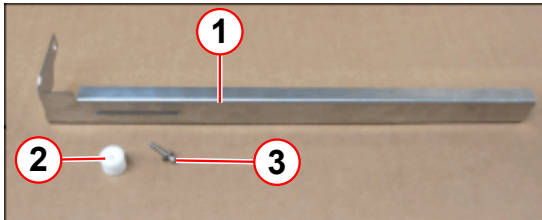
Please refer to "Start-up / Magnet System". Print number M7-010.815.20.

Please refer to "Start-up / Magnet System". Print number M7-010.815.20.

8.1 Adjust the vertical transfer position

- Install the template at the body coil

Fig. 79: Template set

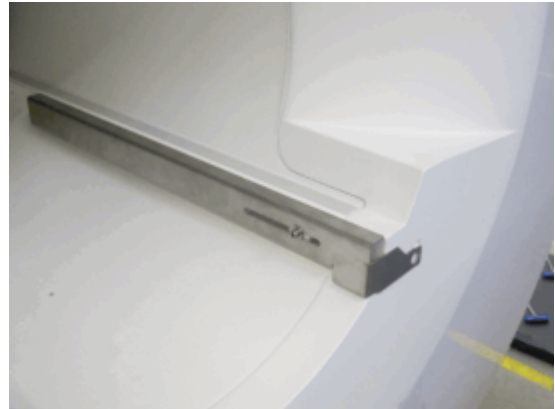


- (1) Template
- (2) Distance piece
- (3) Screw + washer

Fig. 80: Template installation

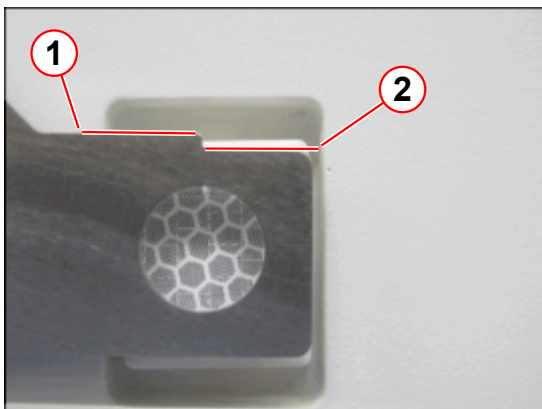


Fig. 81: Template



- Adjust the template in front of the reflector.
- Check the alignment of the reflector with the template reference lines.

Fig. 82: Reference line



- (1) Reference line for the reflector
- (2) Reference line for the reflector film

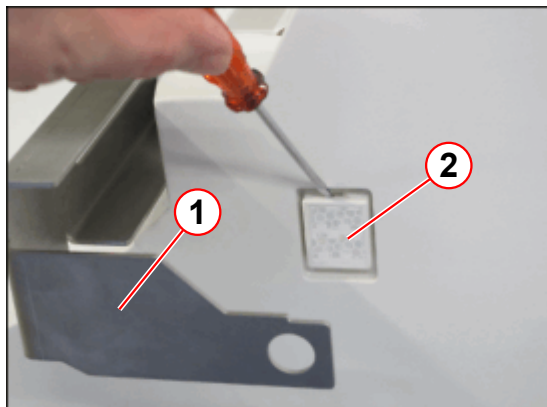
Fig. 83: Reflector - Template



- Align the reflector to the reference lines of the template.

- Remove the reflector by using a screwdriver

Fig. 84: Patient table reflector



- (1) Template
(2) Reflector

- Loosen the screw and adjust the retaining plate.

Fig. 85: Reflector - retaining plate

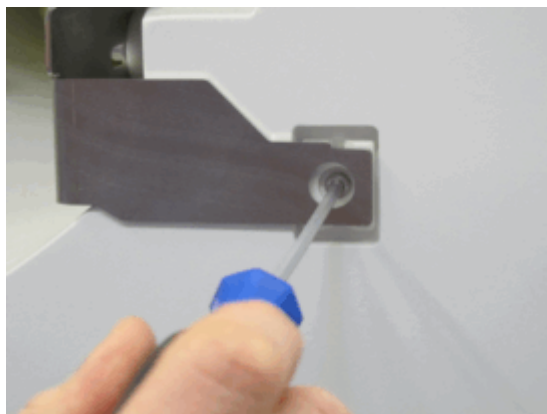
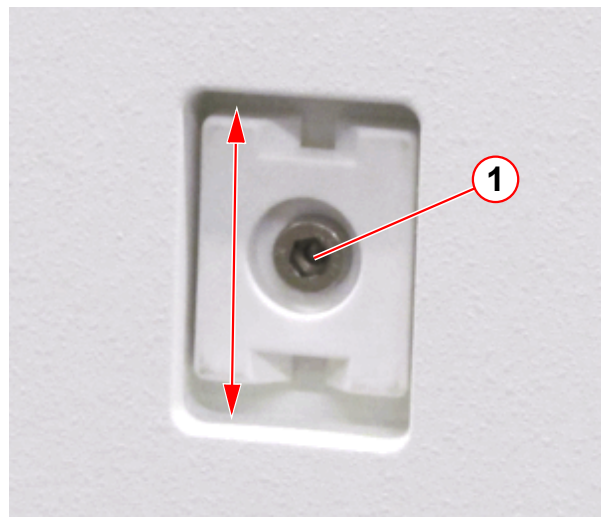


Fig. 86: Reflector - retaining plate

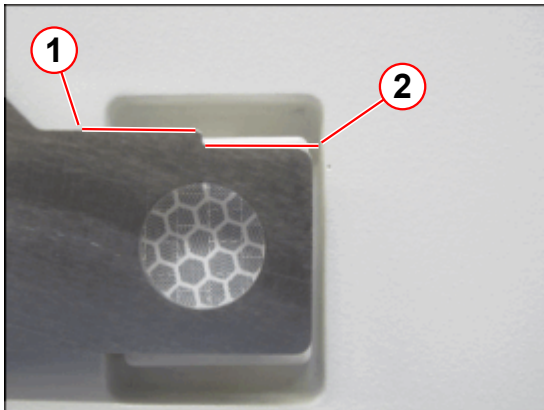


- (1) 3 mm Allen key

- Fasten the screw and install the reflector.

- Check the alignment with the template reference lines.

Fig. 87: Reference line



- (1) Reference line for the reflector
- (2) Reference line for the reflector film

Fig. 88: Reflector - Template



8.2 Setting the laser light localizer (only required for upgrades)

Tools

- Screw driver

Procedure

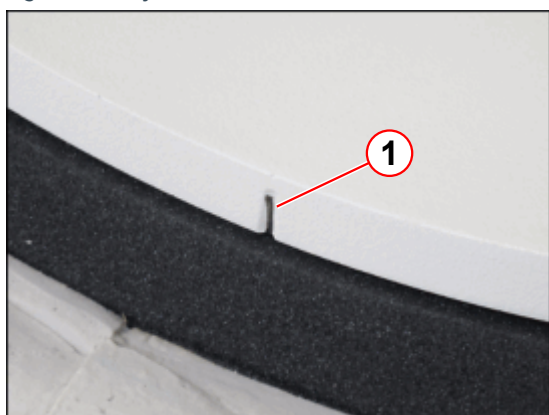
- Remove the funnel.



Use the BC reference point for horizontal alignment. (→ 1/ Fig. 89 Page 80)

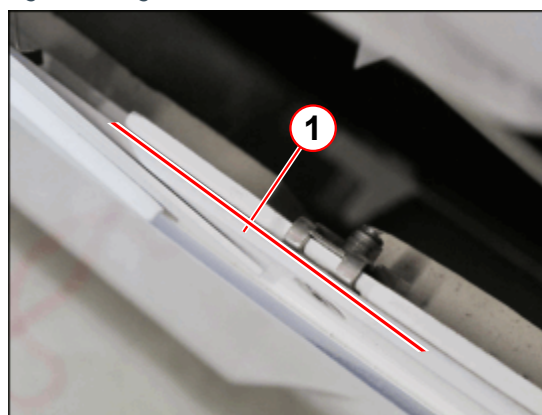
Use the reference edge of the front cover for vertical alignment. (→ 1/ Fig. 90 Page 80)

Fig. 89: Body Coil



(1) BC reference point

Fig. 90: Magnet cover



(1) Reference edge

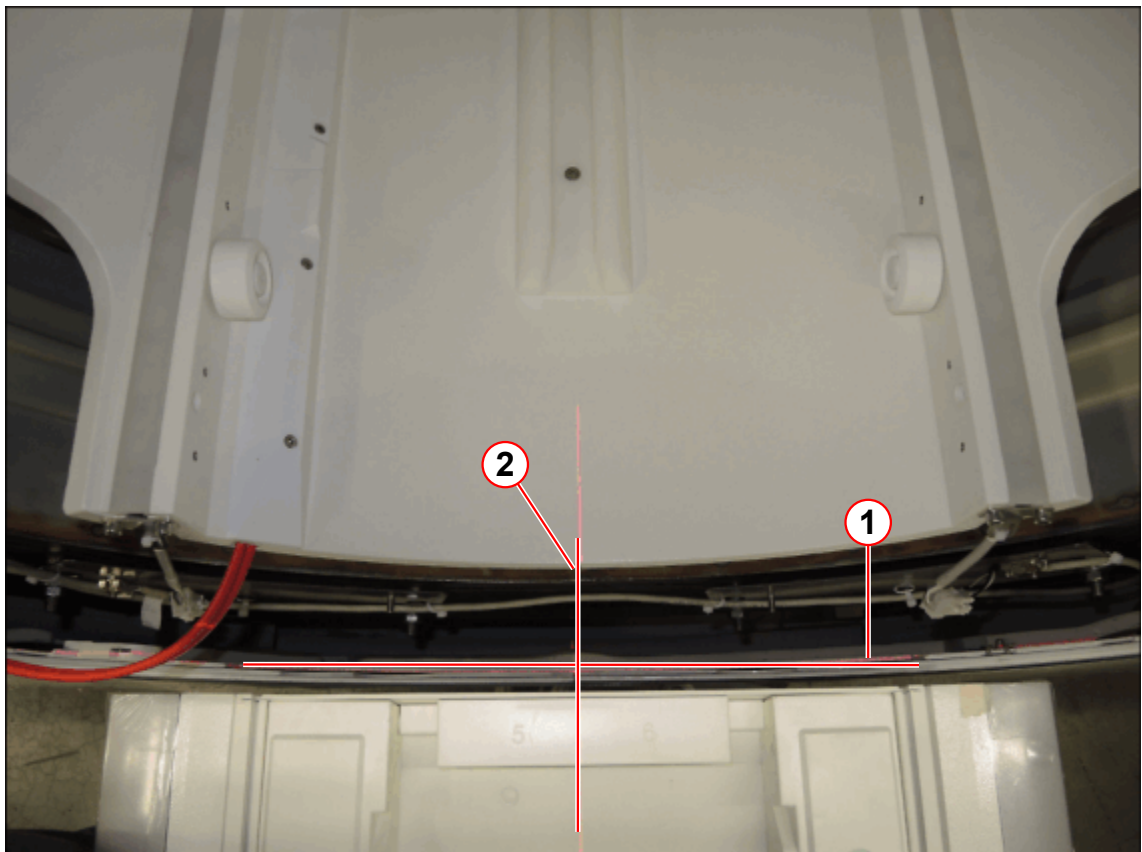


Start with vertical alignment.

1. Vertical alignment of the laser light.

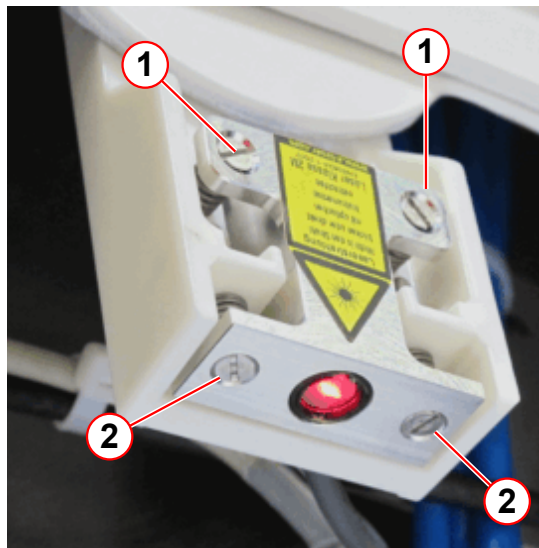
- Switch on the laser light.
- Align the laser light with setting screws (→ 1/ Fig. 92 Page 81)

Fig. 91: Light localizer



- (1) Vertical reference
- (2) Horizontal reference

Fig. 92: Light localizer



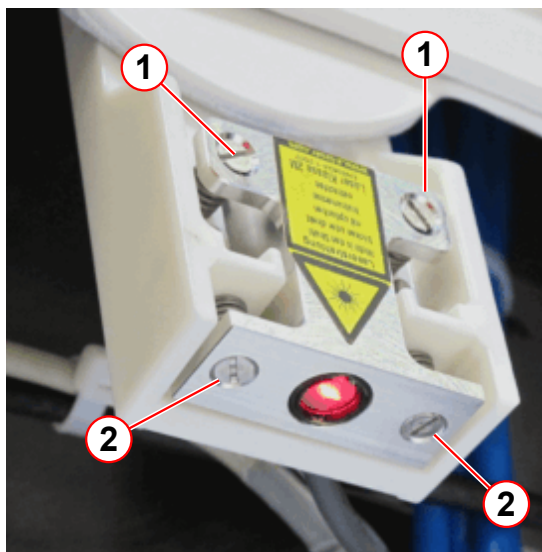
- (1) Vertical setting screws
- (2) Horizontal setting screws

2. Horizontal alignment of the laser light.

- Switch on the laser light.

- Align the laser light with setting screws (→ 2/ Fig. 92 Page 81).

Fig. 93: Light localizer



- (1) Vertical setting screws
- (2) Horizontal setting screws

8.3 Limiting the travel range of the patient table (whole-body option)



If the distance from the fully extended patient table at the service end to the rear wall is not sufficient, the movement range of the patient table has to be limited.

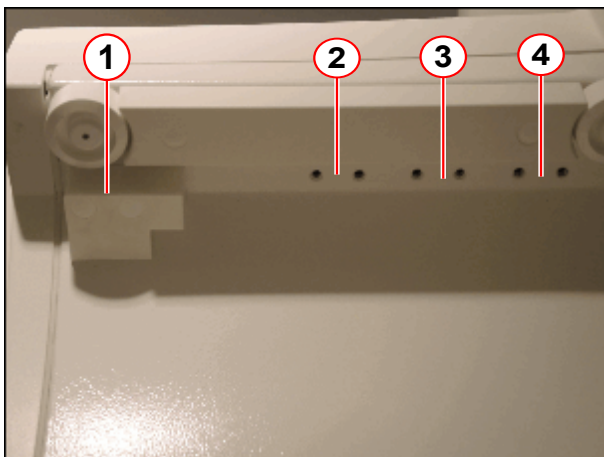
- Set the correct patient table travel range under **Measurement Settings**
- Install the mechanical end stops in the correct position. Refer to the tables below.

Aera:

Position at the rear support, refer to (→ Fig. 94 Page 83) (→ Fig. 95 Page 83)	FWL soft (mm) ¹	SCR (mm) ²	Comment
1	2619	2050	Whole-body
2	2475	1906	Whole-body with limited travel range
3	n.a.	n.a.	n.a.
4	2350	1781	Whole-body with limited travel range
5	2019	1450	Non-whole body

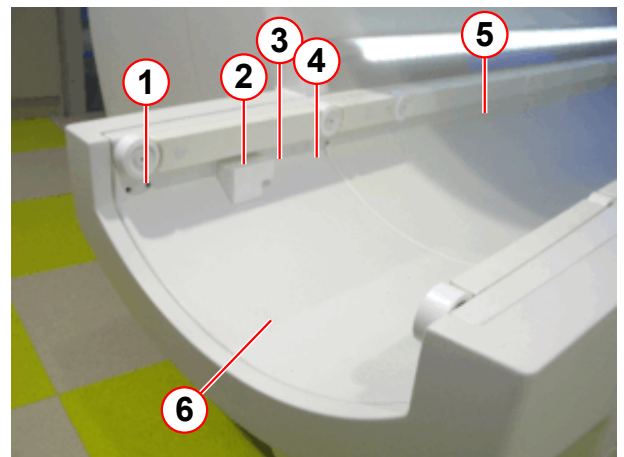
1. Values in the SW configuration
2. Scan range

Fig. 94: Mechanical end stop at the rear support



- (1) Position 1
- (2) Position 2
- (3) Position 3
- (4) Position 4

Fig. 95: MAGNETOM Aera end stop



- (1) Position 1
- (2) Position 2
- (3) Position 3
- (4) Position 4
- (5) Position 5
- (6) Rear support

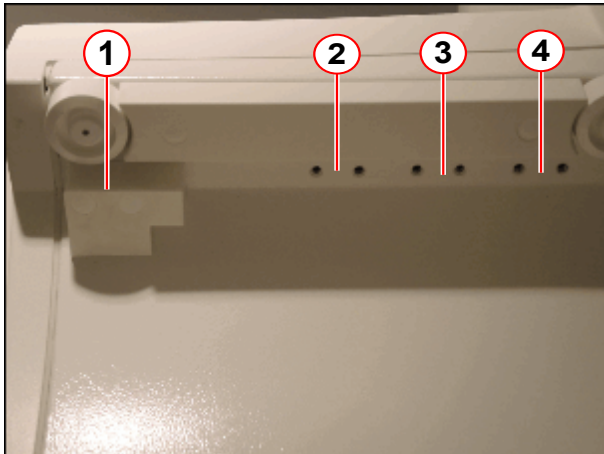
Skyra:			
Position at the rear support, refer to (→ Fig. 95 Page 83) (→ Fig. 94 Page 83)	FWL soft (mm) ¹	SCR (mm) ²	Comment
1	n.a.	n.a.	Only for Aera
2	2757	2050	Whole-body
3	n.a.	n.a.	n.a.
4	2630	1923	Whole-body with limited travel range
5	n.a.	n.a.	n.a.
6	2157	1450	Non-whole body

1. Values in the SW configuration
2. Scan range

Skyra fit			
Position at the rear support, refer to (→ Fig. 94 Page 83) (→ Fig. 97 Page 85)	FWL soft (mm) ¹	SCR (mm) ²	Comment
1	n.a.	n.a.	Only for Aera
2	2757	2050	Whole-body
3	2685	1978	Whole-body with limited travel range
4	2630	1923	Whole-body with limited travel range. ³
4	2540	1833	Whole-body with limited travel range. ⁴
5	2440	1733	Whole-body with limited travel range
6	2157	1450	Non-whole body

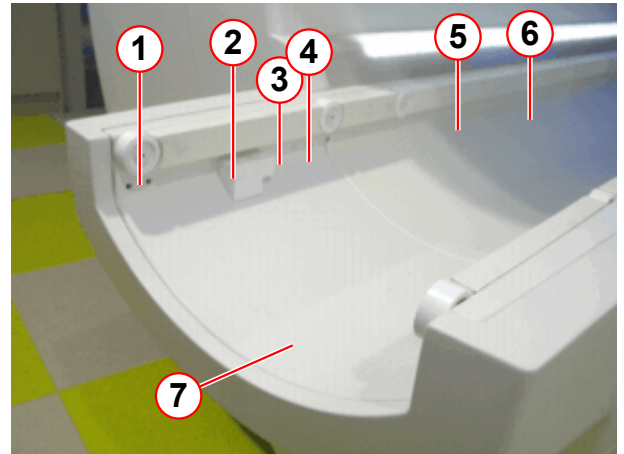
1. Values in the SW configuration
2. Scan range
3. Standard end stop
4. Extended end stop(→ 2/Fig. 98 Page 85)

Fig. 96: Mechanical end stop at the rear support



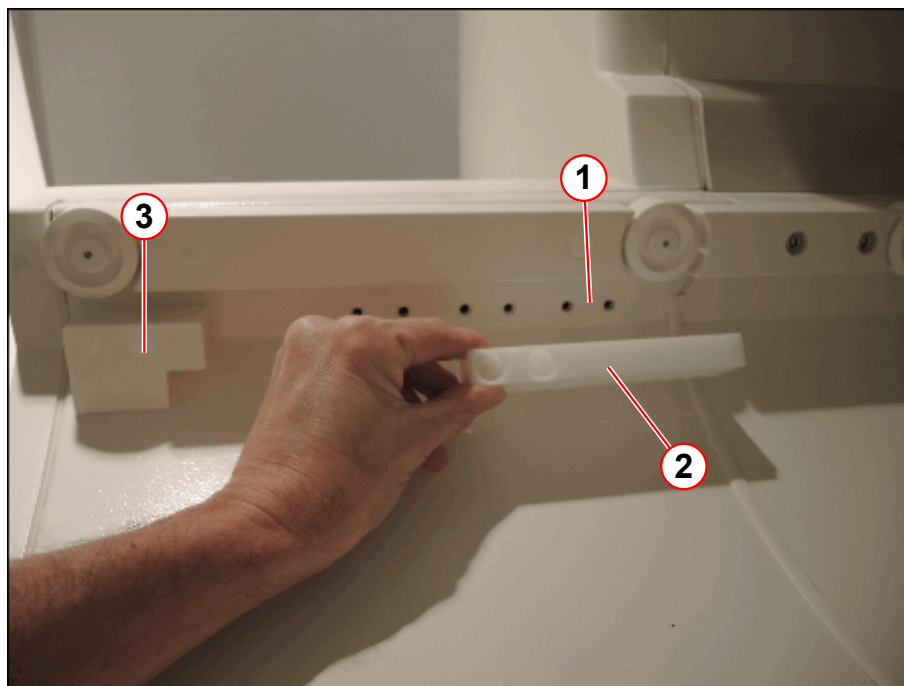
- (1) Position 1
- (2) Position 2
- (3) Position 3
- (4) Position 4

Fig. 97: MAGNETOM Skyra / Skyra fit end stop



- (1) Position 1
- (2) Position 2
- (3) Position 3
- (4) Position 4
- (5) Position 5
- (6) Position 6
- (7) Rear support

Fig. 98: Patient Table Travel Range

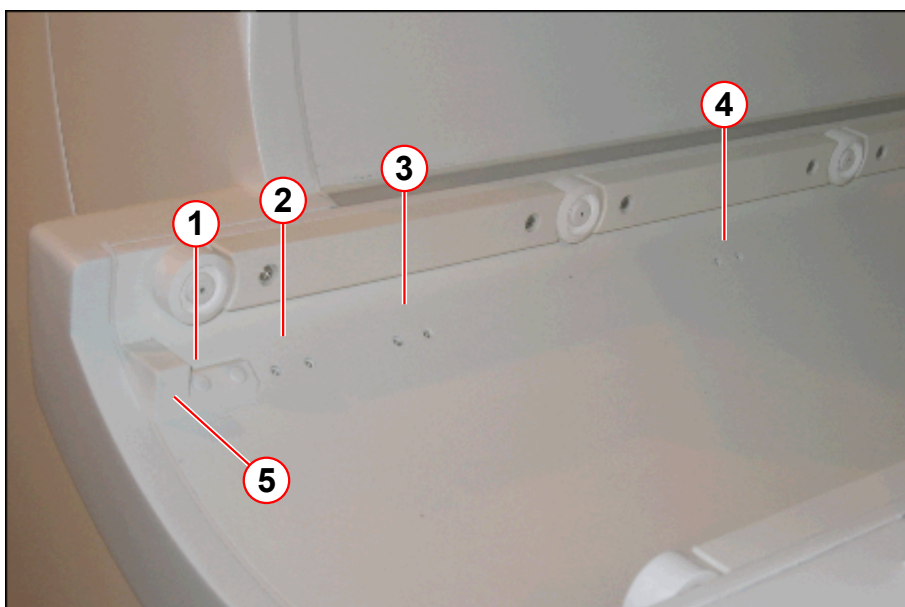


- (1) End stop position
- (2) Extended end stop
- (3) Standard end stop

Avanto fit:			
Position at the rear support, refer to (→ Fig. 99 Page 86)	FWL soft (mm) ¹	SCR (mm) ²	Comment
1	2600	2040	Only for Aera
2	2540	1980	Whole-body
3	2440	1880	Whole-body with limited travel range
4	2240	1680	Non-whole body

1. Values in the SW configuration
2. Scan range

Fig. 99: Patient Table Travel Range



- (1) Position 1
- (2) Position 2
- (3) Position 3
- (4) Position 4
- (5) End stop

- Install the mechanical end stop, refer to the table above for the positions of the end stops.
- Testing the patient table travel:
 - Activate the patient table emergency release.
 - Manually move the table toward the magnet service end until the table stops in the horizontal end position (mechanical end stop).

- Check the safety distance from the patient table (head-end) to the rear wall.
 - » If the distance **is greater than** 500 mm, no further actions are necessary. .
 - » If the distance **is less than** 500 mm, a **warning label** is necessary; refer to (→ Sign for bruising hazards / Page 50)
- Reactivate the emergency release.

8.4 Checking the patient table

- Perform the safety-related tests of the patient table.

The safety-related tests are part of the "Maintenance Instructions Combined Workflow".

- » Refer to "Maintenance Instructions Combined Workflow".



If the patient table must be adjusted, refer to the "Installation Instructions".

9.1 General



Valid for Germany and certain other EU countries.

If the monitor of an MR system is used for medical reporting, a special acceptance test is required in accordance with DIN V6868-57. The monitor must be labeled as such by the installation personnel.

According to the documentation listed below:

1. Acceptance testing for image display devices (→ Acceptance testing for image display devices / MR00-000.820.01).
2. Acceptance test protocol for image display devices. (→ Acceptance test protocol for image display devices / MR00-000.820.02)



The monitor only needs to be checked (and adjusted as necessary) if at least one of the parts listed below is replaced.

- Monitor:
- Graphics board
- PC

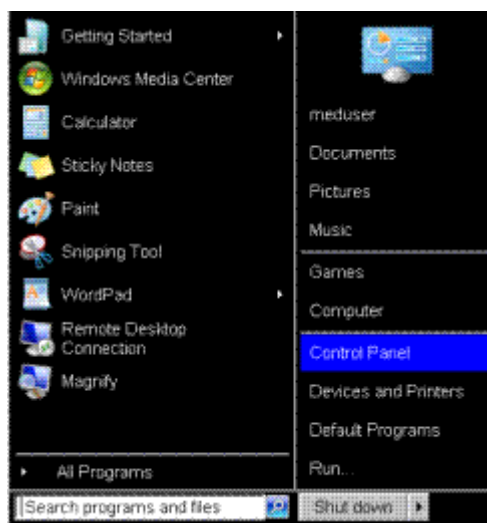
Refer to (→ Host / M7-000.841.12) "Replacement of parts"

10.1 System with N4_VD13A

10.1.1 Settings

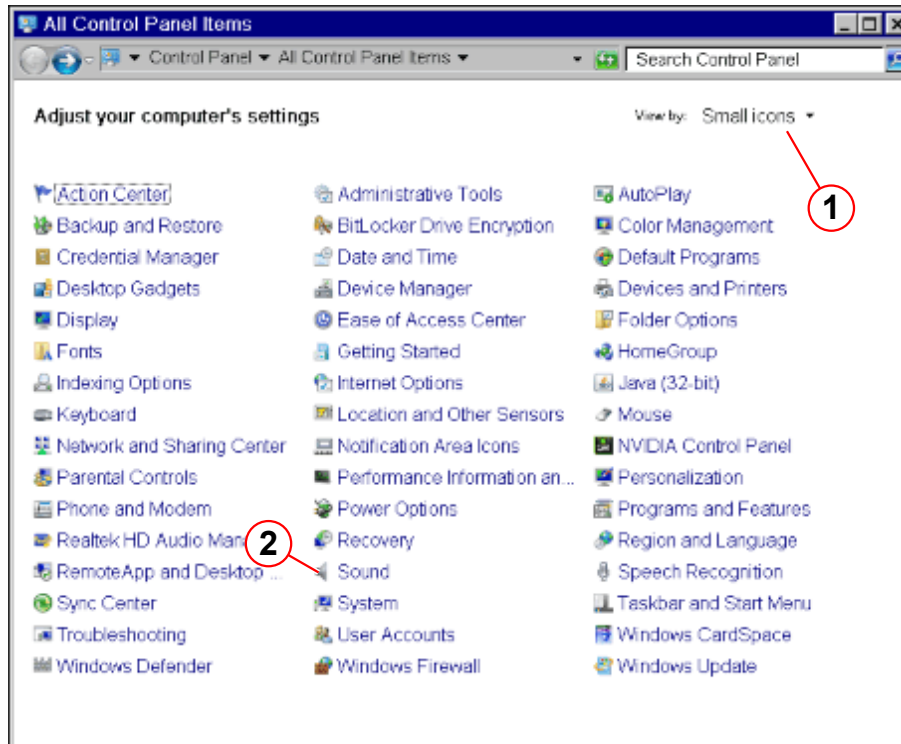
- Switch on and boot the system.
- Close the patient application (Patient/ Close Patient) in case Patient is active.
- Press <Ctrl> and <ESC> and select Programs/ Accessories/ Advanced User and enter the password: meduser1. Click <OK> to exit.
- Press <Ctrl> and <ESC>
- Select Control Panel

Fig. 100: Control Panel



- Select Small Icons (→ 1/ Fig. 101 Page 91)

Fig. 101: Control Panel



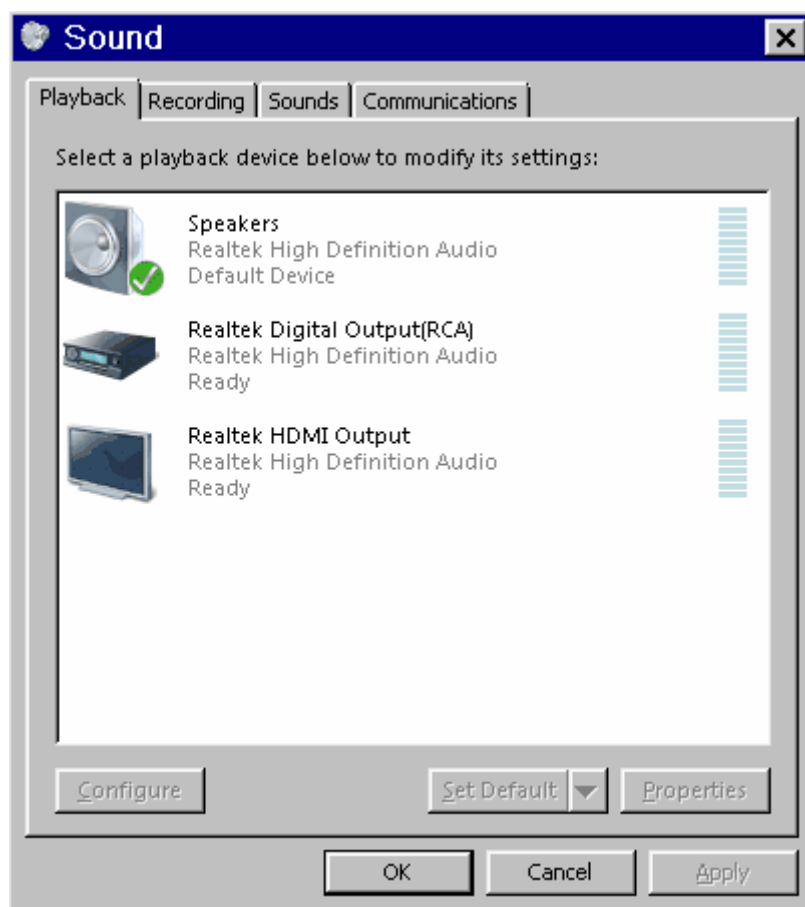
- (1) Small Icons
- (2) Sound



The screenshots can be different. They are dependent on the Host type.

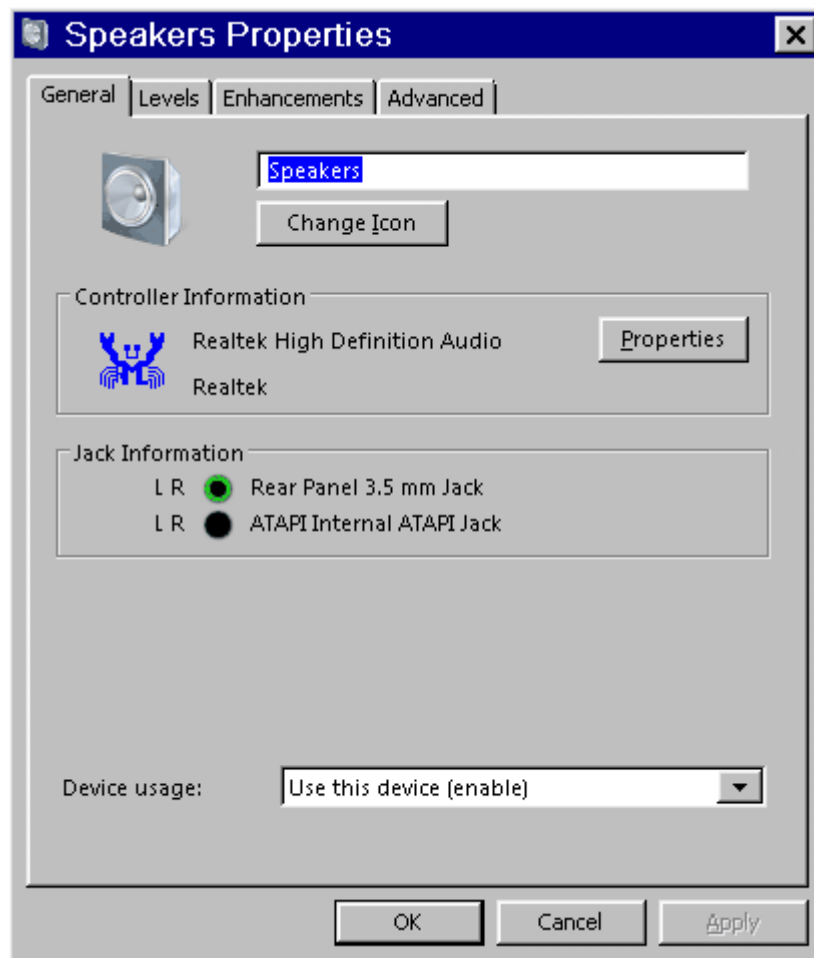
- Click <Sound>” (→ 2/ Fig. 101 Page 91)

Fig. 102: Sound



- Click <Speakers>

Fig. 103: Speakers Properties



- Select <Levels>.
 - Set the volume of "Realtek HD Audio output" to maximum.
 - Select <Balance>
 - Set the Balance volume left and right to maximum.
 - Select <OK>

Fig. 104: Speakers Properties

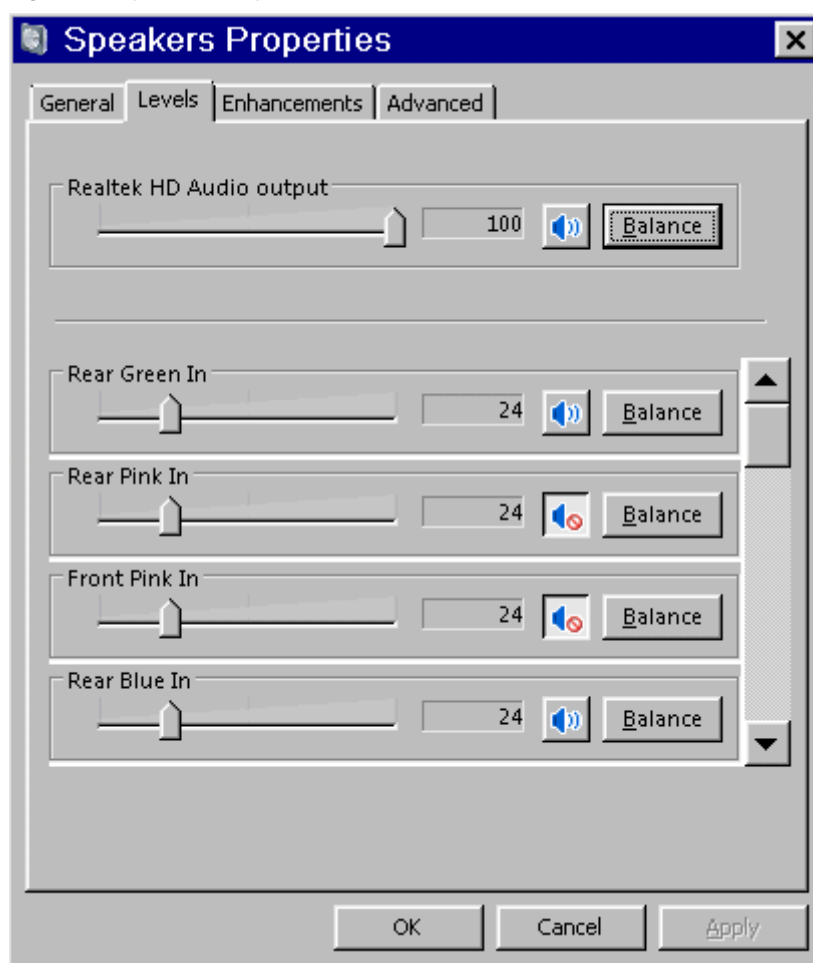
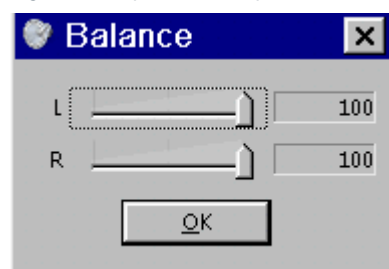
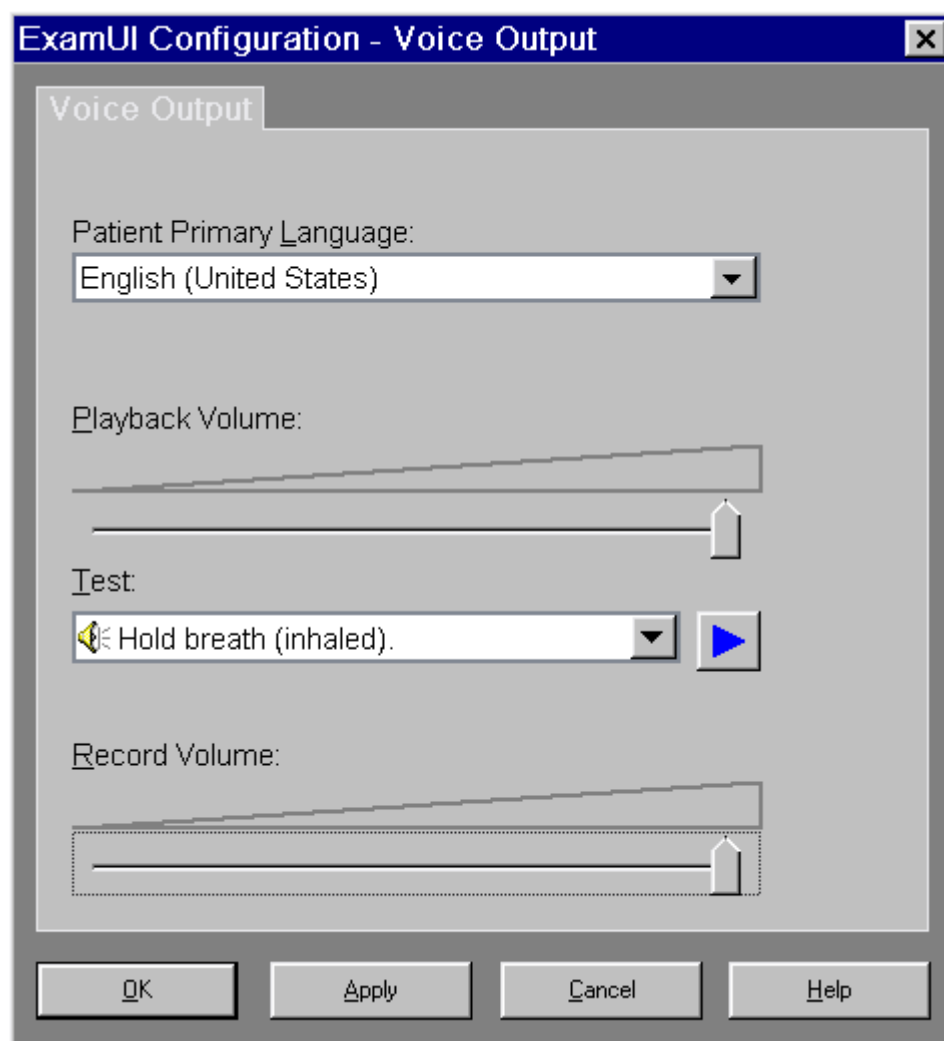


Fig. 105: Speakers Properties



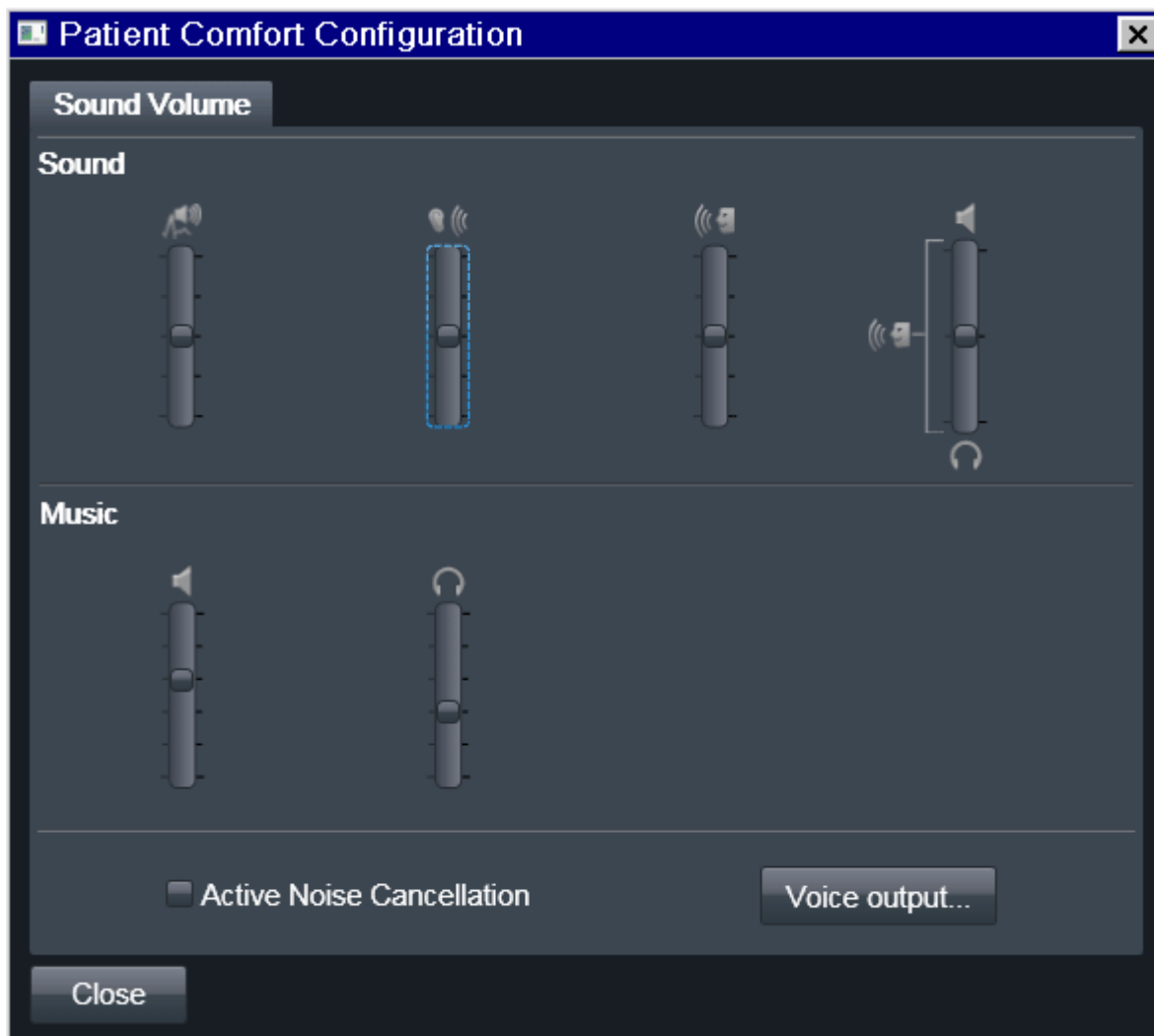
- Select Option / Configuration / Patient Comfort / Voice Output
 - Set the Playback Volume to maximum
 - Select <Apply> and <OK> to save the settings.

Fig. 106: ExamUI Configuration



- Set the regulators

Fig. 107: Patient Comfort Configuration



- Select <Sound> and <Recording>
 - Select "Line in"

Fig. 108: Sound

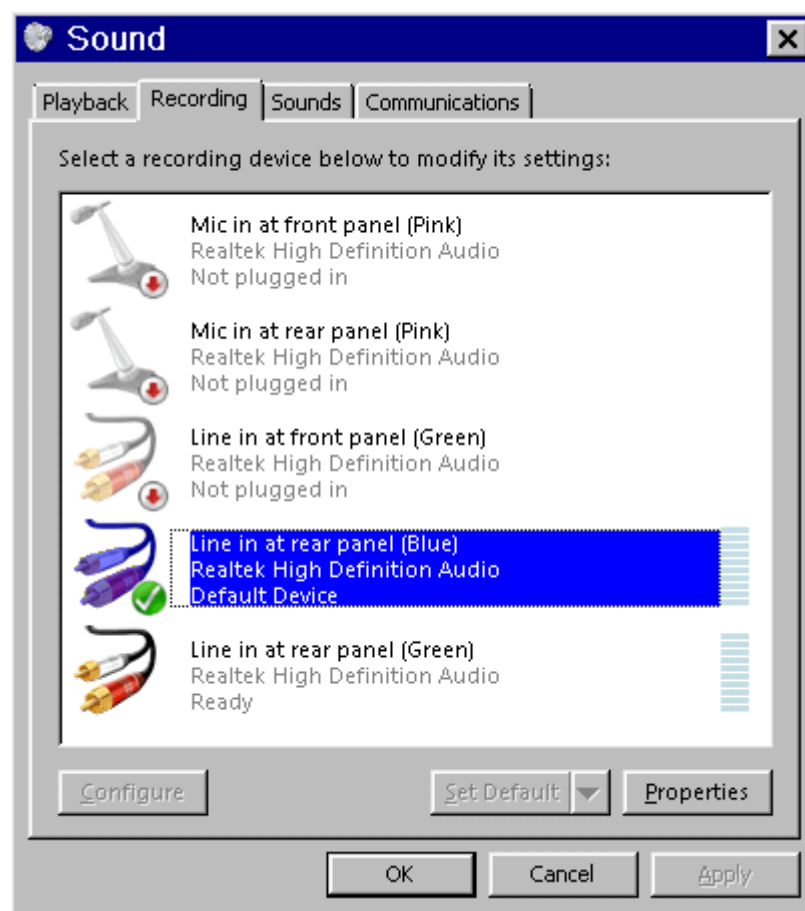
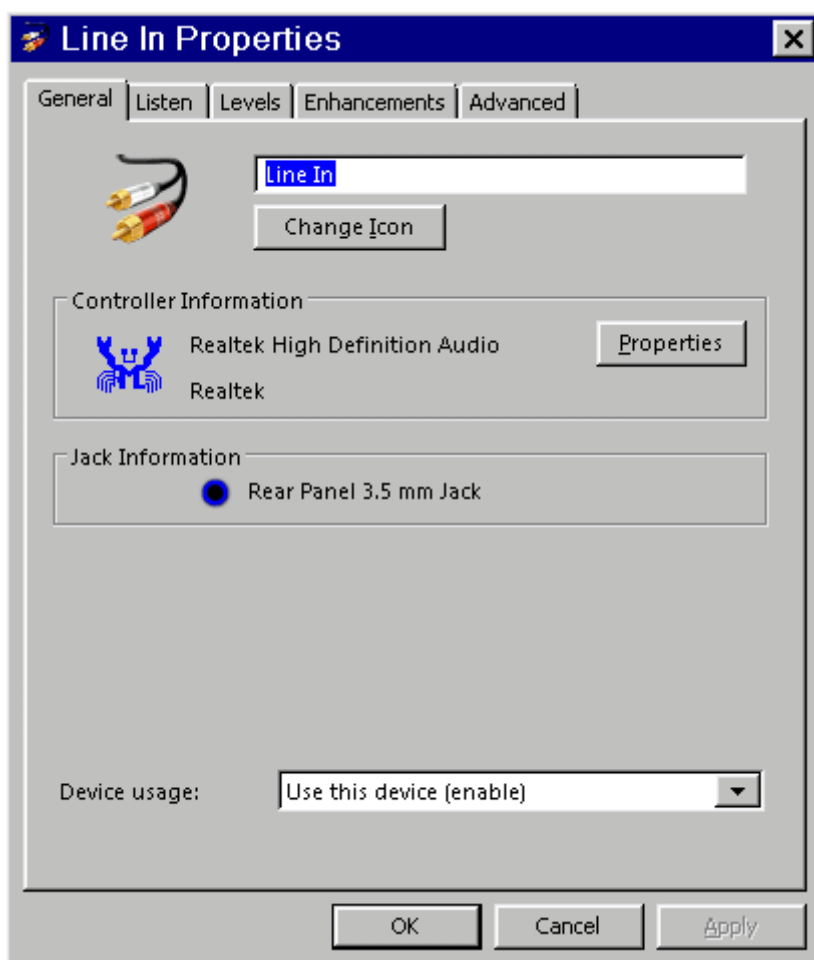
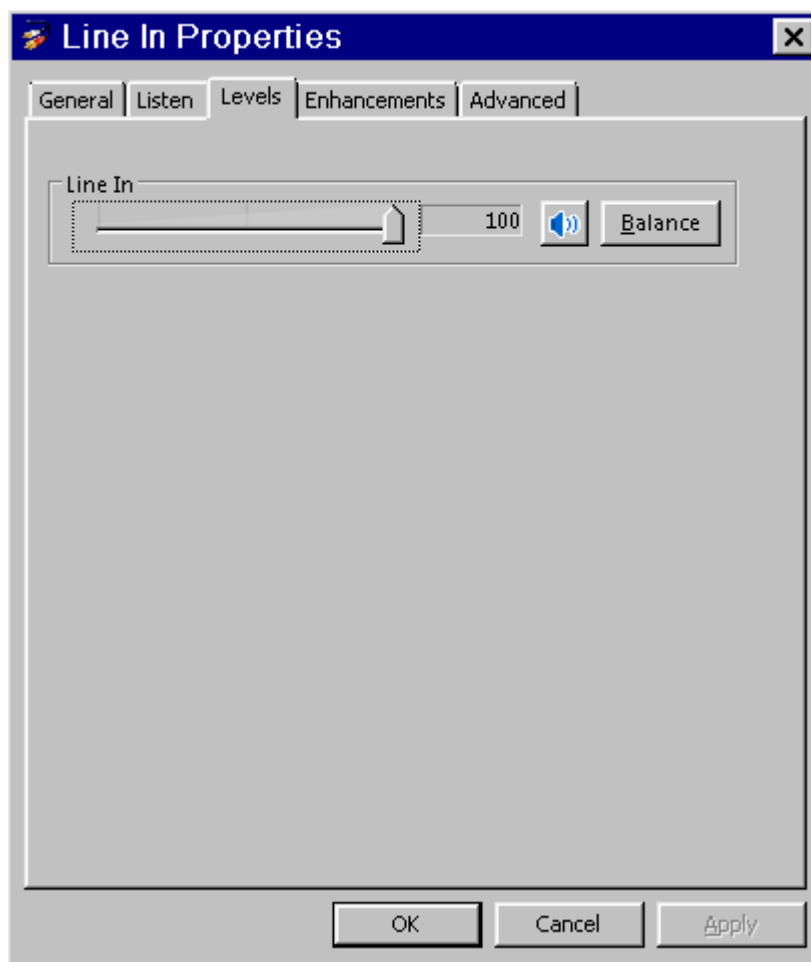


Fig. 109: Line In Properties



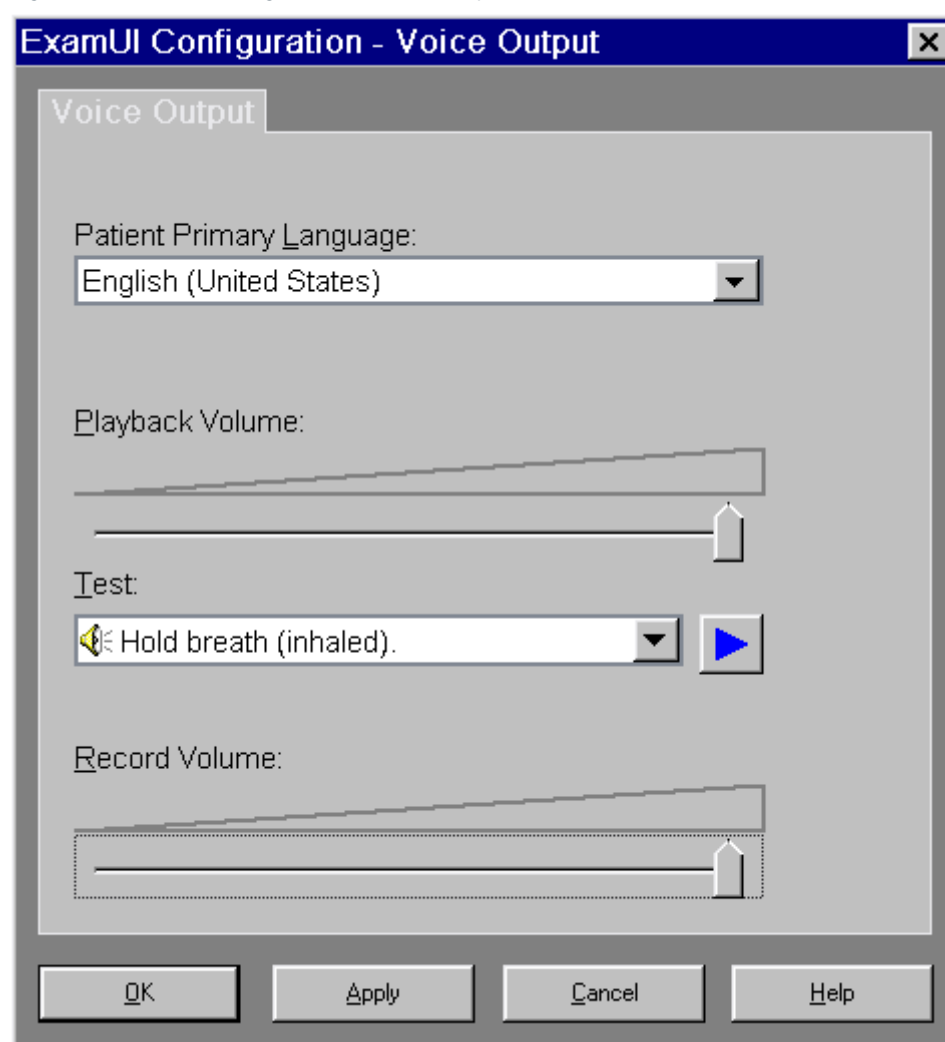
- Select <Levels>
 - Set the "Line In" volume to maximum

Fig. 110: Line In Properties



- Select <ok>
- Select Option / Configuration / Patient Comfort / Voice Output.
 - Set the Record Volume to maximum.

Fig. 111: ExamUI Configuration - Voice Output



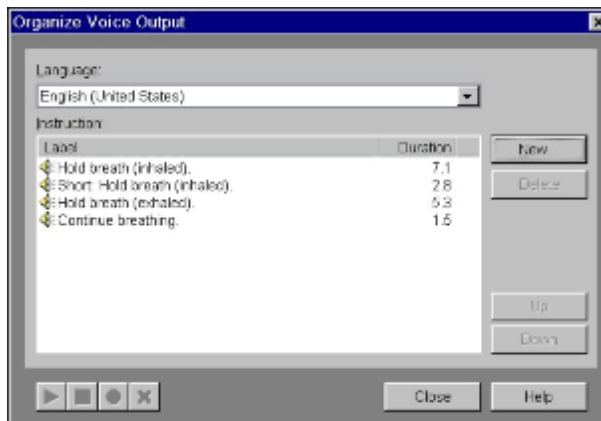
- Select <ok>

10.1.2 Final Check

- Perform a final check for the voice output.
 - Close the patient application (Patient/ Close Patient) if a Patient is active.

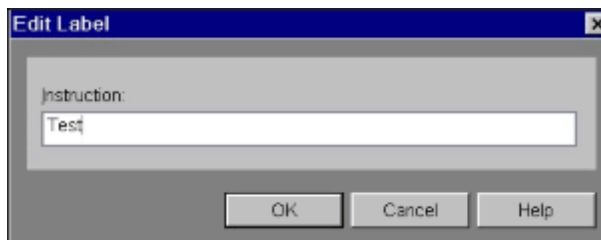
- Select Queue / Voice Output / Organize

Fig. 112: Voice Output



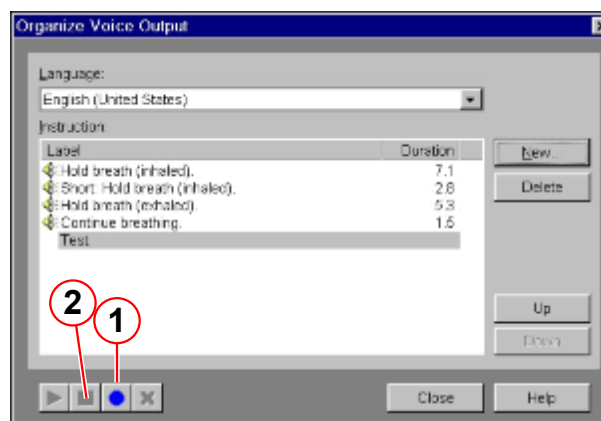
- Click <New>, enter a name, e.g. "Test" and select <OK>

Fig. 113: Organize Voice Output



- Click the start button (→ 1/ Fig. 114 Page 101).

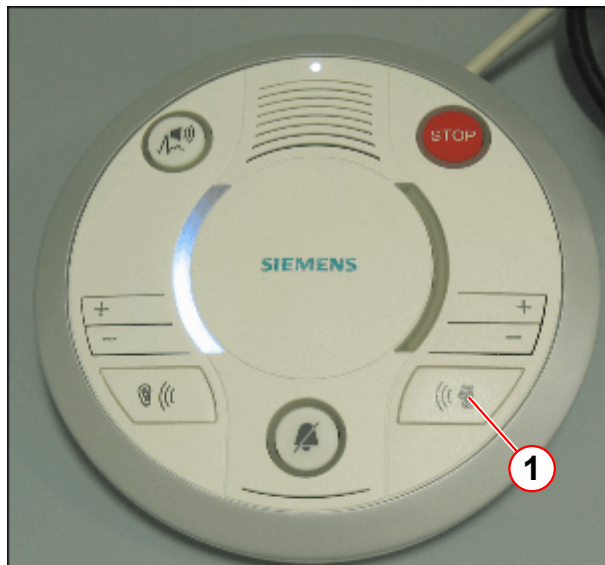
Fig. 114: Voice Output



- (1) Start button
- (2) Stop button

- Press the "Talk button" (→ 1/Fig. 115 Page 102) of the intercoms operating unit and keep it pressed during this procedure.

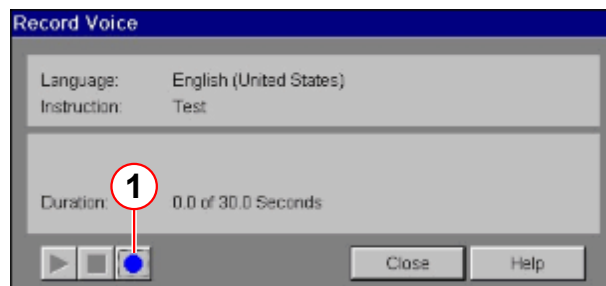
Fig. 115: Intercom operating unit



(1) Talk button

- Click the start button (→ 1/Fig. 116 Page 102) in the Record Voice window.

Fig. 116: Record Voice



(1) Start button

- Speak the command into the operating unit.
- After finishing the text, click the stop button (→ 2/Fig. 114 Page 101) in the Record Voice window.
- Release the talk button on the operating unit.

10.1.3 Informing the customer

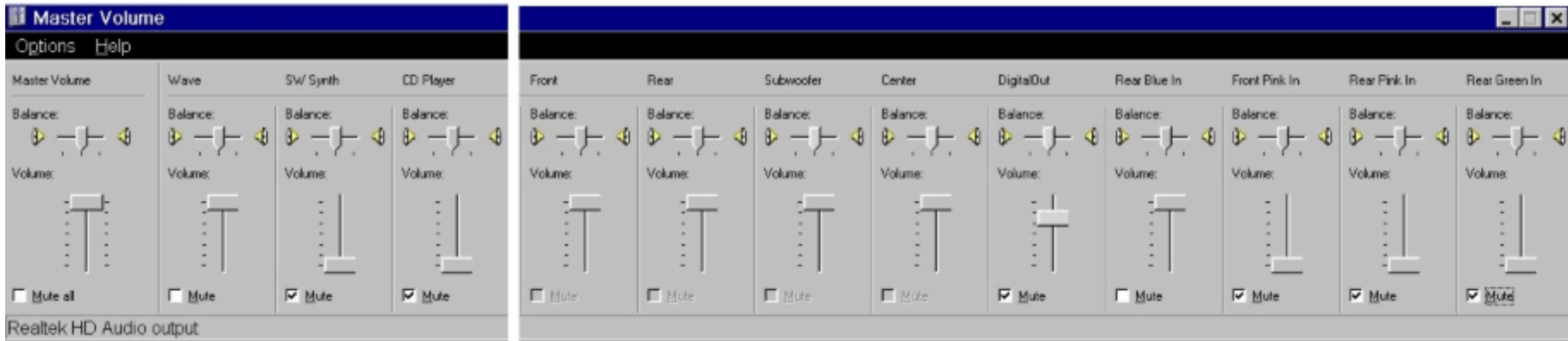
Inform the customer about this new option by explaining how to record automatic voice output and what to keep in mind when using this new option.

10.2 System with N4_VD11

10.2.1 Settings

- Switch on and boot the system.
- Close the patient application (Patient/ Close Patient) in case Patient is active.
- Press <Ctrl> and <ESC> and select Programs/ Accessories/ Advanced User and enter the password: meduser1. Click <OK> to exit.
- Press <Ctrl> and <ESC>
- Select Programs/ Accessories/ Entertainment/ Volume Control.
- Set the regulators in the Master Volume window, as shown in (→ Fig. 117 Page 104).
- Deactivate the Master Volume by clicking mute, as shown in (→ Fig. 117 Page 104)

Fig. 117: Master Volume Window





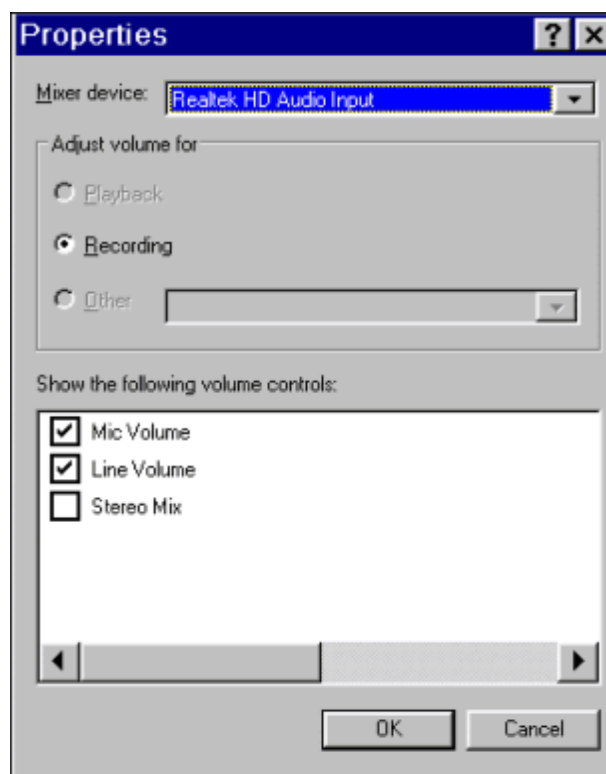
"Mute" does not have to be activated for the controls of Master Volume and Wave.



In case the master volume window, has a different design than the figure (→ Fig. 117 Page 104), set the "Master Volume" and "Wave" to maximum, and set all other controls to "Mute".

- Select Options/ Properties in the same window and click <Recording> (→ Fig. 118 Page 105).

Fig. 118: Properties



- Press <OK>.
- Set the regulators for the "Recording Control" as shown in (→ Fig. 119 Page 106).
- Set the "Line volume" to approx. 100% and the volume.

- Click <Select> for the "Line Volume"

Fig. 119: Recording Control



In case the recording control menu has a different design than the figure (→ Fig. 119 Page 106), set the Line Volume to approx. 25% and all others to 50%

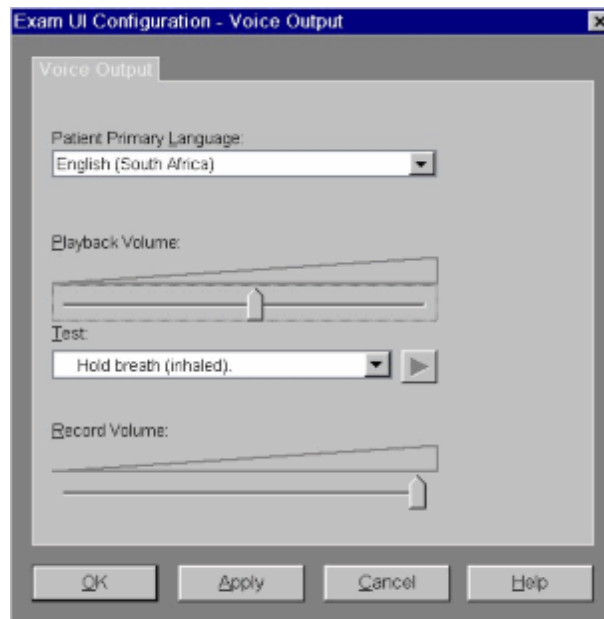
- Click <Select> for the "Line Volume" with all others deselected, as shown in the example.
Refer to online help, if one of the controls for the device is missing.
Close the window via Options/Exit.

Customer setting via Exam UI

- Select Options/ Configuration/ Patient Comfort / Voice Output
 - Set the volume at approx. 50% of the maximum.

- Select <Apply> and <OK> to save the settings.

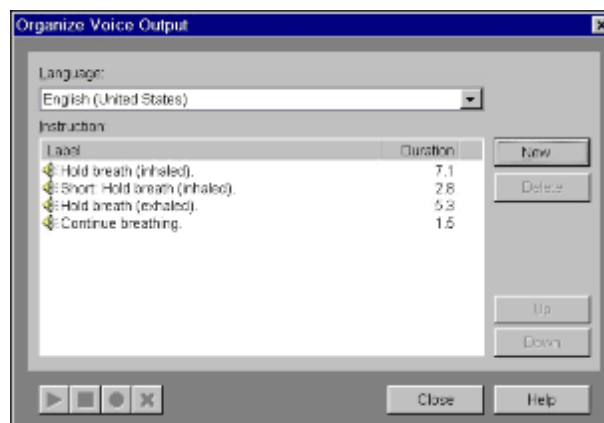
Fig. 120: Voice Output



10.2.2 Final Check

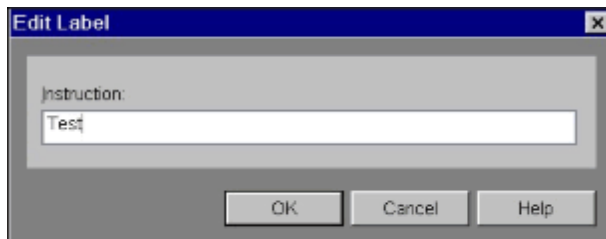
- Perform a final check for the voice output.
 - Close the patient application (Patient/ Close Patient) if a Patient is active.
 - Select Queue / Voice Output / Organize

Fig. 121: Voice Output



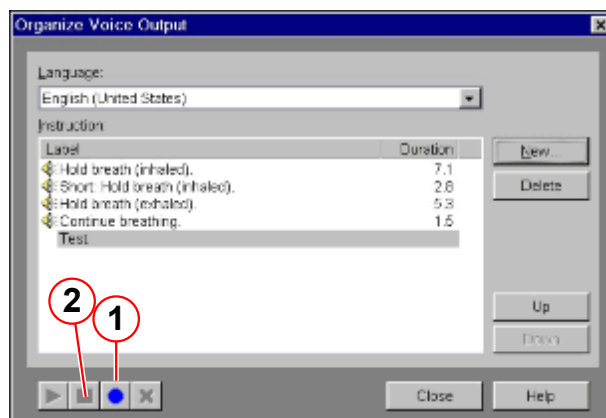
- Click <New>, enter a name, e.g. "Test" and select <OK>

Fig. 122: Organize Voice Output



- Click the start button (→ 1/ Fig. 123 Page 108).

Fig. 123: Voice Output

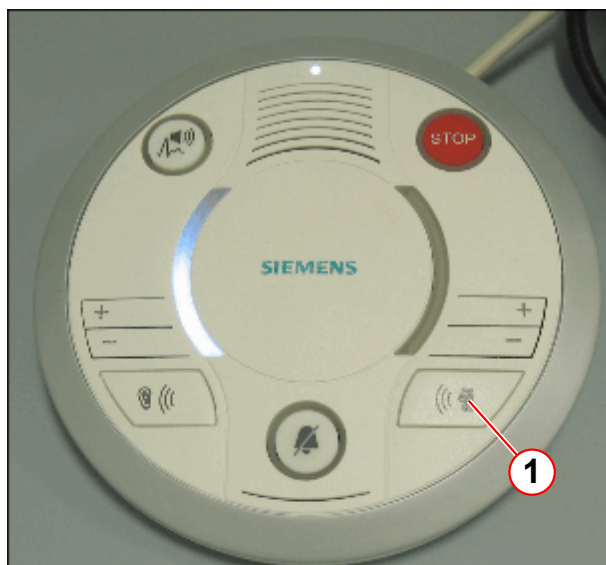


(1) Start button

(2) Stop button

- Press the "Talk button" (→ 1/ Fig. 124 Page 108) of the intercoms operating unit and keep it pressed during this procedure.

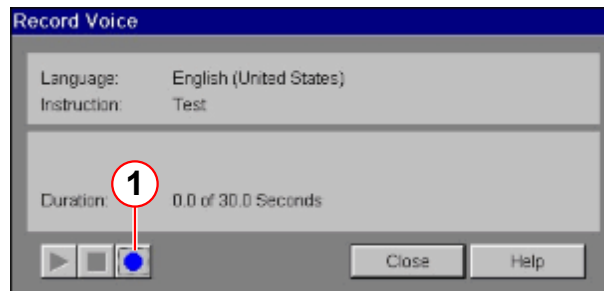
Fig. 124: Intercom operating unit



(1) Talk button

- Click the start button (→ 1/Fig. 125 Page 109) in the Record Voice window.

Fig. 125: Record Voice



(1) Start button

- Speak the command into the operating unit.
- After finishing the text, click the stop button (→ 2/Fig. 123 Page 108) in the Record Voice window.
- Release the talk button on the operating unit.

10.2.3 Informing the customer

Inform the customer about this new option by explaining how to record automatic voice output and what to keep in mind when using this new option.

11.1 General

11.1.1 Functional Description of the VLM VESDA Device

The SIEMENS_VLM incorporates a very sensitive, laser-based smoke detection unit which can detect overheated connections in a very early phase. A little fan continuously draws air through a hose system which connects to the relevant connections which can over-heat in a fault scenario.

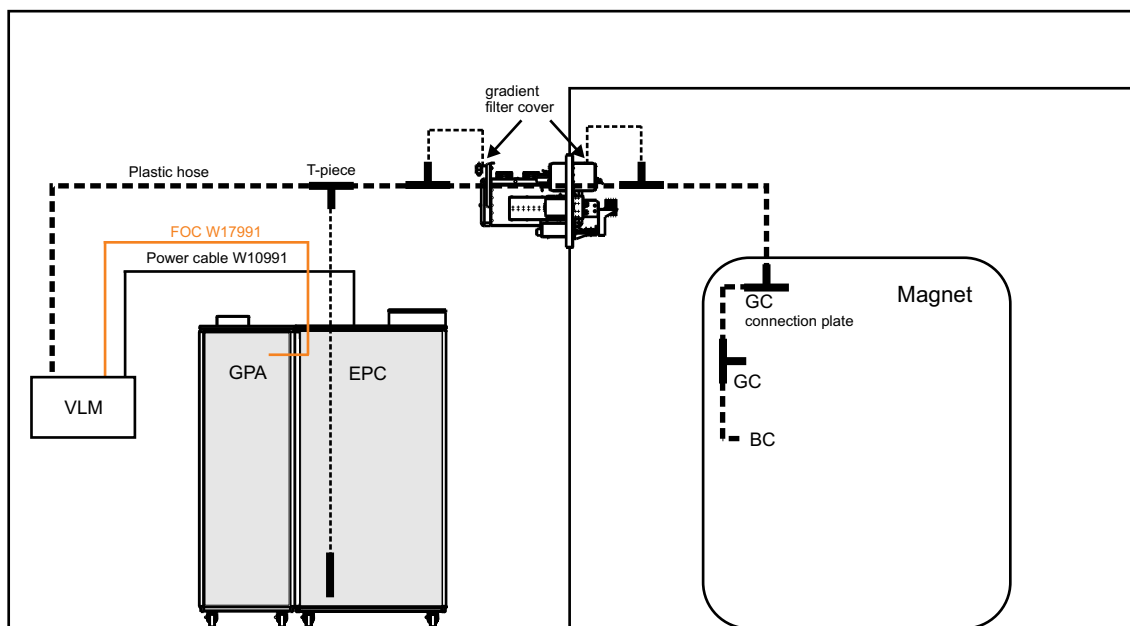
The aspirated air samples are fed through a filter into the laser detection chamber inside the unit.

Inside the calibrated detection chamber the air sample is exposed to a laser light source. If the sample contains small particles (e.g. emitted by overheated gradient connections), light is scattered within the detection chamber and is immediately recognized by the highly sensitive optical receiver system.



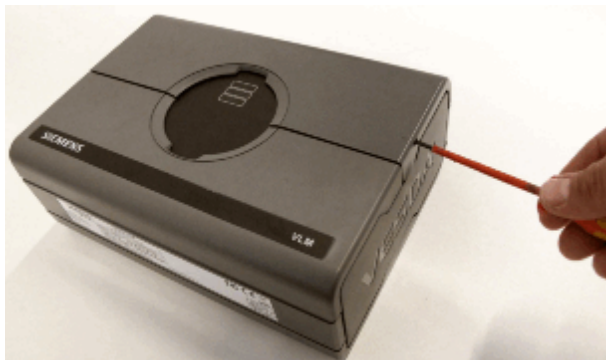
The VLM device is not supposed to work as or to replace a fire warning system, but rather as an additional safety precaution to switch the MR system into "stand by mode" in case of over-heated monitored connection.

Fig. 126: Gradient Connection Supervision (Aera/Skyra/Prisma)



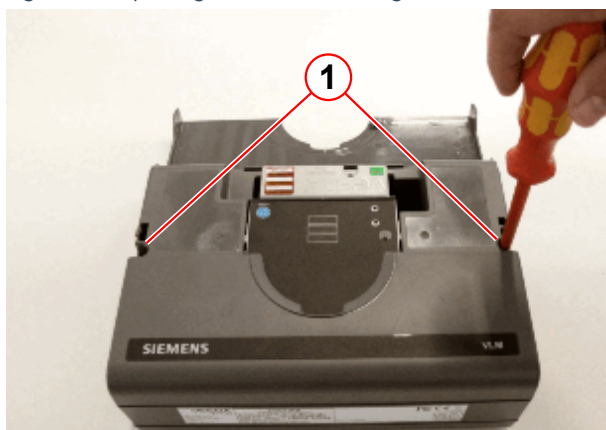
11.1.2 VLM opening

Fig. 127: Opening the upper cover



- Open the upper cover of the device.

Fig. 128: Opening the lower housing



- Remove the 2 Phillips screws (1) and swing out the housing.

(1) 2x Phillips screws

11.2 GPA D70 Configuration

D70 configuration is not necessary if the GPA serial number is higher or equal to those listed in the following table.

GPA 90/22/F	Part number	Serial number	Rev.
GPA 90/22/F	10513350	1239	02
GPA 75/22/F	10513313	1304	02
GPA 62/20/F	10513280	1239	02

- Remove the front cover of the GSSU.

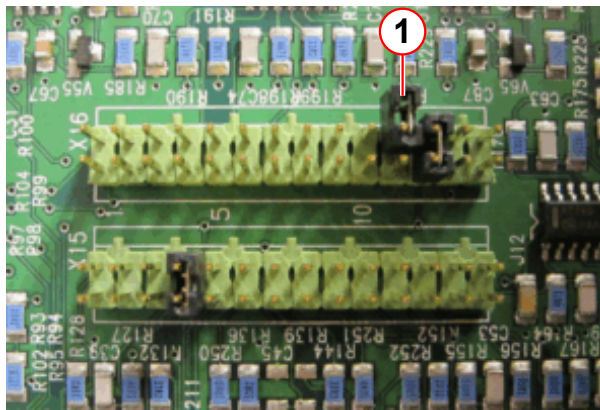
 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad while performing the following work steps.

Fig. 129: Removal of D70 - I/O board



- Remove the upper and the lower front rail of the GSSU.
- Remove D70 X3.
- Remove D70 - I/O board.

Fig. 130: Setting of jumper X16 on D70



(1) Jumper X16 A11-B11 removed

- Remove jumper X16 - 11 (1) on D70.
 - » This enables the supervision via FOC at U1 in front of D70.
- Reinsert D70.
- Reinstall the upper and lower front rail of the GSSU.
- Reinstall the front cover of the GSSU.

11.3 VLM Jumper Settings

Fig. 131: Jumper on 2-3 default setting - correct for Tim systems and Biograph mMR

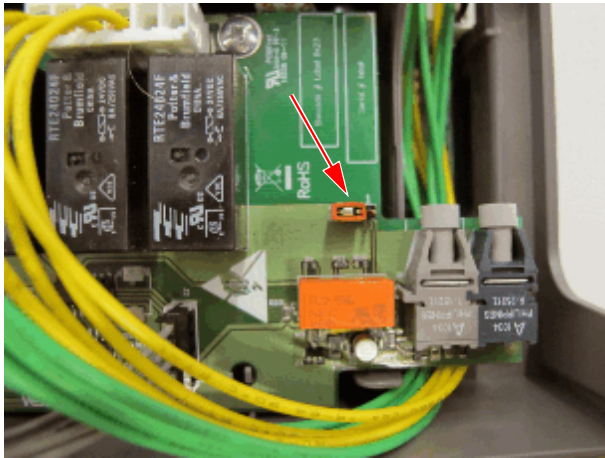
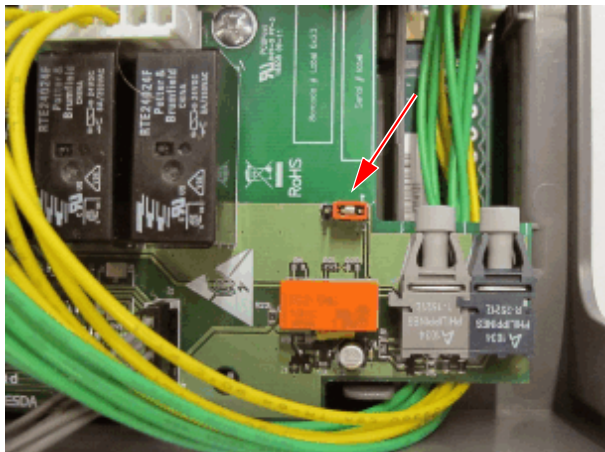


Fig. 132: Jumper on 1-2, correct setting for Aera/Skyra)



■ There is one jumper left from the fiber optic connectors.

- » The default jumper setting is 2-3, which is appropriate for all Tim systems, Biograph mMR, and Terra.
- » On Aera/Skyra/Avanto^{fit}/Skyra^{fit} systems, this jumper must be set to 1-2 as per (→ Fig. 132 Page 114)

11.4 Display

Fig. 133: VESDA device



- (1) OK (green)
- (2) FAULT (orange)
- (3) FIRE (red)
- (4) RESET

Legend	ON	FLASHING	Consequence
OK	Detector is powered and in normal operation		
FAULT	Indicates a major fault	Indicates a minor fault	» Normalize VLM device » Call Service
FIRE			» GPA-Modulator will be disabled » MR scanner will be switched off 1 minute later

11.5 Detector Control Button

There is only one button on front of the VLM device, which is the RESET button.

Icon	Button	Function
	RESET	■ Press and hold down this button to test the function of the LEDs on the unit. ■ Release this button to clear latched faults and alarms. » Alarm and fault lights will switch off, and if the system is still in alarm or fault mode, the lights will reappear after a certain delay. » The reset button does not operate during the delay fire notification interval. ■ Press and hold down this button for 10 sec. until the 3 LEDs start flashing to activate the set-mode for 100% air flow. » For details refer to (→ Start-Up Procedure of the VLM Device / Page 117).

11.6 Start-Up Procedure of the VLM Device

- Close the RF-cabin door.
- If a UPS is installed, re-connect the 2 non-heating apparatus connectors to the UPS distribution box.
- Switch on the UPS (if present) according to it's Operating Manual.
- Switch on F1 in the GPA cabinet, F21 in the EPC cabinet, and F100 on top of the EPC cabinet.
 - » The FAULT lamp is on for 5 sec.
- The VLM device is powered now and needs a 90 sec warm-up time.
- Switch on the MR System.
- Open the front cover of the device and push the RESET button for 10 sec. until all lamps are flashing.
- Release the RESET button and wait until the lamps go off.
- When the lamps start flashing again, press the RESET button once more to confirm the actual setting.
 - » The current air flow through the device (which depends on the individual system configuration and hose length) is now set to 100%.
 - » Deviations from this nominal Air flow $100 \% \pm 20 \%$ will result in an internal error.
 - » A removed, broken, or blocked hose will result in a FAULT now.

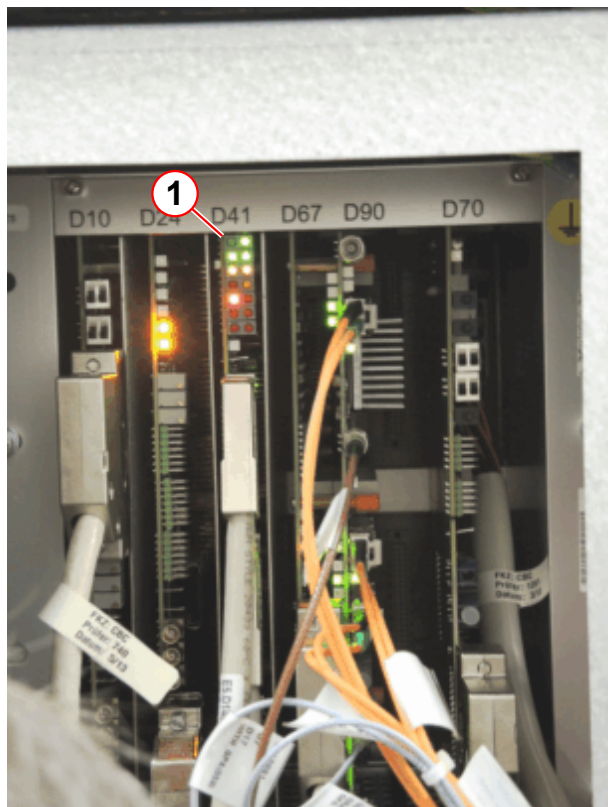


In case the normalization procedure described above does not work properly and the VLM device goes into failure mode, there could be a differential pressure between the RF-cabin and the equipment room, which exceeds the limits of the VLM device. In this case a pressure equalization tube must be fed from the exhaust of the VLM device back into the RF-cabin, according to (→ Pressure Equalization (optional) / Page 126).

11.7 Final Check

Modulator disable test

Fig. 134: Modulator LED in GPA cabinet off



(1) Modulator LED (V120)

- Remove the FOC (W17991) from U1
 - » The GPA modulator should be disabled (scanner signal at MRC is crossed through and modulator LED in GPA cabinet is off, refer to (→ Fig. 134 Page 118))

- » GPA error should be reported in the event log:

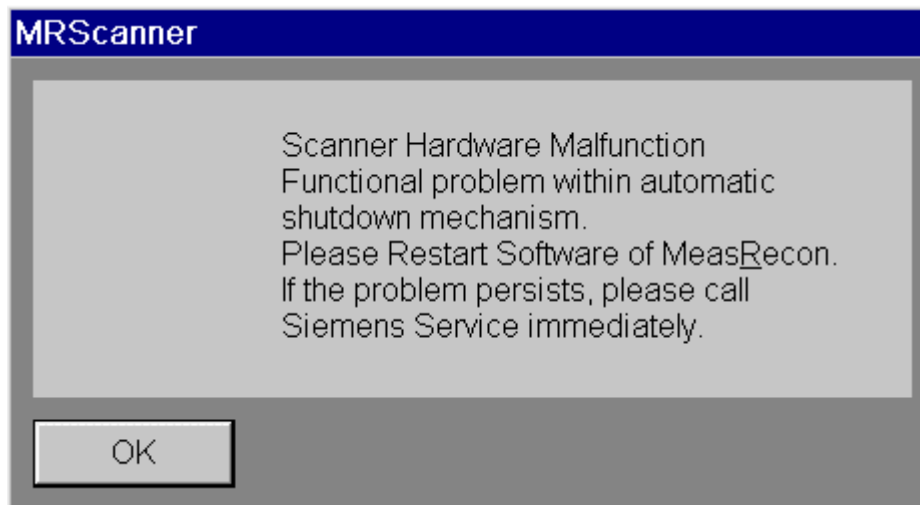
Date/Time	ID	Source	Seq.	Message Text
22-May-15 17:24:25	2016	MRI_GDV	0	Gradient Power Amplifier error: Hardware error. Please reboot

- Reattach the FOC W17991 to U1.
- Reassemble the VLM device.
- Perform a "Reboot Scanner".

Air flow failure simulation

- Remove the hose from the VLM device.
 - » ~30 sec. later a fault should be displayed on front of the VLM device due to high air flow and a pop-up should be displayed at the console (→ Fig. 135 Page 119).

Fig. 135: Pop-up "FAULT" condition



- Refit the hose.
 - » The green OK lamp should go on.

Differential pressure test

- Open the RF-cabin door and wait for 5 minutes.
 - » The green OK lamp on the VLM device should remain on.



In case the VLM device goes into failure mode, there is a differential pressure between the RF-cabin and the equipment room, which exceeds the limits of the VLM device. In this case a pressure equalization tube must be fed from the exhaust of the VLM device back into the RF-cabin, according to (→ Pressure Equalization (optional) / Page 126).



In order to prove the proper function of the detector, a smoke test must be performed.



WARNING

Accumulation of oxygen during helium filling or magnet ramping.

The use of glowing smoke wicks during Helium filling or magnet ramping could result in an open fire in the area of the magnet service turret.

- » Do NOT perform a smoke test during helium filling or magnet ramping!



The smoke wicks, located in a box for the test below, are part of the Update-Kit. Each smoke wick will provide up to 30 minutes of continuous smoke, or 60 individual 30 second tests.

Smoke test

- Take one smoke wick out of the box (→ Fig. 136 Page 120).
- Hold wick at the end with black plastic handle mounted (→ Fig. 137 Page 120).

- Light the tip of the wick and allow it to burn for 10-15 seconds (→ Fig. 137 Page 120).

Fig. 136: Smoke wicks in a box



Fig. 137: Lighted smoke wick



- Extinguish the flame and hold the smoke wick behind the rear cover in the area of the lower sample plate for 5 seconds (→ Fig. 138 Page 120).
- Extinguish the wick by simply placing it in the empty box and close lid carefully. This will extinguish the wick after approximately 15 seconds.
 - » With a delay of about 20-40 seconds (depending on the hose length) the "FIRE" lamp on the VLM device should light up (→ Fig. 139 Page 120).

Fig. 138: Smoke test at the lower sample plate (at the rear side of the magnet)

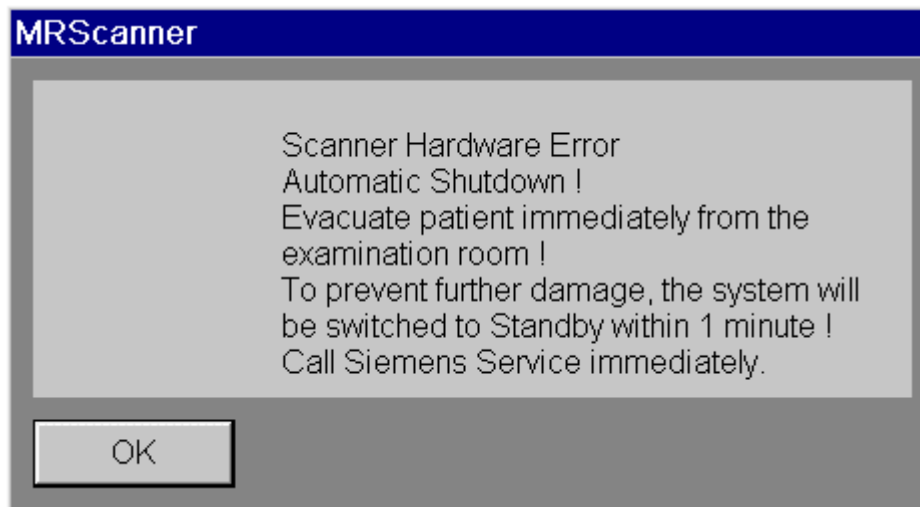


Fig. 139: Fire lamp ON



- » Subsequently the modulator in the GPA cabinet should be disabled and the message (→ Fig. 140 Page 121) should appear on the screen of the host.
- » 1 minute later the scanner should be switched off (to stand-by).

Fig. 140: Pop-up "Automatic Shutdown"



Once the FIRE lamp has been activated, it must be reset with the RESET button (→ 4/ Fig. 133 Page 115). It may take some minutes until the background level of smoke particles has returned to a normal level and a rest is possible. After a successful reset, the system can be switched on again.

- To reuse a smoke wick, simply take it out of the box and follow the steps as described above.
- The smoke wicks have a long shelf time and should be stored in a cool and dry place.



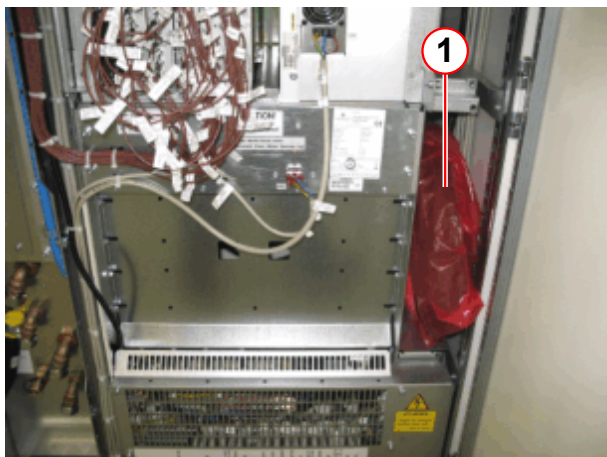
Use only as directed, do not eat and please keep away from children.

11.8 Flexa hose position in the ACC



The remaining length of the Flexa hose can be used for the optional pressure equalization.

Fig. 141: Position of the plastic bag inside the ACC



(1) Remaining Flexa hose with second bushing

Optional for pressure equalization:

The remaining length of the Flexa hose is positioned in the ACC.

Fig. 142: Remaining Flexa hose with second bushing



(1) Second bushing

(2) Remaining Flexa hose

11.8.1 General Information



The maximum hose length is 30 m.



The minimum hose lengths are (this means do not install shorter hose sections than listed below):

- VLM to T-piece on ACC/EPC > 1 m
- T-piece on ACC/EPC to T-piece at filter panel (equipment room) > 1 m
- T-piece at filter panel (examination room) to T-piece magnet > 5 m
- All other lengths are preconfigured and therefore fixed
- Observe the min/max lengths but do not make the hose longer than necessary, because additional length results in a longer reaction time.



Basically a hose system from Flexa is used for all hose connections. There are 10 mm hoses used (inside the ACC/EPC cabinet and for the gradient filter covers) and 22 mm hoses (for all other connections).



Please observe and strictly adhere to the following general hints regarding the proper handling of the Flexa connections being used for the hosing.

Fig. 143: Using the hose cutter



Proper cutting of the Flexa hose

- Prepare a piece of Flexa hose of the required length and use the blue hose cutter which is provided with the Update Kit (→ Fig. 143 Page 124)).
 - » Take care that the blade of the cutter is cutting in the groove of the hose (→ Fig. 144 Page 124) and not on top (→ Fig. 145 Page 124), because the hose can only be plugged into the T-pieces when it is properly cut.

Fig. 144: Proper cut



Fig. 145: Improper cut



Fig. 146: Proper Flexa hose connection



Proper Flexa hose connection

- Push-fit the Flexa hose hard into the T-piece, while twisting it slightly (→ Fig. 146 Page 125).
 - » It should move into the coupling for about 4 ring-section elements of the hose.
 - » In order to verify the proper connection, try to pull out the hose again. This should not be possible.
 - » Twisting the hose in the coupling should not be easy.

Proper removal of the Flexa hose from the T-piece

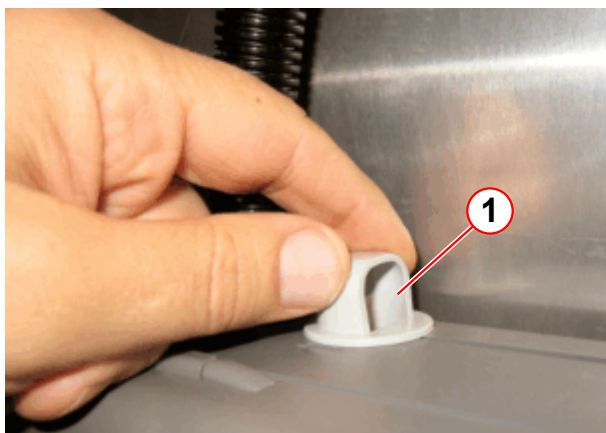
- In case you have to remove the Flexa hose from the T-Piece, just push the upper ring segment of the T-piece down while pulling out the hose.

11.9 Pressure Equalization (optional)



This procedure is only valid if the differential pressure test fails.

Fig. 147: VLM device exhaust cap removal

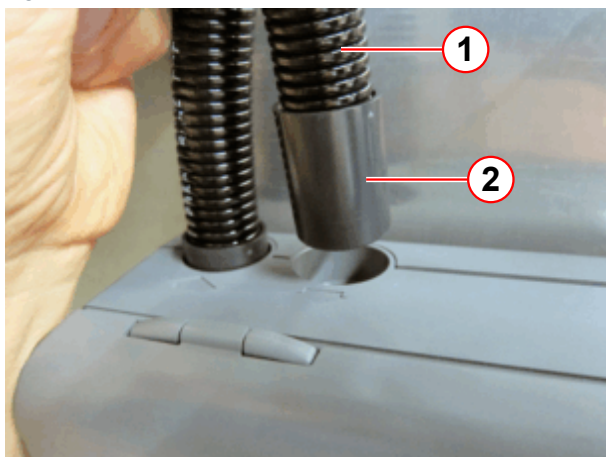


(1) Exhaust cap

Removal of the exhaust cap at the VLM device

- Remove the exhaust cap (1) on top of the VLM device

Fig. 148: VLM exhaust hose connection

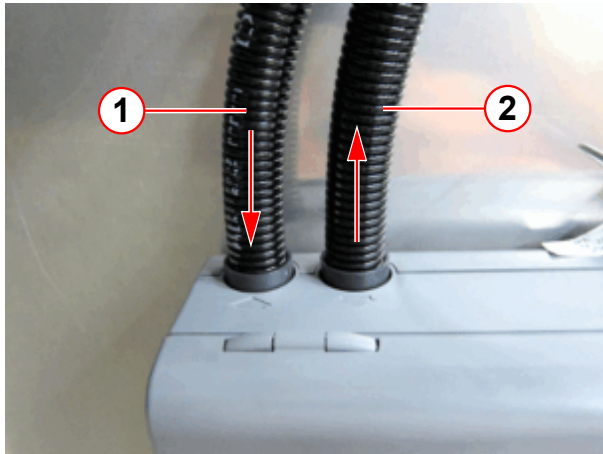


(1) Exhaust hose
(2) Bushing

Mounting of a pressure equalization hose

- Push-fit the bushing (2) onto one end of the remaining 22 mm hose roll.
- Push the bushing with the hose into the exhaust on top of the VLM device (→ Fig. 148 Page 126).

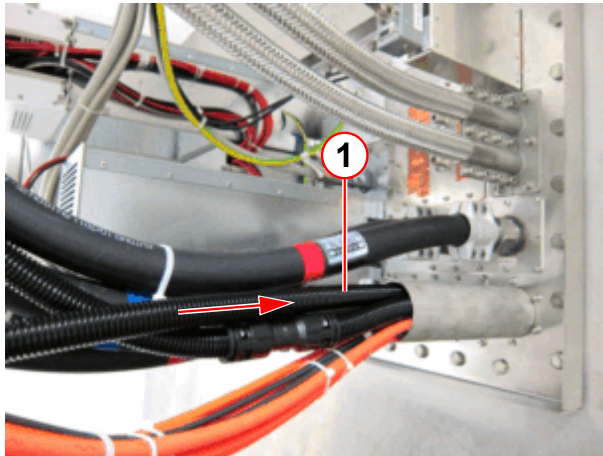
Fig. 149: VLM device with pressure equalization hose



- (1) Inlet
- (2) Exhaust

The air flow is now via hose (1) into the VLM device inlet and into the outlet hose (2).

Fig. 150: Hose fed from the equipment room

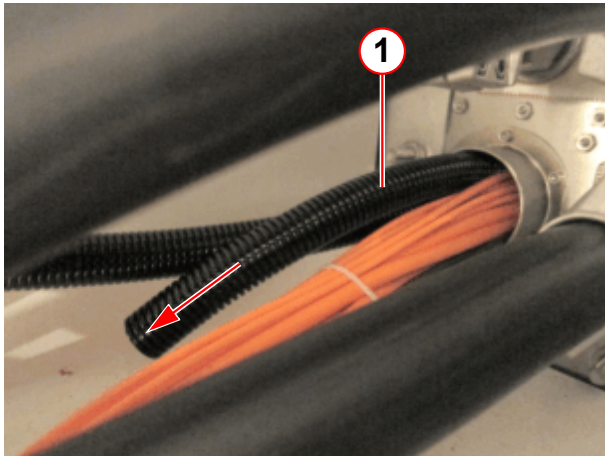


- (1) Pressure equalization hose

Feeding the pressure equalization hose through the filter plate

- Lay the pressure equalization hose (1) through the wave duct into the magnet room.

Fig. 151: Hose fed into the examination room



(1) Pressure equalization hose

Cutting the pressure equalization hose

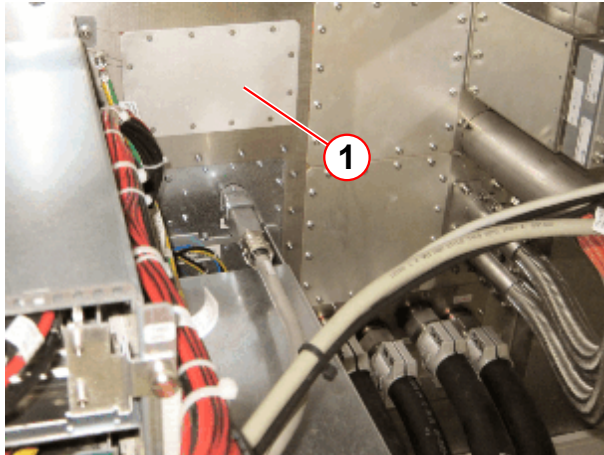
- Cut the pressure equalization hose (1) inside the magnet room at ~20-30 cm behind the wave duct.

11.10 Mounting of an additional Wave Duct (optional)



This procedure is only valid for systems where the original wave duct has not enough space for keeping the hose (-s). In this case, additional hardware is required and must be ordered from CS ML. Part-no.: 10684487

Fig. 152: RF filter plate (equipment room)



(1) Blind plate

Fig. 153: RF feed-through (optional)



Mounting the additional wave duct

- Remove the blind plate (1) from the filter plate.
- Replace it with the new wave duct (which can also be used with the MNO-option)
 - » Now the additional wave duct can be used for the hoses.

11.11 Final Work



Confirm the Test Result

If the system has no deficiencies type the following information under Install&Maintenance / Safety-Related Tests - Test Result / Evaluation / Further Description:

Installation and Start-up of the gradient supervision is completed. The function tests of the gradient supervision were performed successfully.

Fig. 154: Install&Maintenance

Name	Value
Test Result	☒ The system has no deficiencies. The image quality test resulted in no differences from required reference values. ☐ The system / component has slight deficiencies that have no affect on continued operation of the system. However they should be corrected preventively. The image quality tests resulted in no differences from required reference values. ☐ The system / component has serious deficiencies. For safety reasons, continued operation of the system is permitted only after successfully correcting the deficiencies.
Further Description	Installation and start-up of the gradient supervision is completed. The function tests of the gradient supervision were performed successfully.

12.1 Prerequisites for Tune-Up/QA

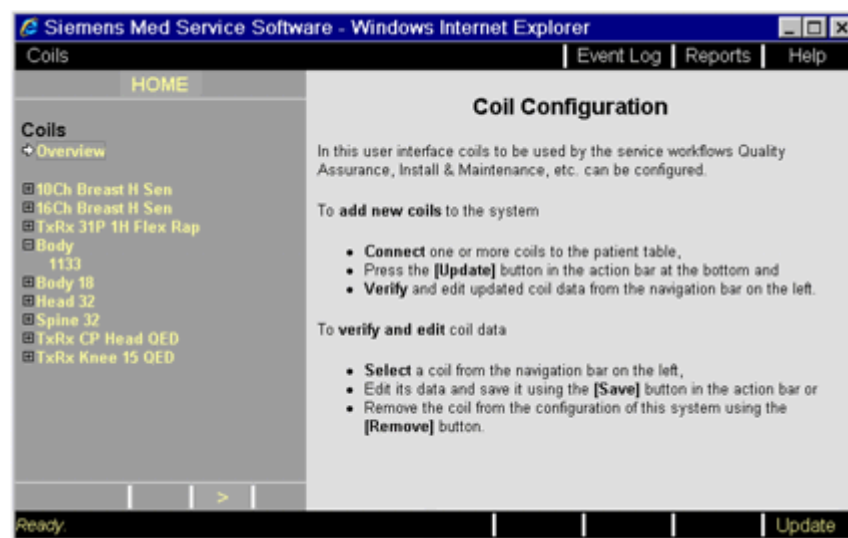
12.1.1 Configuration Check



The status of the Coil QA will be set to "ToDo" after the coil configuration. This is necessary because of the new coil asset mechanism.

- Select **Coils** in Local Service

Fig. 155: Coil Configuration



- Check the correct configuration of the Body Coil and the surface coils.
- Add missing coil configuration.

12.2 Tune-up



The PAA 36 coil may only be selected in the configuration as Peripheal Angio 36 feet (feet first) if:

- the option "Tim Whole Body Suite" is not used.
- there is only limited patient table movement.

This only indicates the coil in the Service Software QA and Installation Protocol.

The Peripheal Angio 36 head may not be used for Quality Assurance without the "Tim Whole Body Suite".

The reason is that the patient table movement distance is too short to measure all of the coil elements.

This would result in an incorrect value in the Installation Protocol.



Tune-up information is not published as hardcopy documentation. For more information, refer to the Online Help.

Performing system tune-up (Options/ Service/ Local Service/ Tune-Up). Most tune-up steps are performed completely automatically. Follow the instructions shown on the screen. In addition, pop-up windows will inform you of the actions required. Refer to the Online Help for detailed information.

12.3 QA (Quality Assurance)



Information regarding QA is not published as hardcopy documentation. For more information, refer to the Online Help.



The Service Software supports only the surface coil that has been configured in the software configuration menu (SeSo/Configuration/Measurement/Avail.Coil).

Perform the QA (Quality Assurance) of the system and of all installed surface coils (Options/ Service/ Local Service/ QA). Most QA steps are performed completely automatically. Follow the instructions shown on the screen. Pop-up windows will inform you of the actions required. Refer to the Online Help for detailed information.

12.4 Coil quality assurance



Perform the Coil Quality Assurance with all surface coils available on the system.
The system can have more than one coil of the same type.

12.5 TestTools



The Phantom Body Loader L500 is no longer delivered with the system.
The TX Power interactive Test is no longer required.



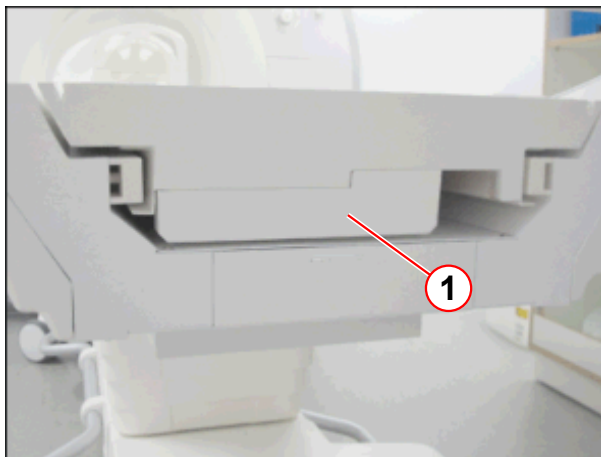
Before handing the system over to the customer, please ensure that all TestTools tests have been performed and that the results are within specifications. Please refer to the chapter "TestTools" in "Software Configuration".

13.1 Mobile Table Adaption



A mobile table from an Aera system can be used on the Avanto^{fit} system.
Only the foot-end cover of the Aera mobile table has to be changed!
The order number of the foot-end cover is 10850473.

Fig. 156: Mobile table



(1) Cover, foot-end

13.2 Safety-related tests

Perform the "Safety-related tests".

13.3 Final work steps

13.3.1 Checking the water pressure

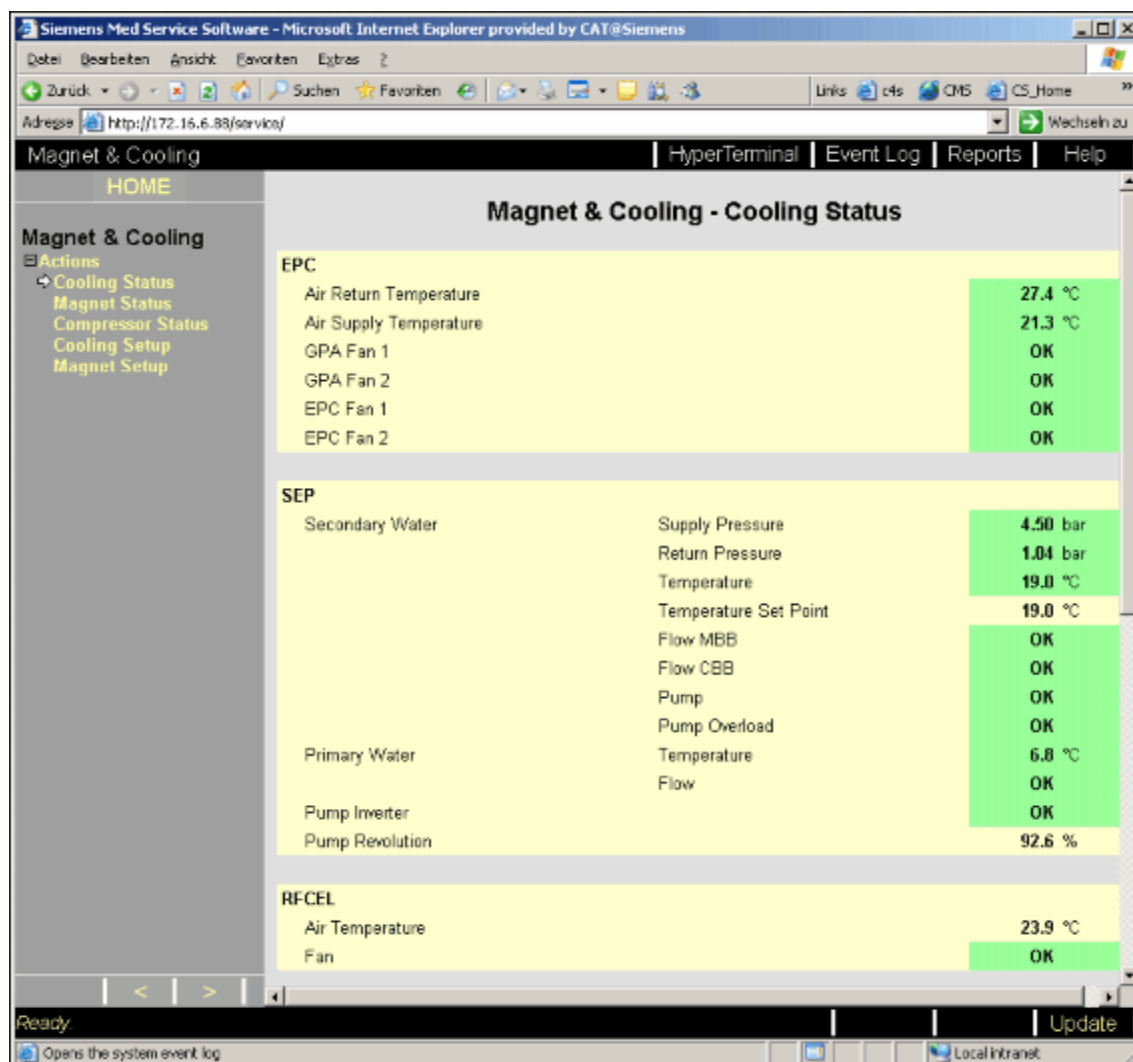


Do not use any connection to a drinking water system if it becomes necessary to refill water.

- Do a check of the water pressure via SeSo / Magnet & Cooling / Cooling Status.
- Check the water connections for leaks.

13.3.1.1 System with SEP

Fig. 157: Cooling status

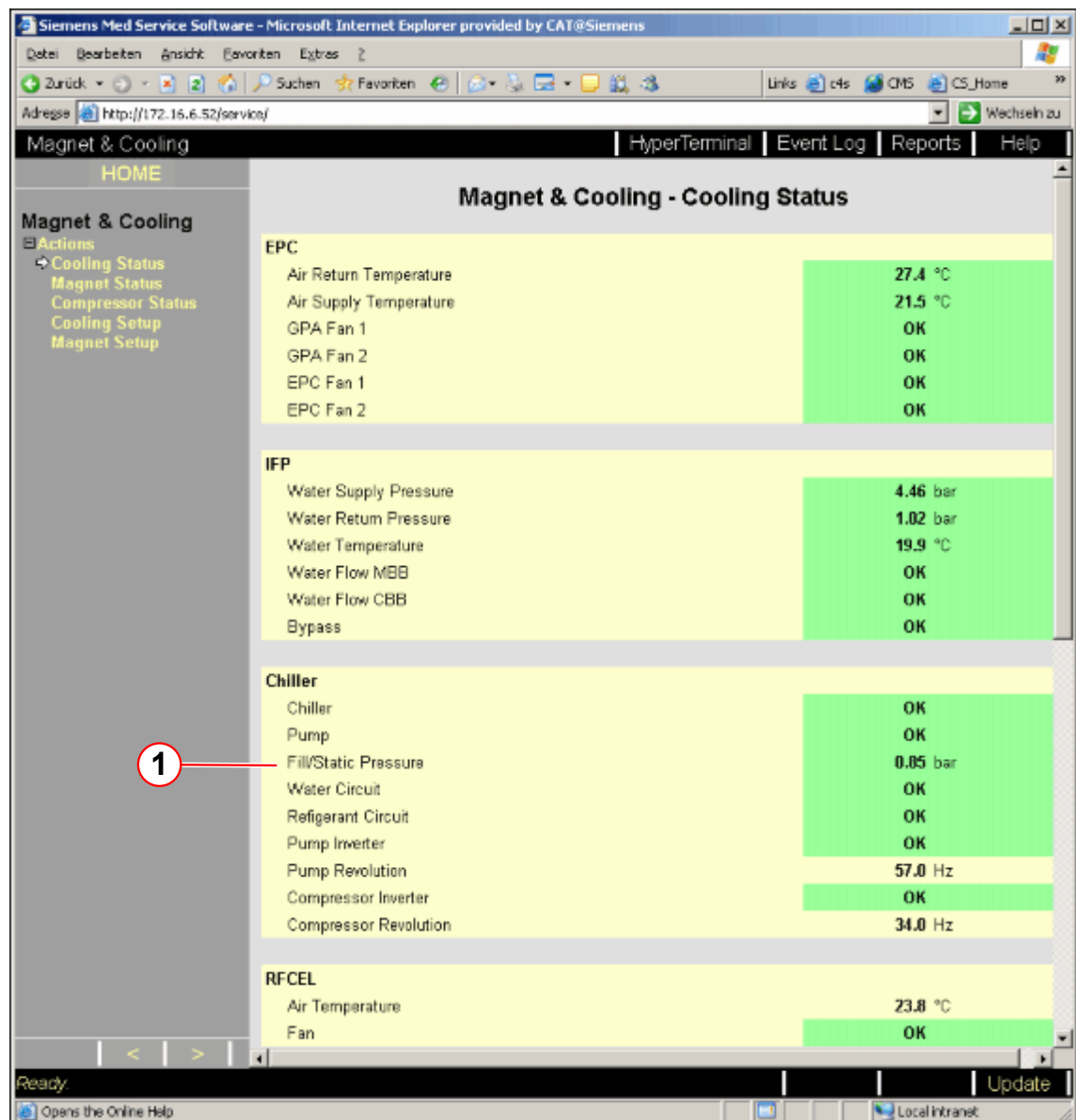


- Check the water pressure of the secondary water circuit:
 - » Return pressure should be 1 bar +/- 0.1 bar
 - » Refill if the return pressure is out of specification.

- Check the water flow and the pump revolution.
 - » Clean the strainer if the water flow or the pump revolution are outside the specifications.
- Clean the strainer if the error message "Inverter overrun" appears.

13.3.1.2 System with Chiller / IFP

Fig. 158: Cooling status



(1) Fill/static pressure

Chiller on the same level as the IFP:

- Check the fill/static pressure of the chiller.
 - » Maximum pressure is 1.5 bar.
 - » If fill/static pressure < 1.5 bar, refill with water.

Chiller higher than the IFP (limitation is 10m above the IFP)

- Check the fill/static pressure of the chiller.
 - » Maximum pressure is 0.5 bar.

13.3.2 Saving the default values

Save the magnet and cooling specific values via SeSo / Magnet & Cooling / Magnet Setup and Cooling Setup as default values.

13.3.2.1 Magnet Setup

- Open Magnet Setup platform.
- Set the checkbox of the "Save values as default". (→ 1/Fig. 159 Page 141)
- Press the **Save** button. (→ 2/Fig. 159 Page 141)
- Perform a download of the MSUP data to the Host
 - » Press **<Download>**

Fig. 159: Magnet Setup

The screenshot shows the 'Magnet & Cooling - Magnet Setup' window. The left sidebar contains a menu with 'Magnet & Cooling' and 'Actions' (Cooling Status, Magnet Status, Compressor Status, Cooling Setup, and Magnet Setup). The main area displays various settings for the magnet, including Helium Level Probes, Helium Level Warning, Shield Temperature Warning, CCR Calibration, Pressure Heater Resistance, Time Setting, MSUP, and Magnet details. At the bottom, there is a checkbox labeled 'Save values as default' (marked with a red circle 1) and a 'Save' button (marked with a red circle 2). The status bar at the bottom shows 'Ready' and buttons for 'Save', 'Default', and 'Update'.

Parameter	Unit	Value
Helium Level Probe 1 Min. (0%)	Ohm	107.0
Helium Level Probe 1 Max. (100%)	Ohm	17.9
Helium Level Probe 2 Min. (0%)	Ohm	107.0
Helium Level Probe 2 Max. (100%)	Ohm	17.9
Helium Level Probe Selection		Probe 1 (selected)
Helium Level Warning Min.	%	45
Shield Temperature Warning 50K Link Max.	K	77.0
Shield Temperature Warning 50K Bore Max.	K	84.0
CCR Calibration Room Temperature	Ohm	937.0
CCR Calibration Nitrogen Temperature	Ohm	1210.0
CCR Calibration Helium Temperature	Ohm	2784.0
Pressure Heater Resistance	Ohm	22.1
Time Setting Daily at	O'Clock	02:00
MSUP Serial Number		1104
MSUP Revision Number		1
MSUP Part Number		1
Magnet Serial Number		21225987
Magnet Revision Number		2
Magnet Part Number		10126959
Magnet Current	A	510.52
Downloads MSUP data from MSUP to host		Download
Uploads MSUP data from host to MSUP		Upload

☒ Save values as default

Ready: Save Default Update

- (1) Checkbox
(2) Save button

13.3.2.2 Cooling Setup

System with SEP

- Open Cooling Setup platform.
- Set the checkbox of "Save values as default". (→ 1/ Fig. 160 Page 142)

- Press **Save** button. (→ 2/Fig. 160 Page 142)

Fig. 160: Cooling Setup

Siemens Med Service Software - Microsoft Internet Explorer provided by Siemen...

Magnet & Cooling | HyperTerminal | Event Log | Reports | Help

HOME

Magnet & Cooling

Actions

- Cooling Status
- Magnet Status
- Compressor Status
- Cooling Setup
- Magnet Setup

Magnet & Cooling - Cooling Setup

EPC

Air Return Temperature	Max.	38.0	°C
Air Supply Temperature	Max.	27.0	°C

SEP

Secondary Water Supply Pressure	Max.	5.8	bar
Secondary Water Return Pressure	Max.	1.8	bar
	Min.	0.7	bar
Secondary Water Temperature	Max.	25.0	°C
Primary Water Temperature	Max.	14.0	°C
	Min.	4.0	°C
Secondary Water Set Point		20.0	°C

☒ Save values as default

Ready. | Save | Default | Update

(1) Checkbox

(2) Save button

System with Chiller / IFP.

- Open Cooling Setup platform.
- Set the checkbox of "Save values as default". (→ 1/Fig. 161 Page 143)

- Press **Save** buttons. (→ 2/ Fig. 161 Page 143)

Fig. 161: Cooling Setup

The screenshot displays the 'Magnet & Cooling - Cooling Setup' window. On the left, a sidebar lists 'Magnet & Cooling' and 'Actions' with sub-items: 'Cooling Status', 'Magnet Status', 'Compressor Status', 'Cooling Setup' (highlighted), and 'Magnet Setup'. The main panel, titled 'Magnet & Cooling - Cooling Setup', features an 'EPC' section with two temperature settings: 'Air Return Temperature' (Max. 38.0 °C) and 'Air Supply Temperature' (Max. 27.0 °C). Below these, a checkbox labeled 'Save values as default' is marked with a red circle '1'. To its right, a 'Save' button is marked with a red circle '2'. The bottom status bar includes the text 'Ready.' and buttons for 'Save', 'Default', and 'Update'.

- (1) Checkbox
- (2) Save button

13.3.3 Installation&Maintenance Protocol

Complete the software-based "Installation&Maintenance Protocol".

13.3.4 Quench Line Acceptance Protocol

Check to ensure that the Quench Line Acceptance Protocol is signed off by the Quench Line Manufacturer if present.

13.3.5 Dynamic Backup

Perform "Backup of dynamic data". For more information, refer to the Online Help (including the receiving address for the print-out).

13.3.6 Cleaning up



This completes the start-up of your MR system. If no other options are being installed in your system, please complete the following final steps. If you are installing other options, complete these final steps after you have installed the options. Please bear in mind that your customer has purchased an excellent, high-quality MR system with a modern design. The visual impression of the system is equally important to your customer's satisfaction.

- Assemble the surface coil cart according to the Installation Instructions.
- Close all magnet covers.
- Close all the doors of the electronics cabinet.
- Touch up paint chips with matching color paint.
- Clean fingerprints off the monitor screens.
- Clean fingerprints off the table and PMU displays.
- Clean fingerprints off the system with a mild cleaner.
- Clean the patient table and the surface coils with a mild cleaner.
- Clean the area of the system, dispose of empty cartons or other packing materials.
- Make sure that the MR system rooms are cleaned by the cleaning crew.

13.3.7 Documentation (For upgrades only)

The Magnet specific data sheet "Magnet Acceptance Documentation" or "Installation Data" should be filed in the new white binder "System Owners Manual", Register 9, Chapter "G".

13.3.8 Documentation for Review

Send back the following mandatory documents to Siemens Healthcare GmbH MR QT department.

immrqpininstallations.healthcare@siemens.com

- Safety-related Test Report
- Start-up Report
- Quench Line Acceptance Protocol incl. pictures of the quench line (provided by the project manager),
if not done by other participants like project manager or mechanical vendor.
- Pictures of the magnet room, shortly before ramping.
- If the System was manufactured in China/Shenzhen, then send the protocols to Siemens Shenzhen Magnetic Resonance Ltd., too.

Version 02

- Chapter 06
 - Renaming of the chapter.
- Chapter 08
 - Checking the patient table added.

Version 03

- Chapter (→ Preparation for Start-up / Page 9)
 - Update of the warning signs.
- Chapter (→ Setting the Water Pressure / Page 58)
 - Section "Pressure at the TX box" added.
- Chapter (→ Checking the Water Circuit / Page 34)
 - Note about MBB water circuit added

Version 04

- Update of chapter (→ Checking the Water Circuit / Page 34)

Version 05

- Update of chapter (→ Preparation for Start-up / Page 9)
 - Checking the gradient connection
- Update of chapter (→ Intercom / Page 90)

Version 06

- Update of chapter (→ Preparation for Start-up / Page 9)
 - Checking the Quench line warning label.
- Update of chapter (→ Installation&Maintenance Protocol / Page 143)
 - Company name changed.

Version 07

- Update of chapter (→ Preparation for Start-up / Page 9)
 - Deflating of the Air-Cushions.
- New chapter added (→ Startup of the Gradient Supervision / Page 110)

Version 08

- Update of chapter (→ System with SEP / Page 58)
 - Auto-refill device
 - Table of water pressure updated.

Version 09

- Update of chapter (→ Intercom / Page 90) with N4_VD13A

Version 10

- Update of the (→ Tune-Up/QA / Page 131) chapter with prerequisites for Tune-up/QA.
- Update of the (→ Posting warning labels / Page 47) chapter.
- Update of the (→ Final Steps / Page 136) chapter with mobile table adaption.

- Update of the (→ Patient Table Adjustments / Page 77) chapter.
- Update of the Software Configuration chapter with (→ ERDU Buttons / Page 55).
- Update of the (→ Setting the Water Pressure / Page 58) chapter.
- Update of the (→ Startup of the Gradient Supervision / Page 110) chapter.
- **The following systems were added:**
 - Avanto^{fit}
 - Skyra^{fit}

Version 11

- Update of the chapter (→ Final work steps / Page 138)
 - Quench Line Acceptance Protocol added.
- Update of chapter (→ TestTools / Page 135)
 - Note with Phantom Body Loader

Version 12

- Update of the chapter (→ Preparation for Start-up / Page 9)
 - Gradient coil position check added.(→ Gradient Coil position check / Page 14)
 - Section "Deflating the air cushions" moved to "Gradient coil position check".
- Update of chapter (→ Patient Table Adjustments / Page 77)
 - Section (→ Setting the laser light localizer (only required for upgrades) / Page 80) added.
- Update of the chapter (→ Final work steps / Page 138)
 - Chapter "Documentation" added.

Version 13

- Update of the chapter (→ Preparation for Start-up / Page 9)
 - Checking the gradient connections

Version 14

- Preparation for switch on :
 - Gradient cable tolerances 28 ± 1 Nm changed to 30 ± 3 Nm, 35 ± 2 Nm changed to 40 ± 4 Nm and 20 ± 2 Nm changed to 21 ± 2 Nm (Charm MR_00459191)
- Update of Chapter Gradient Supervision
 - Start- up of the gradient supervision reworked (Charm MR_00458861)
- Update of Chapter final steps:
 - Changed Siemens AG into Siemens Healthcare GmbH.
 - Quench line acceptance protocol, deleted the sign off by the project manager.

Version 15

- Update of the chapter (→ Final work steps / Page 138)
 - Table groove and belt straps

Version 16

- Update of chapter (→ Final work steps / Page 138)
 - Section, Table groove and belt straps, removed.
 - Minor editorial changes.
 - New section, documentation for review (→ Documentation for Review / Page 144) added.

Version 17

- Update of chapter (→ Checking the water pressure at the TX-box manometer / Page 36)

Version 18

- Update of chapter (→ Emergency shutdown buttons / Page 46)
 - Malfunction of Implant Suite supervision (FPO)

Version 19

- Update of chapter (→ Limiting the travel range of the patient table (whole-body option) / Page 83)

The list shows the Hazard IDs of this document.

- **EGA15.210**
 - Malfunction of the Implant Suite supervision (FPO)! (Page 46)

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